Formalizing AI security

The booklet

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How to read this booklet

This booklet holds the whole theory, from the first definitions to the open targets. Every object has a permanent slug, printed at the right of its first line. The margin says who stands behind it: open, sketch, accepted by a discussant round, checked by Halil, borrowed from a paper, or refuted. A *uses* line links what a statement or a proof relies on, and the graph in the appendix draws the same links. The list below is the queue of results that wait for your check. The chapter History records every round of every thinking space, so the booklet reads the same whatever order the rounds ran in.

Status at a glance

Counts: checked 0, accepted 30, sketch 0, open 2, borrowed 29, refuted 2.
Waiting for Halil's check:

- Definition 3.1 — `def-learning-oracle`
- Definition 3.2 — `def-learning-oracle-model`
- Definition 3.3 — `def-certifier`
- Definition 3.4 — `def-degenerate-oracles`
- Definition 3.5 — `def-isolated-client`
- Lemma 3.6 — `lem-needle-lower-bound`
- Lemma 3.7 — `lem-sphere-code`
- Theorem 3.8 — `thm-coupling-certificate`
- Theorem 3.9 — `thm-black-box-certifiability`
- Theorem 3.10 — `thm-dataflow-composition`
- Theorem 3.11 — `thm-bounded-deviation-transfer`
- Theorem 3.12 — `thm-replay-audit-steer`
- Definition 4.1 — `def-uniform-security-profile`
- Lemma 4.2 — `lem-scrambled-seed-urn`
- Theorem 4.3 — `thm-shadow-run-price-finite-views`
- Proposition 4.4 — `prop-reconstruction-label-sandwich`
- Lemma 5.2 — `lem-sq-learner-post-processing`
- Definition 5.3 — `def-seed-blind-reconstruction-game`
- Lemma 5.4 — `lem-equivariant-view-coupling`
- Lemma 5.5 — `lem-whitening-gives-rotated-records`
- Proposition 5.6 — `prop-sq-blind-reconstruction-bound`
- Proposition 5.7 — `prop-fine-reconstruction-to-hclwe`
- Theorem 5.8 — `thm-blind-split-seed-separation`
- Theorem 5.9 — `thm-reconstruction-from-conditional-mean`
- Proposition 5.10 — `prop-sq-label-relevant-reconstruction-bound`
- Definition 6.1 — `def-seed-access-modes`
- Proposition 6.2 — `prop-drawn-mode-is-fs-oracle`
- Theorem 6.3 — `thm-coupling-free-soundness`
- Proposition 6.4 — `prop-profile-run-coordinate`
- Theorem 6.5 — `thm-exact-twin-price`

Notation

$\mathrm{H}_\gamma(P\|Q)$		hockey-stick divergence, $\sup_E(P(E)-\gamma Q(E))$	Borrowed result 2.1
TV		total variation, H_γ at $\gamma=1$	Borrowed result 2.1
$k^\star$		generative leap exponent of a multi-index model	Borrowed result 2.2
$\mathsf{L}(x;\rho,\xi)$		learning function at input x, seed ρ, hardware randomness ξ	Definition 3.1
$\mathsf{Train}(x,s)$		oracle training call, s a seed name or $\perp$	Definition 3.1
$\epsilon_b(\Delta)$		exceedance probability of the coupled execution under H_b	Definition 3.5
$\beta_\gamma(K)$		coupled divergence of channel K	Theorem 3.8
$\delta_{\mathcal{N}}(\varepsilon;r)$		hockey-stick divergence of two unit Gaussians at distance r	Theorem 3.10
$\mathfrak{e}$		push-to-noise budget $(R-1)B_A^2/(2\sigma^2)$ of a deviating coalition	Theorem 3.11
$g_{\mathfrak{J},\mathsf{P}}(\beta)$		uniform regret of class $\mathfrak{J}$ under policy P at budget β	Definition 4.1
$\bar{\kappa}(\eta)$		label-weighted best ball mass of the record given its label	Proposition 4.4
$\mathcal{P}_H$		split protocols equivariant under a group $H \le O(d)$	Definition 5.3
cost $(\mathsf{work},\mathsf{size})$	$=$	cost meter of one training run: work and examples	Definition 6.1
$\epsilon_b^\Theta(\Delta)$		exceedance probability of the execution coupled by twin rule Θ	Theorem 6.3
$\Lambda_{\alpha,\alpha'}$		$(1-\alpha')\ln((1-\alpha')/\alpha)$, the confidence factor of a twin price	Theorem 6.5

Contents

1 Objective **6**
- 1.1 The question . . . 6
- 1.2 The objective, as proposed on 2026-10-25 . . . 6
- 1.3 Two sides of every question . . . 6
- 1.4 Constraints set by Halil . . . 6

2 Borrowed results **7**
- 2.1 From the space oracle-and-measurement . . . 7
- 2.2 From the space computational-limits . . . 8
- 2.3 From the space statistical-limits . . . 10

3 The learning oracle **13**
- 3.1 Definitions . . . 13
- 3.2 Results . . . 14
- 3.3 Further results, stated . . . 15

4 Statistical limits **16**
- 4.1 The profile of an unbounded adversary . . . 16
- 4.2 The training-run price of an unbounded attacker . . . 16
- 4.3 Halil's reconstruction example, statistically . . . 17

5 Computational limits **19**
- 5.1 A refuted bound, and what survives . . . 19
- 5.2 The reconstruction games . . . 20
- 5.3 Prices of blind reconstruction . . . 21
- 5.4 The label-relevant coordinate . . . 22
- 5.5 The open statement . . . 22

6 The oracle and measurement **24**
- 6.1 One oracle, two seed-access modes . . . 24
- 6.2 Couplings, soundness and accuracy . . . 25
- 6.3 What an experiment certifies about a security profile . . . 25
- 6.4 The price of twins . . . 26

7 Conjectures and targets **27**

8 History **28**

A Dependency graph **30**

B Full statements and proofs from the thinking spaces **31**
B.1 LiRA prices membership inference in shared shadow runs 31
B.2 sample-based membership inference needs order $n + n^2\rho^2$ samples . . . 31
B.3 a noise seed with 100 values is detected by the audit 32
B.4 a GPU rerun at one seed ends far from the original run 32
B.5 logged rounding makes GPU reruns identical 33
B.6 a prover that picks the batches passes replay 33
B.7 statistical query bounds for planted biclique 34
B.8 when statistical queries bind gradient training 34
B.9 the seed-known inverter 35
B.10 homogeneous CLWE is a hard hidden-direction law 35
B.11 pairwise correlations of pancake laws 35
B.12 non-Gaussian component analysis is SQ-hard under moment matching 36
B.13 the price of finding a label-relevant direction 36
B.14 the symmetry mechanism is known 36
B.15 reconstruction security against the narcissus baseline 37
B.16 with the covariance known the sample price falls to order min of n and inverse squared noise 37
B.17 differential privacy bounds the advantage of an informed membership attacker 37
B.18 a loss-threshold attack gains the normalized generalization error . . . 38
B.19 the LiRA game excludes members from the OUT draw 38
B.20 reconstruction robustness and its oblivious baseline 38
B.21 Rényi and pure differential privacy bound reconstruction by the baseline 39
B.22 how many of the trainer's outputs a reconstruction attacker must see . 39
B.23 reconstruction thresholds for subset sums 40
B.24 mutual information bounds the gain of a reconstruction that beats the prior guess 40
B.25 a Gaussian release has mutual information set by its covariance 41
B.26 an accurate representation must leak a correlated attribute 41
B.27 reconstruction by a learned reconstructor costs thousands of shadow runs and needs the initialization 41
B.28 the uniform regret profile of an unbounded adversary in learning units 42
B.29 a random scramble of seed cells turns chosen and opened seeds into fresh draws 44
B.30 the training-run price of distinguishing two training inputs on k views 45
B.31 the trainer's error and the representation's residual information bracket an unbounded reconstructor 50
B.32 the proof note's main theorem 51
B.33 what survives of the proof note 53
B.34 the reconstruction games with and without the seed 53
B.35 the server's view is a function of rotated records 54
B.36 a frozen linear client hands over Haar-rotated records 55
B.37 SQ price of blind reconstruction from a frozen client 56
B.38 polynomial-time price of fine blind reconstruction 57
B.39 the seed separates attackers for pancake records 57
B.40 a good conditional-mean predictor hands over the label direction . . . 58
B.41 what the labels cost the server 59
B.42 withdrawn as stated 60
B.43 the seed is worth the NGCA price of the record law 60
B.44 one learning oracle with two seed-access modes 61
B.45 the two FORMAL_SECURITY oracles are drawn-mode callers 62

B.46 shared seeds buy accuracy, never soundness, for bottom-oracle targets 63
B.47 what a finite experiment certifies about a security profile 65
B.48 certifying leakage at a zero-leakage function costs exact twin runs . . . 67

C The learning-oracle theory in full **71**
C.1 Definitions and prominent results of the learning oracle 71
C.1.1 What the theory shows 71
C.1.2 Definitions 72
C.1.3 Standing facts 75
C.1.4 Needle lower bound [lem-needle-lower-bound] 76
C.1.5 Sphere code [lem-sphere-code] 76
C.1.6 Coupling certificate [thm-coupling-certificate] 77
C.1.7 Black-box certifiability [thm-black-box-certifiability] 78
C.1.8 Dataflow composition [thm-dataflow-composition] 79
C.1.9 Bounded-deviation transfer [thm-bounded-deviation-transfer] . 83
C.1.10 Replay-audit steering [thm-replay-audit-steer] 86
C.1.11 Further accepted results 88
C.1.12 Open problems 89
C.1.13 Prior work 89

Chapter 1

Objective

1.1 The question

The theory measures the security of a learning procedure in the units of learning itself. A learning procedure trains models on data held by one or more parties, and an adversary wants to distinguish, invert or steer. The theory asks how much an adversary must spend to succeed, and in which currency: samples, training runs, queries, or the compute of a stated class of learners.

1.2 The objective, as proposed on 2026-10-25

Price an adversary against a learning procedure in the same units as the procedure's own training. Prove the prices by reduction to learning problems whose thresholds are known, with or without the learning oracle. Measure the same prices by experiment, the oracle supplying the training runs. The headline result would be a natural procedure where the attacker's price exceeds the trainer's by a margin that is named and measured. The proposal is open until Halil adopts it, and the decision is recorded in `lab/theory/STATE.md`.

1.3 Two sides of every question

The statistical side asks what an adversary can do with unlimited computation from the information it sees. Its tools are total variation, hockey-stick divergences, sample complexity of tests and rate distortion. The computational side asks what an adversary that must learn efficiently can do, for instance a learner trained by gradient methods. Its tools are statistical-query and low-degree lower bounds, the exponents that set sharp sample thresholds, and reductions between learning problems. A result of this theory states which side it is on, and the most valuable results relate the two.

1.4 Constraints set by Halil

- A cryptographic plug-in placed into a procedure does not count. Finding that some aspect of a model reduces to or yields a pseudorandom generator or a one-way function would count.
- Every quantity must be realizable, or approximately realizable, by experiment.
- Polynomial-time adversaries are admitted. Definitions alone are not results.
- Beautiful mathematics comes first. Any of the three rules above may be broken for a result whose mathematics is beautiful, and ugly mathematics is not wanted even when it obeys them.

Chapter 2

Borrowed results

Results taken from the literature. Each names its paper, its library key in `lib/literature/`, its theorem number when known, and its entry in `sink/formalizing/facts.md`. The thinking spaces add to this chapter.

Borrowed result 2.1 (Tests and data processing). *fact-test-fact-data-processing*

borrowed from the literature

Let P_0, P_1 be laws on one measurable space and $\gamma \geq 1$. The best advantage of a test, $\sup_E (P_1(E) - P_0(E))$ over measurable E, equals $\mathrm{TV}(P_0, P_1)$. For every Markov kernel K, $\mathrm{H}_\gamma(KP_0 \| KP_1) \leq \mathrm{H}_\gamma(P_0 \| P_1)$, where $\mathrm{H}_\gamma(P\|Q) = \sup_E (P(E) - \gamma Q(E))$. The divergence H_γ is jointly convex under mixtures.

Source. Standard: Neyman and Pearson for the test, data processing for kernels.

Borrowed result 2.2 (Generative leap). *fact-generative-leap-multi-index-rate*

borrowed from the literature

For Gaussian multi-index models with generative leap exponent $k^\star$, detecting the index space takes $n = \Theta(d^{\max(1,k^\star/2)})$ samples. The lower bound holds for low-degree algorithms, and an efficient algorithm meets it.

Source. Damian, Lee and Bruna, NeurIPS 2025, library key DLB25, theorem number to be filled when the paper is read. Facts: fact-generative-leap-multi-index-rate, and fact-generative-exponent-measures-detection, which records that the bound measures detection, not estimation.

2.1 From the space oracle-and-measurement

Borrowed result 2.3 (LiRA prices membership inference in shadow runs). *fact-lira-priced-in-training-runs*

borrowed from the literature

The likelihood-ratio attack trains N shadow models on random halves of a data pool, so each target lies in $N/2$ of them, with $N = 256$ and 64 on ImageNet, and one bank serves every target. Its true-positive rate at low false-positive rate saturates near 64 shadow models.

Source. Carlini, Chien, Nasr, Song, Terzis and Tramèr, IEEE S&P 2022, library key CCNSTT22, §IV-C (Algorithm 1), §V-A with Fig. 5, §VI-B with Fig. 9. Empirical. Accepted by argon in round 00.

Full statement and proof: Section B.1.

Borrowed result 2.4 (Sample price of membership against a noisy mean). *fact-hsu-membership-sample-gap*

borrowed from the literature

Against the noisy empirical mean over all Gaussians, no membership test with m auxiliary samples succeeds with probability 0.51 on each side when $d > c_1(n + n^2\rho^2)$ and $m \leq c_2(n + n^2\rho^2)$. A matching attack exists against symmetric estimators at the critical dimension. With known identity covariance no test exists for $d < c(n + n^2\rho^2 + n^2/m)$.

Source. Haghifam, Smith and Ullman, arXiv 2508.19458 v1 (FOCS 2026 to appear), library key HSU25, Theorems 3.1, 3.3 and 4.1, Definitions 2.1, 2.4 and 3.2. Accepted by argon in round 02.

Full statement and proof: Section B.2.

Borrowed result 2.5 (A small seed space breaks the privacy claim). `fact-small-seed-space-breaks-dp`

borrowed from the literature

A DP-SGD implementation whose noise seed takes 100 values is detected as violating its claimed $\varepsilon = 1.27$ by the Gaussian-DP audit and missed by (ε, δ) audits.

Source. Nasr, Hayes, Steinke, Balle, Tramèr, Jagielski, Carlini and Terzis, USENIX Security 2023, library key `NHSBTJCT23`*, §6.3, Violation 2, Fig. 8. Accepted by argon in round 02.*

Full statement and proof: Section B.3.

Borrowed result 2.6 (Uncontrolled reruns at one seed diverge). `fact-rerun-reproduction-error`

borrowed from the literature

Whole GPU training reruns at one seed end at an ℓ_2 distance comparable to the distance between independently trained models, while a single step nearly reproduces.

Source. Jia, Yaghini, Choquette-Choo, Dullerud, Thudi, Chandrasekaran and Papernot, IEEE S&P 2021, library key `JYCDTCP21`*, Theorem 1 and Table I. Empirical. Accepted by astra in round 01.*

Full statement and proof: Section B.4.

Borrowed result 2.7 (Controlled nondeterminism replicates reruns exactly). `fact-controlled-nondeterminism-exact`

borrowed from the literature

With deterministic settings reruns are identical on one GPU type and diverge across types. Logging the rounding decisions replicates training exactly across three GPU types at 1.2 to 1.7 times the training time.

Source. Srivastava, Arora and Boneh, NeurIPS 2024, library key `SAB24`*, §6. Empirical. Accepted by astra in round 01.*

Full statement and proof: Section B.5.

Borrowed result 2.8 (A prover that picks the batches passes replay). `fact-unlearning-verification-fragile`

borrowed from the literature

A proof of retraining, valid when every step replays within ε and no batch meets the unlearned set, is passed at $\varepsilon = 0$ by retraining with each unlearned point replaced by its nearest same-class neighbour, and at $\varepsilon > 0$ by forging from the logged proof in at most one run's time.

Source. Zhang, Chen, Shen and Li, ICML 2024, library key `ZCSL24`*, Definition 4.1, Propositions 4.2 to 4.5. Accepted by argon in round 02.*

Full statement and proof: Section B.6.

2.2 From the space computational-limits

Borrowed result 2.9 (SQ bounds for planted biclique). `fact-fgrvx17-planted-clique-sq-bound`

borrowed from the literature

For every constant $\delta > 0$, every $k \le n^{1/2-\delta}$ and every $r > 0$, detecting or finding a planted k-biclique with probability $2/3$ needs $n^{\Omega(\log r)}$ queries to $VSTAT(n^2/(rk^2))$, and a 1-bit sampling algorithm needs $\Omega(n^2/k^2)$ calls to 1-STAT. Search problems are bounded through the statistical dimension.

Source. Feldman, Grigorescu, Reyzin, Vempala and Xiao, J. ACM 64(2), 2017, library key `FGRVX17`*, Theorems 2.9, 2.10, 3.2 and 5.4, Lemma 3.10, Corollary 3.12. Accepted by astra in round 00.*

Full statement and proof: Section B.7.

Borrowed result 2.10 (When statistical queries bind gradient training). `fact-sq-simulates-minibatch-sgd`

borrowed from the literature

A mini-batch SGD method with batch size times squared precision above a logarithmic factor is simulated by an SQ method, so SQ lower bounds bind it, under worst-case rounding. At fine precision, constructed models simulate PAC learning, so a generic SQ bound does not cover the whole fine-precision class.

Source. *Abbe, Kamath, Malach, Sandon and Srebro, NeurIPS 2021, library key* `AKMSS21`*, Theorems 1a and 1c. Accepted by astra in round 01.*

Full statement and proof: Section B.8.

Borrowed result 2.11 (Random expansive ReLU layers invert by linear programs)**.** `fact-lei-random-expansive-layers-invertible`

borrowed from the literature

If every layer of a ReLU network has i.i.d. Gaussian weights and expands its input by a factor at least 2.1*, then for any input the input is recovered from the output by layer-wise linear equations with probability* $1 - e^{-\Omega(k)}$ *over the weights. Deciding whether a preimage exists is NP-complete in general.*

Source. *Lei, Jalal, Dhillon and Dimakis, NeurIPS 2019, library key* `LJDD19`*, Assumption 1, Lemma 1, Theorems 1 to 3. Accepted by astra in round 00.*

Full statement and proof: Section B.9.

Borrowed result 2.12 (Homogeneous CLWE is a hard hidden-direction law)**.** `fact-clwe-hardness`

borrowed from the literature

For $\beta \in (0,1)$*,* $\gamma \geq 2\sqrt{d}$ *and* γ/β *polynomial, distinguishing the pancake law* $H_{w,\beta,\gamma}$ *from the Gaussian is as hard as a worst-case lattice problem for quantum polynomial time. Its covariance is* $I/(2\pi)$ *off* w *and* $1/(2\pi) - \Delta$ *along* w *with* $0 \leq \Delta \leq 3(\gamma^2 + 1)e^{-\pi\gamma^2}$*, and the law determines* w *only up to sign.*

Source. *Bruna, Regev, Song and Tang, STOC 2021, library key* `BRST21`*, Definitions 2.19 and 2.21, Corollary 4.2, Lemma 7.2, Corollary 7.3, Theorem 7.5, Corollary 9.4. The covariance clause is proven in the round file. Accepted by argon in round 02.*

Full statement and proof: Section B.10.

Borrowed result 2.13 (Pairwise correlations of pancake laws)**.** `fact-hclwe-sq-correlation-bound`

borrowed from the literature

Pancake laws with nearly orthogonal hidden directions have pairwise correlations, relative to the Gaussian, exponentially small in γ^2*, which gives SQ lower bounds for detecting and finding the direction.*

Source. *BRST21, Proposition 8.3 and Theorem 8.1. Accepted by astra in round 01.*

Full statement and proof: Section B.11.

Borrowed result 2.14 (NGCA is SQ-hard under moment matching)**.** `fact-ngca-sq-lower-bound`

borrowed from the literature

Non-Gaussian component analysis testing is hard for SQ algorithms when the hidden law approximately matches the Gaussian moments to a given degree, with no chi-square condition. The bound covers SQ algorithms only. Lattice reduction solves some noiseless moment-matched instances with $O(n)$ *samples.*

Source. *Diakonikolas, Kane, Ren and Sun, NeurIPS 2023, library key* `DKRS23`*, Theorem 1.5, Definitions 1.2 and 1.3. Accepted by astra in round 01.*

Full statement and proof: Section B.12.

Borrowed result 2.15 (The price of a label-relevant direction)**.** `fact-generative-exponent-sq-recovery-bound`

borrowed from the literature

For Gaussian single-index models with generative exponent k^**, SQ algorithms need of order* $d^{k^*/2}$ *samples to find the direction, and a sample-based algorithm with a label denoiser matches it. The lower bound measures detection first.*

Source. *Damian, Pillaud-Vivien, Lee and Bruna, COLT 2024, library key* `DPLB24`*, Theorem 3.2, Theorem 4.1, Remark 3.3, Lemmas 2.3, E.2 and E.3. Accepted by astra in round 01.*

Full statement and proof: Section B.13.

Borrowed result 2.16 (Orthogonal equivariance of training)**.** `fact-orthogonal-equivariance-of-training`

borrowed from the literature

A rotation-invariant initialization with an equivariant update gives an orthogonally equivariant learner. Gradient descent from i.i.d. Gaussian initialization is orthogonally

equivariant, momentum included, while Adam and AdaGrad are equivariant only under permutations and sign flips.

Source. Li, Zhang and Arora, arXiv 2010.08515, library key `LZA21`*, Definition 3.2, Theorem C.1, Remarks C.1 and C.2, Corollary C.2, Table 1. Accepted by astra in round 01.*

Full statement and proof: Section B.14.

Borrowed result 2.17 (Narcissus resiliency and one-wayness). `fact-cohen-narcissus-equals-one-wayness`

borrowed from the literature

Reconstruction security measured against the baseline of a fresh sample from the same law, Narcissus resiliency, is equivalent, for the preimage relation of an efficiently computable well-spread function, to that function being one-way.

Source. Cohen, Kaplan, Mansour, Moran, Nissim, Stemmer and Tsfadia, ITCS 2025, library key `CKMMNST25`*, Definitions 1.6, C.1 and C.4, Theorem C.5. Accepted by astra in round 00.*

Full statement and proof: Section B.15.

2.3 From the space statistical-limits

Borrowed result 2.18 (With the covariance known the sample price falls). `fact-hsu-gap-prices-covariance`

borrowed from the literature

For identity-covariance Gaussians, $n \geq 4$, $m \geq 2$ and $d < c(n + n^2\rho^2 + n^2/m)$, no sample-based membership attack against the noisy mean exists, so at the critical dimension an attack needs $m \gtrsim \min\{n, 1/\rho^2\}$ reference samples. The price of Borrowed result 2.4 at that dimension is the price of not knowing the covariance.

Source. HSU25, Theorem 4.1 and Remark 4.2, with $\rho > 0$. Accepted by astra and argon.

Full statement and proof: Section B.16.

Borrowed result 2.19 (Differential privacy bounds informed membership advantage). `fact-yeom-dp-bounds-membership-advantage`

borrowed from the literature

An ε-DP learner gives membership advantage at most $e^{\varepsilon} - 1$ against every adversary of the membership experiment, which is handed the data distribution.

Source. Yeom, Giacomelli, Fredrikson and Jha, IEEE CSF 2018, library key `YGFJ18`*, Experiment 1, Definition 4, Theorem 1. Accepted by astra.*

Full statement and proof: Section B.17.

Borrowed result 2.20 (A loss threshold gains the generalization error). `fact-yeom-generalization-error-gives-adva`

borrowed from the literature

For a loss bounded by B, the adversary that outputs one with probability $\ell(A_S, z)/B$ has advantage equal to the average generalization error divided by B, with one model evaluation and no sample or training run.

Source. YGFJ18, Theorem 2 and Definition 3. Accepted by astra and argon.

Full statement and proof: Section B.18.

Borrowed result 2.21 (The LiRA game excludes members from the OUT draw). `fact-lira-low-fpr-yeom-game`

borrowed from the literature

The membership game of LiRA draws its OUT point from the distribution conditioned to lie outside the training set, with the bit renamed against Yeom's experiment, so the two games agree when that event has probability one.

Source. CCNSTT22, Definition 1 in §III-A. Accepted by astra and argon.

Full statement and proof: Section B.19.

Borrowed result 2.22 (Reconstruction robustness and its baseline). `fact-balle-reconstruction-robustness-definit`

borrowed from the literature

A mechanism is (η, γ)-reconstruction-robust for a prior and an error when every deterministic reconstructor from the released model, against an informed adversary holding all other records, succeeds within η with probability at most γ. The baseline $\kappa(\eta)$ is the best chance of a fixed guess under the prior.

Source. *Balle, Cherubin and Hayes, IEEE S&P 2022, library key* `BCH22`*, Definitions 1 and 2, §VII-B. Accepted by astra and argon.*

Full statement and proof: Section B.20.

Borrowed result 2.23 (Differential privacy bounds reconstruction by the baseline)**.**
`fact-balle-dp-bounds-reconstruction`

borrowed from the literature

An (α, ε)-RDP mechanism is reconstruction-robust with $\gamma = (\kappa e^{\varepsilon})^{(\alpha-1)/\alpha}$, an ε-DP one with $\gamma = \kappa e^{\varepsilon}$, and a ρ-zCDP one with $\gamma = \exp(-(\sqrt{\log(1/\kappa)} - \sqrt{\rho})^2)$. Conversely, robustness against every two-point prior gives differential privacy.

Source. *BCH22, Theorem 2, Corollaries 3 and 4, Propositions 6 and 7, Theorem 5. Accepted by astra and argon.*

Full statement and proof: Section B.21.

Borrowed result 2.24 (How many outputs a reconstruction attacker must see)**.**
`fact-reconstruction-query-lower-bounds`

borrowed from the literature

For every (ε, δ)-DP learner on a compact metric space, an adversary that sees n independent outputs has worst-case squared reconstruction error of order at least the squared diameter times $(2(1-\delta)/(e^{\varepsilon}+1))^n$. The claimed matching upper bound and its Rényi extension are not established.

Source. *Mukherjee and Lokam, arXiv 2303.16372v6, library key* `ML24`*, Theorem 9. Theorem 10 rests on an adversary that uses the true target, and the proof of Lemma 16 bounds the total variation from the wrong side. Accepted by argon in round 02.*

Full statement and proof: Section B.22.

Borrowed result 2.25 (Reconstruction thresholds for subset sums)**.** `fact-dinur-nissim-reconstruction-thresholds`

borrowed from the literature

One reconstruction theorem uses all 2^n subset-sum answers and exponential time, and another uses $n(\log n)^2$ random answers and a linear program. The two results change the number of answers as well as the running time.

Source. *Dinur and Nissim, PODS 2003, library key* `DN03`*, Definitions 1 to 4, Theorems 2, 3 and 5. Accepted by astra and argon.*

Full statement and proof: Section B.23.

Borrowed result 2.26 (Mutual information bounds the gain over the prior guess)**.**
`fact-pac-privacy-mutual-information-bound`

borrowed from the literature

For a data law known to the adversary, an adversary whose failure probability is at most the optimal prior failure δ_o satisfies $\mathrm{kl}(\delta \| \delta_o) \le I(X; M(X))$, and every adversary has $\delta_o - \delta \le \sqrt{I(X; M(X))/2}$. Side knowledge conditions both sides.

Source. *Xiao and Devadas, CRYPTO 2023, library key* `XD23`*, Definition 3, Theorem 1, Corollary 1, Remark 1. Accepted by astra and argon.*

Full statement and proof: Section B.24.

Borrowed result 2.27 (A Gaussian release has information set by its covariance)**.**
`fact-pac-privacy-gaussian-release-bound`

borrowed from the literature

For a deterministic M with finite second moment and independent Gaussian noise of positive definite covariance Σ_B, $I(X; M(X) + B) \le \frac{1}{2} \log\det(I + \Sigma_{M(X)} \Sigma_B^{-1})$, and an algorithm that evaluates M on independent draws calibrates the noise with confidence.

Source. *XD23, Theorems 3 and 4, the deterministic specialization of §4.2. Accepted by astra and argon.*

Full statement and proof: Section B.25.

Borrowed result 2.28 (An accurate representation leaks a correlated attribute)**.**
`fact-zhao-accuracy-forces-attribute-leakage`

borrowed from the literature

When the target and a sensitive attribute are correlated, every representation on which some classifier is accurate lets a measurable classifier infer the attribute with an advantage that the accuracy forces, for attribute groups of positive probability.

Source. Zhao, Chi, Tian and Gordon, NeurIPS 2020, library key `ZCTG20`*, Theorems 3.2 and 3.3. Accepted by astra and argon.*

Full statement and proof: *Section B.26.*

Borrowed result 2.29 (A learned reconstructor costs thousands of shadow runs)**.**
`fact-balle-reconnn-priced-in-shadow-runs`

borrowed from the literature

A reconstructor network trained on pairs of shadow models and their target records beats the nearest-neighbour oracle from about 8,000 shadow models on MNIST and 10,000 on CIFAR-10 when the initialization is known, and fails on MNIST with a fresh initialization per model.

Source. BCH22, §IV-B, §VI-B, Fig. 5, Table III. Empirical. One attack in two experiments, not a statement that reconstruction without the initialization is impossible. Accepted by astra and argon.

Full statement and proof: *Section B.27.*

Chapter 3

The learning oracle

The first part of the theory, built in eleven builder and challenger rounds on 16 and 17 October 2026 and audited with FINAL ACCEPT. Statements are shortened here. The full statements and proofs are in the appendix, under the same slugs. The upper bounds hold for any training whose seed can be fixed. The lower bounds and the separation between named and hidden seeds need the oracle.

3.1 Definitions

Definition 3.1 (Learning oracle)**.** `def-learning-oracle`

accepted in a round, waits for Halil

A *learning function* is a measurable map $\mathsf{L}(x;\rho,\xi)$ of a training input x (learner name, starting model, dataset, hypothesis class), a seed ρ with atomless law μ, and hardware randomness ξ with law π. A *learning oracle* answers $\mathsf{Train}(x,s)$: with $s=\perp$ it draws a fresh seed and stores it under a new name, with s a seed name it reuses that seed, and in both cases it draws a fresh ξ and returns a name for $\mathsf{L}(x;\rho,\xi)$. It also answers $\mathsf{Eval}(h,u)$, the evaluation map at a stored model, and $\mathsf{Open}(h)$, which returns the stored object. Every drawn seed, every ξ and every evaluation draw is independent of all other randomness, and a call needs the label open, or eval for Eval, on every name it uses.

Definition 3.2 (Learning-oracle model)**.** `def-learning-oracle-model`

accepted in a round, waits for Halil

A *procedure* is a dataflow graph whose learning nodes call Train and whose evaluation nodes call Eval. Each node value reaches a coalition through the plain channel, the query channel (a name with label eval, a query set U and an allowance Q), or no channel. An adversary is an oracle algorithm with a budget (q_T,q_E) of its own Train and Eval calls. It distinguishes, by the hockey-stick divergence of its view under two hypotheses, or steers, by landing a node's value in a target set.

Statement uses: Definition 3.1.

Definition 3.3 (Certifier)**.** `def-certifier`

accepted in a round, waits for Halil

A *certifier* of a target $\theta(\mathsf{L})\in[0,1]$ is an experimenter that makes at most N Train calls and outputs $\hat{t}$. It is *sound* at level α over a class of learning functions when $\Pr[\theta(\mathsf{L})\le\hat{t}]\ge 1-\alpha$ for each of them. It works in the *drawn-seed discipline* when it never passes a seed and never opens one, and in the *coupled discipline* otherwise.

Statement uses: Definition 3.1.

Definition 3.4 (Degenerate oracles)**.** `def-degenerate-oracles`

accepted in a round, waits for Halil

The *bottom oracle* grants the adversary nothing beyond the procedure, with no bound on its Train calls. The *top oracle* also grants it the seed name of every honest learning node. For an experimenter or an auditor the two cases are the drawn-seed and the

coupled discipline. A result is non-trivial only if it fails or becomes vacuous in one degenerate case.

Statement uses: Definitions 3.2 and 3.3.

Definition 3.5 (Isolated client). `def-isolated-client`

accepted in a round, waits for Halil

A procedure *isolates client j behind Gaussian releases* when, for R rounds, client j trains with drawn seeds from a choice c_t and its state, no other node reads its dataset, its round-t output reaches the coalition only through the release $y_t = O_t + f_t(c_t, W_t) + Z_t$ with $Z_t \sim \mathcal{N}(0, \sigma^2 I_d)$, and its dataset node and models reach the coalition through no channel. The *coupled execution* runs a twin of client j on the other dataset with the same seeds, and the *exceedance probability* $\epsilon_b(\Delta)$ is the chance that some round's twin contribution differs by more than Δ.

Statement uses: Definition 3.2.

3.2 Results

Lemma 3.6 (Needle lower bound). `lem-needle-lower-bound`

accepted in a round, waits for Halil

If learning functions $\mathsf{L}_1, \dots, \mathsf{L}_J$ agree with a reference L° outside disjoint sets $X_1, \dots, X_J$ of training inputs and have $\theta(\mathsf{L}_k) \geq \vartheta$, then every certifier with at most N Train calls, sound at level α over a class containing them, outputs at least ϑ at L° with probability at least $1 - \alpha - N/J$.

Statement uses: Definition 3.3.

Lemma 3.7 (Sphere code). `lem-sphere-code`

accepted in a round, waits for Halil

For every $d \geq 1$ the unit sphere of $\mathbb{R}^d$ holds at least $2^{0.18d}$ points with pairwise distances at least one, and nearest-point decoding of a point of such a code under $\mathcal{N}(0, \sigma'^2 I_d)$ noise errs with probability at most $\frac{J}{2} e^{-\mathfrak{s}^2/(8\sigma'^2)}$ for J points at distance $\mathfrak{s}$.

Theorem 3.8 (Coupling certificate). `thm-coupling-certificate`

accepted in a round, waits for Halil

For one learning node seen through a channel K, every learning function and every $\gamma \geq 1$, $\mathrm{H}_\gamma(P_0 \| P_1) \leq \beta_\gamma(K) \leq \Pr_\Gamma[W_0 \neq W_1]$, where $\beta_\gamma(K)$ is the mean channel divergence over same-seed pairs. Coupled Train calls certify $\beta_\gamma(K)$ soundly for every learning function when the channel divergence at two models is computable. Through a deterministic channel with atomless view laws, no drawn-seed certifier sound for every learning function certifies the total variation below one.

Statement uses: Definition 3.3.
Proof uses: Borrowed result 2.1 and Definition 3.1.

Theorem 3.9 (Black-box certifiability). `thm-black-box-certifiability`

accepted in a round, waits for Halil

For one learning node behind the query channel with a finite query set and finitely many outputs, the best advantage of q label queries is certifiable from above and from below by drawn-seed runs, with an error that falls as the inverse square root of the number of runs, for every learning function. Drawn seeds can refute, and never certify, a claim that label queries reveal all the model reveals.

Statement uses: Definitions 3.2 and 3.3.

Theorem 3.10 (Dataflow composition). `thm-dataflow-composition`

accepted in a round, waits for Halil

If a procedure isolates client j behind Gaussian releases, every adversary under the bottom or the top oracle, with any strategy, has $\mathrm{H}_\gamma(P_0 \| P_1) \leq \delta_{\mathcal{N}}(\varepsilon; \sqrt{R}\Delta/\sigma) + \min\{\epsilon_0(\Delta), \gamma\epsilon_1(\Delta)\}$. Coupled runs of a fixed strategy certify $\epsilon_0(\Delta)$ with a number of runs free of the dimension, while drawn-seed certifiers pay a number of Train calls exponential in the smaller of d and B^2/σ^2 in the clipped case.

Statement uses: Definition 3.5.
Proof uses: Borrowed result 2.1 and Lemma 3.7.

Theorem 3.11 (Bounded-deviation transfer)**.** *thm-bounded-deviation-transfer*

accepted in a round, waits for Halil

In the same setting, a coalition whose additive inputs to the releases stay within B_A of the prescribed ones raises the exceedance probability at most to $\Upsilon_{\mathfrak{e}}$ of the protocol's own, with $\mathfrak{e} = (R-1)B_A^2/(2\sigma^2)$. Coupled runs of the honest protocol certify the bound with about $e^{(1+o(1))\mathfrak{e}}$ runs, and every certifier sound for every learning function needs $e^{(1-o(1))\mathfrak{e}}$ Train calls in the clipped aggregation instance when $\mathfrak{e}/d \to 0$.

Statement uses: Theorem 3.10.
Proof uses: Lemma 3.6.

Theorem 3.12 (Replay-audit steering)**.** *thm-replay-audit-steer*

accepted in a round, waits for Halil

When the seed is the output of a random oracle on a dataset commitment and a nonce, a prover with q random-oracle queries passes an audit of n_{rep} replays with a model in a target set with probability at most $(q+1)n_{\mathrm{rep}}$ times the chance that an honest run lands within the tolerance of the target. Grinding attains this bound within a constant factor under deterministic kernels, and an auditor without seed control cannot verify at all.

Statement uses: Definition 3.1.

3.3 Further results, stated

The appendix proves or points to each of these.

- `thm-perfect-substitution`: the real procedure and the bottom oracle give the same joint law of honest values and transcript.
- `thm-reference-access`: under deterministic kernels the top oracle's best advantage is the coupled divergence.
- `thm-active-choice-dichotomy`: certifying against a coalition with admissible choices costs runs linear in the number of choices, and is impossible for infinitely many.
- `thm-record-level-certificate`: one shared base certifies the fraction of a dataset's records whose leakage exceeds a threshold, at a cost free of the dataset size.
- `thm-smoothed-choice-certificate`: a covering certifies a norm-bounded coalition at a cost exponential in d.
- `thm-bank-soundness`: a bank on random halves estimates the random-subset game, which can differ by one from the fixed-dataset game.

Chapter 4

Statistical limits

Filled from the thinking space `statistical-limits`. It asks what an adversary with unlimited computation achieves from the information a learning procedure releases, how many samples, training runs or queries it needs, and which of these limits are known, with their exact hypotheses. Three rounds on 25 October 2026 (discussants astra in place of argon, astra, argon) ended with every object below accepted and no blocking challenge. The prices are stated in the trainer's units, so they can be set against the computational prices of Chapter 5.

4.1 The profile of an unbounded adversary

Definition 4.1 (Uniform security profile)**.** `def-uniform-security-profile`

accepted in a round, waits for Halil

Fix a procedure, a coalition, a channel interface, a game and a policy, bottom or top oracle. An instance fixes the learning function, the seed and hardware laws, the data and a reference law. A budget $\beta = (m, N)$ allows m reference samples and N Train calls. The *informed optimum* of an instance is the best success of any finite-budget adversary that may depend on the instance. Over a class $\mathfrak{J}$ of instances, the *uniform regret* at budget β is the least, over adversaries of that budget chosen before the instance, of the largest gap to the informed optimum over the class. The *attained price* at accuracy a is the set of minimal budgets at which one adversary has gap at most a on every instance, finite by Dickson's lemma. The *threshold price* of a criterion, such as the two conditional probabilities of HSU25, is the set of minimal budgets at which one adversary meets it on every instance.

Statement uses: Definitions 3.2 and 3.4.
Proof uses: Theorems 3.8 and 3.9 and Borrowed result 2.4.

Source. On one instance the regret is zero as an infimum, since an adversary that knows the instance simulates every finite-budget adversary. On the plain channel the optimum is the total variation under the bottom oracle and the top divergence under the top oracle, attained with no call. Every positive regret is therefore a price of not knowing which member of the class is the instance. For a bounded adversary the profile is not degenerate on a singleton (Chapter 5).

Full statement and proof: Section B.28.

4.2 The training-run price of an unbounded attacker

Lemma 4.2 (A scramble of seed cells makes chosen seeds fresh)**.** `lem-scrambled-seed-urn`

accepted in a round, waits for Halil

Partition the seed interval into M cells, and let a learning function return at x_0 the label that a uniformly random bijection π of the cells, composed with a labelling of law p, assigns to the seed's cell. Fix a learning setting shared across the compared laws. For any adversary with at most $N < M$ Train *calls and any* Eval *and* Open *calls, with*

chosen inputs, seed values, names and draws, its transcript law is within total variation $N^2/(2(M-N))$ of the lazy oracle that answers each new cell with a fresh draw of p. Two lazy oracles built from p and p' give transcripts within Kullback–Leibler divergence $N\,\mathrm{KL}(p\|p')$.

Statement uses: Definition 3.1.

Source. The statistical form of the switch from a random permutation to a random function. The proof is a hybrid bound integrated along the paths of the scrambled run, since the index of the t-th new-cell call varies with the path.

Full statement and proof: Section B.29.

Theorem 4.3 (Shadow-run price on finite views). `thm-shadow-run-price-finite-views`
One learning node trains x_0 or x_1, and its model reaches the adversary only through a deterministic map onto $k \geq 4$ views. Let $k' = 2\lfloor k/2 \rfloor$. (a) For every learning function, $\lfloor N/2 \rfloor$ drawn-seed calls at each input give a plug-in test within $2\sqrt{k/\lfloor N/2 \rfloor}$ of the total variation. (b) Under the bottom oracle every adversary with at most N calls has, on some learning function, a gap of at least $\min\{\frac{1}{8}, \frac{1}{32}\sqrt{3k'/N}\}$ to the optimum, and under the top oracle the same against the top optimum. (c) The same holds over learning functions with deterministic kernels, up to any $\zeta > 0$, through Lemma 4.2. (d) Under the top oracle with deterministic kernels one call at the honest seed reaches the top optimum. (e) So under the bottom oracle a regret at most $a < \frac{1}{8}$ everywhere costs at least $3k'/(1024a^2)$ Train calls, and $2\lceil 4k/a^2 \rceil$ drawn-seed calls suffice. (f) For label-only access with C outputs and allowance q_E the bounds hold with $k = C^{q_E}$, and the known upper and lower bounds differ by a factor of order $\ln|U|/(C-1)+1$ in the query set U.

accepted in a round, waits for Halil

Statement uses: Definitions 4.1 and 3.4.
Proof uses: Lemma 4.2 and Theorems 3.8 and 3.9.

Source. Chosen, opened and reused seeds and calls after the challenge do not beat the passive rate under the bottom oracle, while the honest seed collapses the price to one call under deterministic kernels. The passive rate is classical (Daniely, Sabato, Ben-David and Shalev-Shwartz, JMLR 2015, Theorem 5, as the discussants cite it, not read). The oracle part has no published statement known to the space, after one query. Without the oracle, `thm-perfect-substitution` makes (b) and (c) lower bounds on the shadow training runs of every attack that treats training as a black box. The integrator re-derived (a) to (f) and the lemma on 2026-10-25.

Full statement and proof: Section B.30.

In the trainer's units. The trainer spends one training run. An attacker that must come within a of the best advantage on every learning function with k views spends of order k/a^2 runs. LiRA's 64 to 256 runs per benchmark (Borrowed result 2.3) sit below this worst case because its Gaussian model restricts the class. The thousands of shadow runs of the learned reconstructor, and its failure without the initialization (Borrowed result 2.29), point the way part (d) does, in a different game. This paragraph is heuristic.

4.3 Halil's reconstruction example, statistically

Proposition 4.4 (The label sandwich of an unbounded reconstructor). `prop-reconstruction-label-sandwich`
Let (X, Y, Z) be a record in a Polish metric space, a finite label and a representation. Let $\bar{\kappa}(\eta)$ be the label-weighted best chance of a fixed centre falling within η of X given Y, $S^\star(\eta)$ the best chance of any reconstructor of Z, and $\varepsilon^\star$ the Bayes error of predicting Y from Z. (1) Floor. $S^\star \geq \bar{\kappa} - \varepsilon^\star$, attained by decoding the predicted label to its class centre. (2) Ceiling. If $S^\star \geq \bar{\kappa}$ then $\mathrm{kl}(S^\star\|\bar{\kappa}) \leq I(X; Z \mid Y)$, hence

accepted in a round, waits for Halil

$S^\star \le \bar{\kappa} + \sqrt{I(X;Z \mid Y)/2}$. *(3) For a Gaussian release* $Z = f(X) + B$ *the conditional mutual information is at most the label average of* $\frac{1}{2}\ln\det(I + \Sigma_{f|y}\Sigma_B^{-1})$. *(4) Both ends are tight.* $Z = Y$ *gives* $S^\star = \bar{\kappa}$, *and an injective representation gives* $S^\star = 1$.

Statement uses: Borrowed result 2.22.
Proof uses: Borrowed results 2.26 and 2.27.

Source. The floor is the lower bound of Halil's example in the statistical regime. The reconstructor recovers at least the class prior of the label the head predicts, less the head's error. Only the gain above $\bar{\kappa}$ *needs information beyond the label. For a deterministic injective representation the statistical baseline is total, so the content of the example lies in the computational regime (Theorem 5.9), or in noisy, quantized or many-to-one representations. The ceiling is XD23 Theorem 1 with side knowledge.*

Full statement and proof: Section B.31.

Chapter 5

Computational limits

Filled from the thinking space `computational-limits`. It asks three things. What can an adversary achieve that must learn efficiently, as a learner trained by gradient methods or a statistical-query or low-degree algorithm? Which known learning problems does an attack task reduce to or from? Where does the computational price depart from the statistical one? Three rounds on 25 October 2026 (discussants astra, astra in place of argon, argon) ended with no blocking challenge. The space made Halil's reconstruction example formal in split learning. The reconstructor is a learner from representations to records. After a symmetry argument its problem is non-Gaussian component analysis on the record law when the server has no labels, and the trainer's own single-index problem for the label-relevant part. No result of this chapter uses the learning oracle, and each reduction costs one honest training run.

5.1 A refuted bound, and what survives

Theorem 5.1 (The proof note's black-box reduction). *`thm-fs-sq-black-box-reduction`*
The FORMAL_SECURITY proof note `.tools/proof.tex` claims, as its Theorem 2, that a black-box adversary against a model trained by a deterministic statistical-query learner on the vertex-row encoding of a graph has advantage at most the SQ advantage for planted clique, so that with polynomially many queries of constant tolerance no black-box adversary has constant advantage. The claim is false. With the exact oracle, or an honest oracle averaging each query over $\lceil 32n^2 \ln n \rceil$ fresh draws, a learner with $O(n^2)$ queries of tolerance $1/3$ stores every row of the graph, and an unbounded adversary reading the model decides the planted clique with advantage $1 - o(1)$ at $k = n^{1/2-\delta}$.

refuted

Proof uses: Borrowed result 2.9.

Source. *Counterexample by the space, accepted line by line by astra in round 00, re-derived by the integrator. The note applies FGRVX17 to the empirical law of the n rows of one symmetric matrix, and chains a worst-oracle lower bound with the run of one particular oracle. Its quoted bound also grows with the tolerance. `prob-attacker-lower-bound-from-hard-learning` rested on this note. The registers of `sink/formalizing/` hold it as `thm-sq-hardness-black-box-attacks`, with this slug as an alias.*

Full statement and proof: Section B.32.

Lemma 5.2 (An SQ learner cannot hand over what it cannot detect). *`lem-sq-learner-post-processing`*
Let $k \le n^{1/2-\delta}$, let a learner reach its training law, uniform or planted biclique, only through m calls to 1-STAT with coins independent of the law, and let any randomized map, unbounded and white-box, read its output. If the composition distinguishes uniform from every planted law with probability $2/3$, then $m = \Omega(n^2/k^2)$. The VSTAT form holds against every valid oracle, and by Borrowed result 2.10 it covers mini-batch SGD learners in the coarse-precision regime.

accepted in a round, waits for Halil

Statement uses: *Borrowed result 2.9.*
Proof uses: *Borrowed result 2.10.*

Source. *The bound is priced in the trainer's fresh samples and binds unbounded attackers. An SQ bound on the learner binds every attacker, while an SQ bound on the attacker binds only SQ attackers.*

Full statement and proof: *Section B.33.*

5.2 The reconstruction games

Definition 5.3 (Seed-blind reconstruction games)**.** `def-seed-blind-reconstruction-game`
The protocol class $\mathcal{P}_H$, for a closed subgroup $H \leq O(d)$, holds split protocols whose client applies an H-equivariant pre-map, a first layer $W\pi(x) + b$ and any further map, whose initialization law is invariant under $W_0 \mapsto W_0 Q$ for $Q \in H$, whose update of the first layer is equivariant under right multiplication by H, and whose server, honest or not, never sees first-layer weights, their gradients, records, or record-representation pairs. Side data are invariant under H. A target is a map g of the record with finite second moment and a finite symmetry group K. In the *sample game* the attacker sees the view of one run on N records, runs in time T, and its fine success at scale ϵ is the chance that some $\kappa \in K$ puts every output within ϵ of $\kappa g(x_i)$, minus the same chance for fresh records (the narcissus baseline of Borrowed result 2.17). In the *population SQ game*, for frozen clients, the attacker has q queries of tolerance τ to the law of (representation, side data) and its weak success is the correlation of its output map with κg. The sample game reconstructs the observed records. The SQ game evaluates a reconstruction map on the record law. An experiment fixes finite representations of queries and of returned maps. Seed-known variants also give the attacker the initialization, and the *price of the seed* at a success level is the pair of budget sets at which the two attackers reach it.

accepted in a round, waits for Halil

Statement uses: Borrowed results 2.16 and 2.17.

Source. Accepted by astra in round 01. The integrator applied the wording correction of round 02 to the sentence on observed records.

Full statement and proof: Section B.34.

Lemma 5.4 (The view is a function of rotated records)**.** `lem-equivariant-view-coupling`
For a protocol in $\mathcal{P}_H$ and $Q \in H$, the view of the run with coins ω on records x equals, pathwise, the view of the run with W_0 replaced by $W_0 Q^\top$ on records Qx. Hence for Q Haar on H a simulator that runs the protocol from fresh coins on the rotated records $u_i = Qx_i$ reproduces the joint law of every attacker's output with the targets, at the cost of one training run. If the hidden part of the record law is P_V^A off a label subspace and H fixes that subspace, the attacker's task becomes NGCA coordinate recovery with a uniformly random frame.

accepted in a round, waits for Halil

Statement uses: *Definition 5.3.*
Proof uses: *Borrowed result 2.16.*

Source. *The induction of LZA21 Theorem C.1, carried to the whole transcript, an adaptive server and side data, stated as a pathwise coupling. Adam, sign updates, coordinate-wise normalization, convolutional first layers and published weights each break a hypothesis.*

Full statement and proof: *Section B.35.*

Lemma 5.5 (Whitening a frozen linear client)**.** `lem-whitening-gives-rotated-records`
For a frozen client $z = Wx + b$ with W of full column rank and right rotation invariant in law, and records of mean zero and identity covariance, the population whitening map

accepted in a round, waits for Halil

T satisfies $T(z-b) = Rx$ with $R \in O(d)$ a function of W alone, Haar-distributed and independent of the records.

Statement uses: Definition 5.3.

Full statement and proof: Section B.36.

5.3 Prices of blind reconstruction

Proposition 5.6 (SQ price of blind reconstruction)**.** `prop-sq-blind-reconstruction-bound`

accepted in a round, waits for Halil

Let the client be frozen in $\mathcal{P}_{O(d)}$, with no side data and the target $\langle v, x\rangle$. (a) For pancake records $H_{v,\beta,\gamma}$ (homogeneous CLWE, Borrowed result 2.12) with $\gamma \geq 2$ and $\beta \in (0,1)$, an attacker reaching weak success c_0 with q queries of tolerance τ makes at least $\frac{11}{288} M\tau_*^2\beta^2/\gamma^2 - d$ queries, with $M = \exp(\Omega(c_0^4 d))$ and $\tau_* = \min(\tau, \tilde{\Theta}(c_0 d^{-1/2}))$, under two margins that hold for d large and $\gamma^2 \geq C\log(d/c_0)$. (b) For records whose hidden coordinate law matches moments under the hypothesis of DKRS23 Theorem 1.5, with a fixed witness degree, the attacker needs $2^{d^{\Omega(c)}}$ queries or a tolerance polynomially small in d, provided at least four moments match.

Statement uses: Definition 5.3 and Borrowed result 2.12.

Proof uses: Lemma 5.4 and Borrowed results 2.9, 2.13 and 2.14.

Source. Part (b) is proven conditional on DKRS23 Theorem 1.5. The reducer turns a reconstruction into a direction with d truncated queries and then meets the statistical dimension of FGRVX17 Theorem 3.2. The bounds are SQ only, and lattice-reduction attackers lie outside them.

Full statement and proof: Section B.37.

Proposition 5.7 (Fine blind reconstruction is as hard as hCLWE)**.** `prop-fine-reconstruction-to-hclwe`

accepted in a round, waits for Halil

For a protocol in $\mathcal{P}_{O(d)}$, frozen or trained, on pancake records with $\gamma \geq 2\sqrt{d}$ and γ/β polynomial, an attacker of time T that reconstructs every record's hidden coordinate within c_1/γ, up to sign, with probability p gives an hCLWE distinguisher of time T plus one protocol run plus $\mathrm{poly}(d)$ and advantage $p - e^{-\Omega(d)}$. So p is negligible for every polynomial-time attacker unless a worst-case lattice problem has a polynomial-time quantum algorithm (BRST21 Corollary 4.2).

Statement uses: Borrowed result 2.12.

Proof uses: Lemma 5.4.

Full statement and proof: Section B.38.

Theorem 5.8 (The seed separates attackers for pancake records)**.** `thm-blind-split-seed-separation`

accepted in a round, waits for Halil

Let the client's first layer be $\mathrm{ReLU}(Wx)$ with i.i.d. Gaussian W of expansion at least 2.1, further layers meeting the same assumption, an orthogonally equivariant update, and pancake records with $\gamma \geq 2\sqrt{d}$. (i) A seed-known server recovers every record exactly in polynomial time, for a frozen client and, for a trained one, for the records it sees under the initial weights. (ii) A seed-blind SQ server of a frozen client pays the exponential price of Proposition 5.6 (a). (iii) A seed-blind polynomial-time server with the record-level view, frozen or trained client, reconstructs finely with negligible probability unless a worst-case lattice problem is quantum polynomial-time.

Statement uses: Borrowed result 2.11.

Proof uses: Propositions 5.6 and 5.7 and Lemma 5.4.

Source. The price of the seed is one linear solve per record against exponentially many statistical queries or a quantum lattice algorithm. Astra judges this an artificial-distribution reduction. The Gaussian layer is no plug-in, but the hardness enters through the record law. Whether that meets Halil's rule on cryptographic plug-ins is a decision for Halil. Tracking a trained client's weights from a known seed is heuristic and is not claimed.

Full statement and proof: Section B.39.

5.4 The label-relevant coordinate

Theorem 5.9 (A good predictor hands over the label direction). *thm-reconstruction-from-conditional-mean*

accepted in a round, waits for Halil

Let the client be frozen and linear of full column rank, records Gaussian, labels a noisy function of $\langle u, x\rangle$ with conditional mean m and first Hermite coefficient $\beta_1 \neq 0$. A server holding a top model whose predictor has excess square risk ε, the population whitening, and N held-out representations, recovers u up to sign within $2(\sqrt{\varepsilon}+\eta_N)/|\beta_1|$ with probability $1-\delta$, where $\eta_N = B\sqrt{d/(N\delta)}$. It then reconstructs the hidden coordinate of any records, training records included, with mean square error at most that squared distance times the top eigenvalue of their empirical second moment. This is the floor on attack success in Halil's example. The attacker's error is bounded above by a function of the attacked model's excess risk.

Statement uses: Definition 5.3.
Proof uses: Lemma 5.5.

Source. The integrator applied the wording correction of round 02 to the sentence on Halil's example. The bound holds for any server and costs one pass over N representations. It is an upper bound on attack cost.

Full statement and proof: Section B.40.

Proposition 5.10 (What the labels cost the server). *prop-sq-label-relevant-reconstruction-bound*

accepted in a round, waits for Halil

For a frozen linear client, Gaussian records and labels from a single-index law with generative exponent $k^* \geq 3$ and finite χ^2 information, all fixed as d grows. (a) An SQ attacker reaching weak success c_0 with at most $d^r - d$ queries to $VSTAT(n)$ needs $n \geq c\, d^{k^*/2}$. (b) With population whitening, the algorithm of DPLB24 on the whitened representations and denoised labels reaches the direction from $O(d^{k^*/2})$ labelled representations and reconstructs the coordinate.

Statement uses: Definition 5.3.
Proof uses: Lemmas 5.4 and 5.5 and Borrowed results 2.9 and 2.15.

Source. The server's price for the label-relevant coordinate is the generative exponent of the task, whatever the trainer's model learned. The lower bound measures detection first and the upper bound is sample-based, so the two sides are not one SQ price. The off-diagonal correlation bound uses $\lambda_k^2 \leq 1$, which holds by conditional Jensen for normalized Hermite coefficients (the integrator's reading, DPLB24 not re-read).

Full statement and proof: Section B.41.

Conjecture 5.11 (Reconstruction at the cost of training). *conj-reconstruction-at-training-cost*

refuted

Round 00 conjectured that blind reconstruction of the label-relevant coordinate is equivalent to learning the task. Part (b) is false. The labels $y = \langle u, x\rangle \xi$ with an independent sign ξ have Bayes square-loss predictor zero, so learning needs nothing, while the label law reveals $\langle u, x\rangle$ up to sign at generative exponent two. Part (a) became Theorem 5.9.

Full statement and proof: Section B.42.

5.5 The open statement

Conjecture 5.12 (The seed is worth the NGCA price). *conj-seed-price-is-ngca-price*

open

For frozen clients of $\mathcal{P}_{O(d)}$ from which a seed-known attacker recovers every record exactly in polynomial time, and records from P_V^A with A of mean zero, identity covariance and fixed dimension, the budget sets of seed-blind attackers in the SQ game (weak success) and the sample game (fine success) equal, up to polynomial factors, those of NGCA coordinate recovery for A with a uniformly random hidden frame. Settled so

far are the lower inclusion for pancakes in both games and for moment-matched laws in the SQ game, and the upper inclusion for frozen linear clients given the population whitening.

Statement uses: Definition 5.3.
Proof uses: Lemmas 5.4 and 5.5 and Propositions 5.6 and 5.7.

Source. Open: the upper inclusion for ReLU and deeper clients, where the blind server must learn the client map without the seed, the cost of estimating the whitening, and coarse reconstruction in polynomial time. Accepted in round 02 as a well-posed conjecture.

Full statement and proof: Section B.43.

Decisions proposed for Halil. The proposed first target is the seed separation Theorem 5.8, with Conjecture 5.12 as its general form and Theorem 5.9 with Proposition 5.10 as the label side. The proposed attacker class for lower bounds uses two games and never mixes them. Population SQ attackers of frozen clients give unconditional bounds, and polynomial-time attackers with the record-level view give bounds conditional on lattice hardness. The open question is whether a separation whose hard instance is a record law meets the rule on cryptographic plug-ins. The procedure is standard and no primitive is placed in it, yet the hardness comes from the data.

Chapter 6

The oracle and measurement

Filled from the thinking space `oracle-and-measurement`. Its question is how the learning oracle and experiments realize or approximate the prices of the two previous chapters, what a measured security profile can certify, and how knowing the training seed changes the learning problem an adversary faces. Three rounds on 25 October 2026 (discussants argon, astra, argon) ended with every object below accepted and no blocking challenge. The full statements and proofs are in the appendix under the same slugs.

6.1 One oracle, two seed-access modes

Definition 6.1 (Seed-access modes)**.** `def-seed-access-modes`

accepted in a round, waits for Halil

Fix a learning oracle for (L, μ, π). A game assigns each caller four things. (1) A *seed-access mode*. In the *named mode* its Train calls are those of Definition 3.1. In the *drawn mode* the oracle accepts $\mathsf{Train}(x, s)$ only with $s = \bot$, returns the model's name only, keeps the seed under no name the caller holds, and refuses Open on any seed. (2) A *return label*, open or eval, for the names its own calls return. (3) The learner names and starting models its calls may use. (4) A budget for each operation. A *cost meter* $\mathsf{cost} = (\mathsf{work}, \mathsf{size})$ maps a training input to the work of one run in a fixed machine model and the number of examples in its dataset, and a caller's *oracle cost* is the coordinatewise sum over its Train calls. The bottom oracle is the named mode with label open and no grant, the top oracle adds grants of the honest seeds, and an experimenter in the drawn mode follows the drawn-seed discipline.

Statement uses: Definitions 3.1 to 3.4.

Source. Contains, as drawn-mode callers, the passive learning oracle and the generic training oracle of FORMAL_SECURITY (Proposition 6.2). A remark of the round file turns Lemma 3.6 into a bound in oracle cost: a certifier of work at most C makes at most $C/\inf_{x \in X} \mathsf{work}(x)$ calls in the needle set X, vacuous when that infimum is zero, and a bound in oracle cost is a bound in machine compute only under FORMAL_SECURITY's no-skipping hypothesis.

Full statement and proof: Section B.44.

Proposition 6.2 (The FORMAL_SECURITY oracles are drawn-mode callers)**.** *`prop-drawn-mode-is-fs-oracle`*

accepted in a round, waits for Halil

(a) A drawn-mode caller with return label open, restricted to the recipes a game names, one fixed empty starting model and each recipe's declared initialization, has call by call the transcript law of a caller of FORMAL_SECURITY's passive learning oracle, and its oracle cost is that oracle's metered cost. (b) A drawn-mode caller with return label eval, an evaluation map for outputs, input gradients and parameter gradients, and budgets per operation realizes an access class of the generic training oracle, with reveal realized by Open *exactly when the class contains it.*

Statement uses: Definition 6.1.

Source. *FORMAL_SECURITY,* `exp/survey/paper/parked/formalization/oracle.tex`, *definitions* `def:passive` *and* `def:cost`, *and the generic training oracle of its report of 2026-10-13. The integrator checked (a) against* `oracle.tex` *and (b) against the report's operation table. Part (b) needs the starting model fixed as in (a).*

Full statement and proof: Section B.45.

The proposal for Halil. Both repositories use the one oracle of Definition 6.1, and each game fixes the four items per caller. Drawn coins lose no soundness for any bottom-oracle target and lose the accuracy that the separations of Chapter 3 measure (Theorem 6.3). Named seeds are needed for the top-oracle threat model, for replay audits, and for the accuracy of coupled certificates. Halil decides among three options. He can adopt the two modes, keep drawn coins only, which makes those three inapplicable, or keep named seeds only, which needs declared starting models whenever the meter is work.

6.2 Couplings, soundness and accuracy

Theorem 6.3 (Coupling-free soundness). `thm-coupling-free-soundness`

accepted in a round, waits for Halil

A twin rule Θ *is a sequence of laws on pairs of (seed, hardware randomness) whose two marginals are* $\mu \otimes \pi$. *In the setting of Theorem 3.10, let the* Θ*-coupled execution drive the node by one coordinate and its twin by the other, with the draws independent of the procedure's other randomness and of the release noise, and let* $\epsilon_b^{\Theta}(\Delta)$ *be its exceedance probability. (a) For one node, every coupling of the two model laws bounds* $\mathrm{H}_\gamma(P_0 \| P_1)$ *by the mean channel divergence over its pairs, and a coupling with a common seed bounds the top-oracle divergence the same way. (b) Under the bottom oracle, for every twin rule, every adversary,* $\varepsilon \geq 0$ *and* $\Delta > 0$, $\mathrm{H}_\gamma(P_0 \| P_1) \leq \delta_{\mathcal{N}}(\varepsilon; \sqrt{R}\Delta/\sigma) + \min\{\epsilon_0^{\Theta}(\Delta), \gamma\epsilon_1^{\Theta}(\Delta)\}$, *and symmetrically. Under the top oracle the same holds for twin rules that share the seed. (c) The certificates of Theorem 3.8 and Theorem 3.10 computed on* Θ*-coupled runs are sound for these targets for every* Θ, *and certify the coupled tail* $\epsilon_0(\Delta)$ *itself only under the identity coupling. (d) Neither the independence from the release noise nor the shared seed for top-oracle targets can be dropped.*

Statement uses: Theorems 3.8 *and* 3.10 *and* Definition 3.5.

Proof uses: Theorem 3.11.

Source. *The integrator re-derived (a) to (d) on 2026-10-25 against the proof of Theorem 3.10 (b): only the coordinates that drive the two chains change. Independence from the release noise is what lets the step-4 kernel reproduce the hybrid. Part (d) gives two counterexamples,* $\mathrm{TV} = 0.474$ *against a claimed* 0.276 *for a coupling that correlates with the noise.*

Full statement and proof: Section B.46.

6.3 What an experiment certifies about a security profile

Proposition 6.4 (The run coordinate of a profile). `prop-profile-run-coordinate`

accepted in a round, waits for Halil

Fix a procedure, a distinguishing goal under the bottom oracle and a learning function L, *and let* $\epsilon_{\mathsf{L}}(q_T, q_E)$ *be the best advantage at budget* (q_T, q_E). *(a)* $\epsilon_{\mathsf{L}}(q_T, q_E) = \epsilon_{\mathsf{L}}(0, q_E)$. *An unbounded adversary that knows* L *simulates its own* Train *calls, so the profile of an unrestricted class is flat in training runs. (b) For a family of adversaries declared before* L, *indexed by their number of shadow runs, independent trials give a lower band on its profile, sound for every learning function at level* α, *with width* $\sqrt{2\ln(r/\alpha)/n_i}$ *at the* i*-th of* r *budgets. A two-phase family reuses one set of fresh shadow runs for*

both hypotheses of a trial. A general family pays two complete executions. One shadow bank reused across trials gives no band. (c) The coupled certificate of Theorem 3.8 is an upper band at every budget at once, and through a deterministic channel with atomless view laws no drawn-mode certificate gives an upper band below one.

Statement uses: Definition 3.2.
Proof uses: Theorem 3.8.

Source. The flatness of (a) is a statement about measurable strategies, not an efficient simulation. It says that the profile, the curve from a budget to the success named in FORMAL_SECURITY's terminology, is a measured object only once its family is named, uniformly over a class of learning functions (Chapter 4), or for a computationally bounded class (Chapter 5).

Full statement and proof: Section B.47.

6.4 The price of twins

Theorem 6.5 (Exact twins have a price)**.** `thm-exact-twin-price`

accepted in a round, waits for Halil

Take one Gaussian release $y = W + Z$ with a fixed choice, training inputs $x_0 \neq x_1$, and three targets: the coupled tail $\epsilon_0(\Delta)$, the total variation of the release, and its one-sided divergence $\mathrm{H}_{e^\varepsilon}(P_1 \| P_0)$. Let L° have deterministic kernels and the same model at x_0 and x_1 for every seed, so all three targets vanish. Let $c_{\mathrm{acc}} = -\ln(1-\epsilon_{\mathrm{acc}})$ and $\Lambda_{\alpha,\alpha'} = (1-\alpha')\ln((1-\alpha')/\alpha)$. (a) Every certifier, in either mode, sound at level α over all learning functions with deterministic kernels and below ϵ_{acc} at L° with probability $1-\alpha'$, makes at least $\Lambda_{\alpha,\alpha'}/c_{\mathrm{acc}}$ expected Train *calls at x_1, and also at x_0 for the coupled tail and the total variation. Calls at any other training input, fine-tunes and warm starts included, count for nothing. Coupled certificates match the bound up to a ceiling. (b) Certifying every record of a dataset of n_D records costs n_D times as much at the twin inputs, and $\lceil \ln(1/\alpha)/c_{\mathrm{acc}} \rceil$ common seeds achieve it with no union bound. (c) For the one-sided target no bound holds at the base input. A sound certifier makes four calls at x_0 where the bound of (a) would ask for 298.*

Statement uses: Theorem 3.10 and Definition 6.1.
Proof uses: Lemma 3.6 and Theorem 6.3.

Source. At $\alpha = \alpha' = 10^{-3}$ and $\epsilon_{\mathrm{acc}} = 10^{-2}$ the bound is 686.5 expected calls, and for 60,000 records 4.12×10^7 twin calls (computed in the round file). The integrator re-derived (a) to (c) and the numbers on 2026-10-25. The lower bound is a rare-event change of measure over a finite partition of the seed space, which handles seeds passed by value.

Full statement and proof: Section B.48.

Realization. For the total variation and the one-sided divergence, Theorem 6.3 makes a certificate sound under any implementable coupling, so realizing it needs each GPU run to have the deployed law at its input and repetitions to be independent. Accuracy at a zero-leakage function needs the pairs to coincide, which needs deterministic kernels. Whole uncontrolled GPU reruns at one seed do not coincide (Borrowed result 2.6). Logged rounding makes them coincide at 1.2 to 1.7 times the training time, for a modified pipeline that must then be the deployed one (Borrowed result 2.7). This paragraph is heuristic.

Chapter 7

Conjectures and targets

Conjecture 7.1 (Attacker and trainer in one unit)**.** `conj-same-units-gap`

open

Some natural fine-tuning procedure, a low-rank adapter trained on data whose labels depend on few directions, has a security profile for membership adversaries trained by gradient methods whose sample exponent exceeds the trainer's sample exponent.

Statement uses: *Borrowed result 2.2.*

Source. *The objective of 2026-10-25, problem* `prob-same-units-matched-budget-separation` *in* `sink/formalizing/problems.md`*. A placeholder statement until a space sharpens it.*

Chapter 8

History

One entry per round of every thinking space, in date order, written by the integrator when it merges the round.

2026-10-16 to 2026-10-17, learning-oracle rounds 00 to 10
The learning oracle and its certificates, eleven builder and challenger rounds, audit FINAL ACCEPT with seven minor findings not applied. Source: `sink/2026-10-16-formal-tuning/`.

2026-10-25, objective and context
The objective proposed, the shared context of the line built in `sink/formalizing/`, this booklet opened, thinking spaces started.

2026-10-25, oracle-and-measurement/round-00, argon
Twelve facts on auditing, proof of learning, unlearning and reruns, read from eight library papers, and three aims. ACCEPT 7, REVISE 7, REJECT 1, BLOCKING 12. Argon rejected the paired-advantage conjecture as common random numbers and gave the lower-bound construction of the twin price.

2026-10-25, oracle-and-measurement/round-01, astra
Five objects answering every challenge. ACCEPT 4, REVISE 7, BLOCKING 7. Astra refuted the base-input bound for the one-sided target with a certifier of four calls.

2026-10-25, oracle-and-measurement/round-02, argon
The seven blocking challenges answered. ACCEPT 8, BLOCKING 0. Merged: `def-seed-access-modes`, `prop-drawn-mode-is-fs-oracle`, `thm-coupling-free-soundness`, `prop-profile-run-coordinate`, `thm-exact-twin-price`, and six borrowed facts. Argon's review of this round is short (gemini-4-argon-medium, after two argon-high calls died on the backend's stream timeout), so the integrator re-derived every proof.

2026-10-25, computational-limits/round-00, astra
Thirteen facts read from the library, the judgement of FORMAL_SECURITY's `.tools/proof.tex`, and three aims. ACCEPT 6, REVISE 12, BLOCKING 12. Astra accepted the counterexample to the proof note's main theorem line by line and pointed to Li, Zhang and Arora for the symmetry mechanism.

2026-10-25, computational-limits/round-01, astra
Run with astra after three argon calls failed. The protocol class, the view coupling, whitening, the conditional-mean theorem and the first prices. ACCEPT 13, REVISE 6, REJECT 1, BLOCKING 10. Astra refuted part (b) of the reconstruction-at-training-cost conjecture.

2026-10-25, computational-limits/round-02, argon

The ten blocking challenges answered, with the covariance of the pancake law corrected to a deficit. ACCEPT 7, BLOCKING 0 (gemini-4-argon-medium). Merged twelve objects, among them `thm-blind-split-seed-separation` and the refuted `thm-fs-sq-black-box-reduction`, and nine borrowed facts. The integrator re-derived the arguments and took the cited lattice and SQ theorems as the space read them.

2026-10-25, statistical-limits/round-00, astra

Run with astra after argon's headless call could not read files. Thirteen facts read from eight library papers, five gaps, three aims. ACCEPT 2, REVISE 14, BLOCKING 14. Astra broke the first construction of the shadow-run price with one chosen-seed call, which led to the scrambled seed cells.

2026-10-25, statistical-limits/round-01, astra

Run with astra after two argon timeouts and a wait for astra's quota. The uniform profile, the scrambled-seed lemma, the shadow-run price and the label sandwich. ACCEPT 14, REVISE 4, BLOCKING 3.

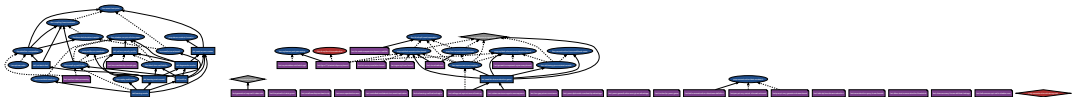
2026-10-25, statistical-limits/round-02, argon

The three blocked statements repaired, and one double-counting error in the lemma's hybrid bound found by the builder. ACCEPT 18, BLOCKING 0 (gemini-4-argon-medium). Merged `def-uniform-security-profile`, `lem-scrambled-seed-urn`, `thm-shadow-run-price-finite-views`, `prop-reconstruction-label-sandwich`, and twelve borrowed facts. The integrator re-derived every proof.

Appendix A

Dependency graph

Solid edges are uses of a statement, dashed edges are uses of a proof. Colours follow the margin marks.



Appendix B

Full statements and proofs from the thinking spaces

Each section is the object's text in the round that accepted it, converted at build time from the round file named in its chapter. Mathematics, labels and proofs are the builder's, as the discussant accepted them.

B.1 LiRA prices membership inference in shared shadow runs

`fact-lira-priced-in-training-runs`, from `lab/theory/spaces/oracle-and-measurement/round-00/builder.md`.

status: borrowed uses: none

CCNSTT22 (Carlini, Chien, Nasr, Song, Terzis and Tramèr, IEEE S&P 2022). The online attack of Algorithm 1 trains N shadow models on random halves of the data distribution's sample, fits two Gaussians to the target's logit-scaled confidence under the IN and OUT models, and applies a likelihood ratio test (Section IV-C). The same N shadow models serve every target, since each target lies in $N/2$ of the shadow training sets (Section IV-C, paragraph "Our complete online attack"). The evaluation uses $N = 256$, and $N = 64$ for ImageNet (Section V-A and the caption of Fig. 5). At false-positive rate 0.1%, the true-positive rate gains little beyond 64 shadow models and falls quickly below that number (Fig. 9, Section VI-B). The shadow runs are drawn independently, with no shared seed.

Proof. Read in the paper, Sections IV to VI. The section numbers agree with the note that the space `statistical-limits` wrote from its own reading on the same day.

Other regime. The price is in runs of the trainer's own pipeline, so it is a price in the units of this theory. The attack is a bottom-oracle attack in the drawn mode. The paper proves nothing, and every number is empirical on CIFAR-10, CIFAR-100, ImageNet and WikiText-103.

B.2 sample-based membership inference needs order $n + n^2\rho^2$ samples

`fact-hsu-membership-sample-gap`, from `lab/theory/spaces/oracle-and-measurement/round-02/builder.md`.

status: borrowed uses: none

HSU25 (Haghifam, Smith and Ullman, FOCS 2026 to appear, arXiv 2508.19458 v1, the version the library holds). Definition 2.1 asks a test, given m auxiliary samples, the noisy mean and a candidate, to answer correctly with probability at least 0.51 on

each side. Theorem 3.1 proves that for $d > c_1(n + n^2\rho^2)$ and $m \le c_2(n + n^2\rho^2)$ no such test exists against the noisy empirical mean of Definition 2.4 over all Gaussians, and Theorem 1.1 is its informal version. Theorem 3.3 is a matching attack, for $0 < \rho < 1$, $m \ge cd$ and $d \ge c(n + n^2\rho^2)$, against the symmetric estimators of Definition 3.2, so the orders match at the critical dimension $d \asymp n + n^2\rho^2$. Theorem 4.1, whose informal version is Theorem 1.2, proves for identity covariance, $n \ge 4$ and $m \ge 2$, that no test exists when $d < c(n + n^2\rho^2 + n^2/m)$. The matching attack for $d \gtrsim n + n^2\rho^2 + n^2/m$ is attributed by the paper to Dwork, Smith, Steinke, Ullman and Vadhan (FOCS 2015) and is not proven there. With known covariance and $m = 2$ the dimension at which a test exists is therefore of order $n + n^2\rho^2 + n^2/2$, which is of order n^2 when ρ is bounded and of order $n^2\rho^2$ when ρ grows. §1.2.1 notes that a test with a statistic free of the population and a calibrated threshold needs $O(1)$ trained models, that is $O(n)$ samples.

Proof. Read in the paper, §1.2, §2, §3 and the statement of Theorem 4.1. The proofs were not re-derived.

Other regime. The fact prices auxiliary samples for an attacker that knows the estimator and computes without bound, and it prices no training runs. Divided by n, the price of Theorem 3.1 is $1 + n\rho^2$ datasets' worth of samples. The result belongs to the space statistical-limits. For measurement, §1.2.1 says that an audit built from restricted tests can sit far below the profile it lower-bounds.

B.3 a noise seed with 100 values is detected by the audit

`fact-small-seed-space-breaks-dp`, from `lab/theory/spaces/oracle-and-measurement/round-02/builder.md`.

status: borrowed uses: none

NHSBTJCT23 (USENIX Security 2023), §6.3, Violation 2. A DP-SGD implementation whose noise seed takes only 100 values, so that only 100 noise vectors occur, is detected as violating its claimed $\varepsilon = 1.27$ by the Gaussian-DP audit and by Lu et al.'s method, and is missed by (ε, δ) audits (Fig. 8).

Proof. Read in the paper, §6.3.

Other regime. Enumerating a finite seed space gives the view laws P_0 and P_1 exactly only when every other source of randomness, hardware randomness, batching drawn outside that seed and channel noise, is integrated exactly, for instance when the seed drives all of it under deterministic kernels. It gives the top oracle exactly only when one decoder $\hat{s}$ of the seed from the view satisfies $\hat{s}(Y) = s$ under both hypotheses. If instead $\Pr[\hat{s}(Y) \ne s] \le e$ under each hypothesis, an adversary holding $(\hat{s}(Y), Y)$ is within total variation e of one holding (s, Y) under each hypothesis, so its best advantage is at least $\mathrm{D}_1^{\top} - 2e$ (proven, by the triangle inequality for total variation). Argon's example $\mathsf{L}(x_b; s) = s + bM/2 \bmod M$ has $\mathrm{D}_1^{\top} = 1$ and $\mathrm{TV}(P_0, P_1) = 0$, and no decoder works under both hypotheses there. In Violation 2 the 100 noise vectors are far apart in high dimension, which explains the detection (heuristic). This is the empirical side of open problem 9 of the oracle theory.

B.4 a GPU rerun at one seed ends far from the original run

`fact-rerun-reproduction-error`, from `lab/theory/spaces/oracle-and-measurement/round-01/builder.md`.

status: borrowed uses: none

JYCDTCP21 (IEEE S&P 2021). Theorem 1 (Entropy Growth) states that the entropy of the training process grows linearly in the number of steps, under the assumption of i.i.d. hardware and library noise at each step (§VI-A). Table I (§VI-C) reports the reproduction error of retraining from a checkpoint with the same initial parameters, batching, architecture and strategy, normalized by d_{ref}, the distance between two independently trained models, as a mean with a spread. For a whole run the normalized ℓ_2 error is 0.955 ± 0.004 for ResNet-20 on CIFAR-10 and 0.815 ± 0.002 for ResNet-50 on CIFAR-100. For one step it is 0.001 and 0.002. PyTorch's deterministic operations give 0.569 ± 0.004 for ResNet-20 and 0.816 ± 0.001 for ResNet-50.

Proof. Read in the paper, §IV to §VI and Table I.

Other regime. Theorem 1 is proven inside its noise model, heuristic for real hardware, and Table I is empirical. Read as a distance concentrated near its mean, Table I puts the spread sp_τ of a whole run in parameter space near one for τ below about $0.95\, d_{\text{ref}}$ for ResNet-20, below about $0.81\, d_{\text{ref}}$ for ResNet-50, and below about $0.57\, d_{\text{ref}}$ for ResNet-20 with deterministic operations, and near zero for one step (heuristic, the table gives no tail probabilities).

B.5 logged rounding makes GPU reruns identical

`fact-controlled-nondeterminism-exact-replication`, from `lab/theory/spaces/oracle-and-measurement/round-01/builder.md`.

status: borrowed uses: none

SAB24 (Srivastava, Arora and Boneh, NeurIPS 2024). With the same software version, seed and deterministic library flags, models trained on A40, Titan XP and RTX 2080 Ti GPUs differ (Fig. 2), while runs on one GPU architecture repeat identically (§6). Training in FP64, rounding each layer output to FP32, and logging the rounding direction of values within an adaptive threshold of the boundary make full training of ResNet-50 on CIFAR-10 and fine-tuning of GPT-2 replicate exactly across all pairs of the three GPU types (§6, Fig. 4). The prototype costs 1.2 to 1.4 times the training time for the trainer and 1.3 to 1.7 times for the auditor (Table 5).

Proof. Read in the paper, §1, §3 to §6 and §8.

Other regime. Empirical, on two models and three GPU types. In the oracle theory's terms, the named-seed Train call with deterministic kernels is realizable on real hardware at a stated time factor, which settles in this setting the hardware part of prob-coupled-reruns-rarely-differ. The data part, how far a leave-one-out twin moves from the base run, is a property of training that no hardware control changes.

B.6 a prover that picks the batches passes replay

`fact-unlearning-verification-fragile`, from `lab/theory/spaces/oracle-and-measurement/round-02/builder.md`.

status: borrowed uses: none

ZCSL24 (ICML 2024). Definition 4.1 calls a proof of retraining valid when every step is reproducible within ε and no batch meets the unlearned set D_u. Algorithm 1 retrains while replacing each unlearned point of the original batch by its nearest same-class neighbour in the retained data, and Proposition 4.2 proves that it passes at $\varepsilon = 0$. Algorithm 2 forges the retraining proof from the logged training proof, editing only the steps whose batch met D_u, and Proposition 4.4 proves that it passes at $\varepsilon > 0$ under the assumptions of Proposition 4.3, vanilla SGD for the replay update and the stated step-size bounds. Proposition 4.5 states that Algorithm 1 takes the time $\mathcal{T}(n)$ of naive retraining and Algorithm 2 the time $(p_u/P + (1 - p_u)/(mP))\, \mathcal{T}(n)$ on P parallel

processes, with $p_u = 1 - (1 - |D_u|/|D|)^m$ and batch size m. The text calls these results Theorems 4.2 to 4.5.

Proof. Read in the paper, §3 and §4.

Other regime. The time factor of Algorithm 2 is at most one, equals one at $P = m = 1$, and is below one when $P \geq 2$, or when $m \geq 2$ and $p_u < 1$ (proven, from the formula). Its work factor is the value at $P = 1$, $p_u + (1 - p_u)/m$. The prover's freedom is the batch order, a part of the seed. Algorithm 1 costs one run and Algorithm 2 at most one, so a meter in runs separates neither from honest retraining.

B.7 statistical query bounds for planted biclique

`fact-fgrvx17-planted-clique-sq-bound`, from `lab/theory/spaces/computational-limits/round-00/builder.md`.

status: borrowed uses: none

Feldman, Grigorescu, Reyzin, Vempala and Xiao, J. ACM 64(2), 2017, held copy arXiv 1201.1214v6, key FGRVX17. Problem 1.1 fixes $S \subseteq [n]$, $|S| = k$, and the law D_S on $\{0,1\}^n$ that is uniform with probability $1 - k/n$ and, with probability k/n, sets the coordinates of S to 1 and the rest uniform. U is uniform on $\{0,1\}^n$. Definition 2.2: STAT(τ) returns any value within τ of $E_D[h]$ for $h : X \to [-1, 1]$. Definition 2.3: VSTAT(t) returns any value in $[p - \tau', p + \tau']$ with $p = E_D[h]$ and $\tau' = \max\{1/t, \sqrt{p(1-p)/t}\}$. Definition 2.4: 1-STAT evaluates a Boolean h on a fresh sample. Theorem 2.9: for every constant $\delta > 0$, every $k \leq n^{1/2-\delta}$ and every $r > 0$, solving the distributional planted k-biclique problem with probability 2/3 needs $n^{\Omega(\log r)}$ queries to VSTAT($n^2/(rk^2)$), and the same holds for distinguishing any D_S from U. Theorem 5.4 is the randomized decision version. Theorem 2.10: a 1-bit sampling algorithm that distinguishes U from every D_S with probability 2/3 needs $\Omega(n^2/k^2)$ calls to 1-STAT. Theorem 3.2 and Corollary 3.12 bound search problems through the statistical dimension, the latter from pairwise correlations (Definitions 3.8 and 3.9). Section 2.4 gives the matching upper bound: n queries $h_i(x) = x_i$ to VSTAT($16n^2/k^2$), tolerance $k/(4n)$, recover S. Appendix A relates the average-case bipartite problem to the distributional one with $t = n$ samples, a statement about samples, not about statistical access to one graph.

Two corrections to the registers (proven by reading). The held copy has no Theorem 1.1 of the form $q\tau^2 \geq \Omega(n^{2\delta})$, which `fact-fgrvx17-theorem-one-quoted-form` records from the proof note. Every bound is against an oracle that may answer adversarially within its tolerance, except Theorem 2.10, which is against the honest 1-STAT.

Other regime. These are query bounds against statistical algorithms with unbounded computation between queries. They price computation only through access, so they speak about efficient adversaries only through the heuristic that natural algorithms are statistical.

B.8 when statistical queries bind gradient training

`fact-sq-simulates-minibatch-sgd`, from `lab/theory/spaces/computational-limits/round-01/builder.md`.

status: borrowed, revised uses: none

AKMSS21 Theorems 1a and 1c as stated in round 00, with the error of a bSGD method defined as $E[\sup L_D]$ over valid roundings. Revised reading. A generic SQ lower bound does not cover the whole fine-precision bSGD class, because Theorem 1a lets constructed models simulate PAC learning when $\rho < 1/(8b)$. Bounds for narrower architectures, dynamics or attacker classes remain possible. That single-precision

training sits in the regime of Theorem 1a is heuristic: it needs an arithmetic model that the paper's uniform rounding lattice does not supply.

Other regime. As in round 00.

B.9 the seed-known inverter

`fact-lei-random-expansive-layers-invertible`, from `lab/theory/spaces/computational-limits/round-00/builder.md`.

status: borrowed uses: none

Lei, Jalal, Dhillon and Dimakis, NeurIPS 2019, held copy arXiv 1906.07437v2, key LJDD19. Assumption 1: a weight matrix $W \in \mathbb{R}^{n_i \times n_{i-1}}$ has i.i.d. Gaussian entries and $n_i = c_0 n_{i-1}$ with $c_0 \geq 2.1$. Theorem 3: if every layer of a ReLU network G satisfies Assumption 1, then for any input z^* and $x = G(z^*)$, with probability $1 - e^{-\Omega(k)}$ over the weights, z^* is recovered by layer-wise linear equations (k the input dimension). Lemma 1 gives injectivity of one layer: at least k coordinates are positive with probability $1 - \exp(-\Omega(k))$, and any k Gaussian rows are invertible. Theorems 1 and 2: deciding whether a preimage exists is NP-complete for constructed two-layer (binary input) and four-layer (real input) networks. The probability is over W for each fixed input, so N records cost a union bound $Ne^{-\Omega(k)}$.

Other regime. The result needs the weights. It is the seed-known side of any seed separation and says nothing about an attacker without W.

B.10 homogeneous CLWE is a hard hidden-direction law

`fact-clwe-hardness`, from `lab/theory/spaces/computational-limits/round-02/builder.md`.

status: borrowed, revised uses: none

BRST21, in its normalization. Definitions 2.19 and 2.21, Claim 2.22, Corollary 4.2, Theorem 7.5 and Corollary 9.4 as in round 01. Covariance. On $w^\perp$ the covariance of $H_{w,\beta,\gamma}$ is $I/(2\pi)$. Along w it is $1/(2\pi) - \Delta$ with $0 \leq \Delta \leq 3(\gamma^2+1)e^{-\pi\gamma^2}$ for $\gamma \geq 1$ and $\beta \in (0,1)$.

Proof of the covariance clause. The proof of Lemma 7.2 writes the noiseless law $H_{w,\gamma'}$ as $D_{(1/\gamma')\mathbb{Z}} \times D_{\mathbb{R}^{n-1}}$, and Poisson summation with the transform $(1/(2\pi) - y^2)\rho(y)$ gives $E[x_1^2] = 1/(2\pi) - \Delta'$ with $\Delta' = \sum_{y \in \gamma'\mathbb{Z}} y^2 \rho(y)/\rho(\gamma'\mathbb{Z}) \in [0,\, 2.01\gamma'^2 e^{-\pi\gamma'^2}]$ for $\gamma' \geq 1$, since $\rho(\gamma'\mathbb{Z}) \geq 1$ and the terms with $|k| \geq 2$ add less than $0.01\gamma'^2 e^{-\pi\gamma'^2}$. The proof of Corollary 7.3 writes $H_{w,\beta,\gamma}$ as $H_{w,\gamma'}$ plus independent Gaussian noise of parameter $\beta'/\gamma' = \beta/\gamma$, rescaled by $1/\sqrt{1+(\beta/\gamma)^2}$, with $\gamma' = \sqrt{\beta^2+\gamma^2}$. Along w this gives variance $(1/(2\pi) - \Delta' + (\beta/\gamma)^2/(2\pi))/(1+(\beta/\gamma)^2) = 1/(2\pi) - \Delta$ with $\Delta = \Delta'/(1+(\beta/\gamma)^2) \leq \Delta'$, and $\gamma^2 \leq \gamma'^2 \leq \gamma^2 + 1$. On $w^\perp$ the same rescaling of $1/(2\pi) + (\beta/\gamma)^2/(2\pi)$ gives $1/(2\pi)$. □ (Proven.)

The homogeneous law determines w only up to sign.

Other regime. Computational, conditional on worst-case lattice hardness.

B.11 pairwise correlations of pancake laws

`fact-hclwe-sq-correlation-bound`, from `lab/theory/spaces/computational-limits/round-01/builder.md`.

status: borrowed, revised uses: fact-fgrvx17-planted-clique-sq-bound

BRST21 Proposition 8.3 and Theorem 8.1 as in round 00, now explicitly with the reference D_0 of covariance $I_n/(2\pi)$ and the widths and spacing of BRST21's normalization.

Other regime. Unconditional for SQ.

B.12 non-Gaussian component analysis is SQ-hard under moment matching

`fact-ngca-sq-lower-bound`, from `lab/theory/spaces/computational-limits/round-01/builder.md`.

status: borrowed, revised uses: none

DKRS23 Theorem 1.5 and Definitions 1.2 and 1.3 as in round 00, which astra checked against the PDF. Revised interpretation. The theorem bounds SQ algorithms only. Lattice-reduction algorithms solve some discrete or noiseless moment-matched instances with $O(n)$ samples while SQ bounds stay formally true (ZSWB22, COLT 2022, Theorem 1.1, `fact-lll-beats-sos-noiseless`, recorded), so an SQ separation does not predict security against every polynomial-time attacker, and experiments must include a non-statistical attacker. The round-00 sentence on statistical identifiability is withdrawn.

Other regime. SQ only.

B.13 the price of finding a label-relevant direction

`fact-generative-exponent-sq-recovery-bound`, from `lab/theory/spaces/computational-limits/round-01/builder.md`.

status: borrowed, revised uses: fact-fgrvx17-planted-clique-sq-bound

DPLB24 as in round 00, with these scope lines. Theorem 4.1 needs the denoiser T it constructs and pays $\tilde{\lambda}_k = \lambda_k / \log^k(3/\lambda_k)$, and its sample bound has the term $d/(\tilde{\lambda}_k^2 \epsilon^2)$ besides $d^{k/2}/\tilde{\lambda}_k^2$. The partial-trace algorithm is sample-based, not an SQ algorithm, so the paper gives no matching SQ estimation upper bound, and Remark 3.3 leaves the SQ detection-estimation gap open. The proof of Theorem 3.2 uses Lemma B.6, which is FGRVX17 Corollary 3.12, through Corollary B.7, and the pairwise bound $\chi_0^2(D_{w_1}, D_{w_2}) \le \lambda_{k^*}^2 m^{k^*} + m^{k^*+1}/(1-m)$ for $m = w_1 \cdot w_2$ (equation 53, from Lemma E.2).

Other regime. As in round 00.

B.14 the symmetry mechanism is known

`fact-orthogonal-equivariance-of-training`, from `lab/theory/spaces/computational-limits/round-01/builder.md`.

status: borrowed uses: none

Li, Zhang and Arora, arXiv 2010.08515 (ICLR 2021 as astra gives it, the arXiv page names no venue), key LZA21. Definition 3.2: an algorithm is G-equivariant if training on $\{(g(x_i), y_i)\}$ and composing with g gives, in distribution, the hypothesis trained on $\{(x_i, y_i)\}$. Theorem C.1: if a group G_W acts on weights through an isomorphism τ with $M[\tau(g)W](g(x)) = M[W](x)$, the update rule is invariant under the joint action, and the initialization law is G_W-invariant, the algorithm is G-equivariant, by induction on the iterates. Remark C.1 extends it to stochastic and trajectory-dependent updates. Corollary C.2: (stochastic) gradient descent from i.i.d. Gaussian initialization is orthogonally equivariant. Remark C.2: momentum. Table 1: Adam and AdaGrad are permutation and sign-flip equivariant only. Theorem 5.2: orthogonally equivariant learners need $\Omega(d^2)$ samples on a quadratic task where a convolutional network needs $O(1/\varepsilon \log(1/\varepsilon))$.

Other regime. A statement about the law of the learned function. It says nothing about attackers or about the joint law of a transcript and the data.

B.15 reconstruction security against the narcissus baseline

`fact-cohen-narcissus-equals-one-wayness`, from `lab/theory/spaces/computational-limits/round-00/builder.md`.

status: borrowed uses: none

Cohen, Kaplan, Mansour, Moran, Nissim, Stemmer and Tsfadia, ITCS 2025, arXiv 2405.15753, key CKMMNST25. Definition 1.6: M is $(\varepsilon, \delta, \mathcal{F})$- R-Narcissus-resilient if for all $D \in \mathcal{F}$ and all attackers A, $\Pr[R(S, z) = 1] \leq e^{\varepsilon} \Pr[R(T, z) = 1] + \delta$ with $S, T \sim D$ independent, $y \leftarrow M(S)$, $z \leftarrow A(y)$. Definition C.1 is the computational version against polynomial-time attackers. Definition C.4: f is well-spread if every output has negligible probability. Theorem C.5: for an efficiently computable well-spread f and $R(x, x') = 1$ iff $f(x) = f(x')$, f is one-way iff M_f is computationally Narcissus-resilient for (U_λ, U_λ).

Other regime. The relation of Theorem C.5 is equality of images, which is preimage finding. A reconstruction relation, closeness to the record, is a different relation, and the theorem says nothing about it.

B.16 with the covariance known the sample price falls to order min of n and inverse squared noise

`fact-hsu-gap-prices-covariance`, from `lab/theory/spaces/statistical-limits/round-01/builder.md`.

status: borrowed

uses: fact-hsu-membership-sample-gap

Statement. Key HSU25, Theorem 4.1. For the identity-covariance family $\mathcal{P}_d^{\mathrm{id.cov}}$, a universal c, $n \geq 4$, $m \geq 2$ and $d < c\,(n + n^2\rho^2 + n^2/m)$, no $(\mathcal{P}_d^{\mathrm{id.cov}}, \{\mathcal{A}_{n,\rho}\}, n, m)$ sample-based attack exists. The paper attributes the converse, an attack once $d \gtrsim n + n^2\rho^2 + n^2/m$, to Dwork et al. without proof. At $d \asymp d^\star$ Theorem 4.1 therefore forces $m \gtrsim \min\{n, 1/\rho^2\}$. Remark 4.2 states that for $\rho > 1/\sqrt{n}$ knowing the covariance lowers the required samples from $\Omega(n^2\rho^2)$ to $O(n)$, an upper bound.

Proof. Borrowed (proven in HSU25, appendix not read).

The other regime. In the experiment that both theorems analyse, the price of Theorem 3.1 at $d \asymp d^\star$ is the price of not knowing Σ (proven, by Theorems 3.1 and 4.1 in that regime). An attacker that knows the data law and the mechanism needs no sample, so in this experiment every sample price is a price of not knowing part of the experiment.

B.17 differential privacy bounds the advantage of an informed membership attacker

`fact-yeom-dp-bounds-membership-advantage`, from `lab/theory/spaces/statistical-limits/round-01/builder.md`.

status: borrowed

uses: none

Statement. Unchanged from round 00. Key YGFJ18, Experiment 1, Definition 4 and Theorem 1. An ε-DP learner, with DP over replacement of one record, gives $\mathrm{Adv}^M \leq e^\varepsilon - 1$ for every adversary of Experiment 1, which receives $\mathcal{D}$.

Proof. Borrowed (proven in YGFJ18, proof read).

The other regime. Unchanged. The bound holds against unbounded adversaries and prices no learning resource.

B.18 a loss-threshold attack gains the normalized generalization error

`fact-yeom-generalization-error-gives-advantage`, from `lab/theory/spaces/statistical-limits/round-01/builder.md`.

status: borrowed

uses: fact-yeom-dp-bounds-membership-advantage

Statement. Key YGFJ18, Theorem 2. Let the loss be nonnegative with $\ell(A_S, z) \leq B$ for a constant $B > 0$, all S and all z drawn from S or $\mathcal{D}$. Adversary 1 evaluates the model at z, computes $\ell(A_S, z)$ and outputs 1 with probability $\ell(A_S, z)/B$. Its advantage equals $R_{\mathrm{gen}}(A)/B$, with R_{gen} the average generalization error of Definition 3. For the 0/1 loss and $B = 1$ it is the generalization error.

Proof. Borrowed (proven in YGFJ18, proof read).

The other regime. The attack uses one model evaluation, one loss computation and one biased coin, and no sample or training run. It shows what one attack achieves, not a bound on all attacks.

B.19 the LiRA game excludes members from the OUT draw

`fact-lira-low-fpr-yeom-game`, from `lab/theory/spaces/statistical-limits/round-01/builder.md`.

status: borrowed

uses: fact-yeom-dp-bounds-membership-advantage

Statement. Key CCNSTT22, Definition 1 in §III-A. The challenger samples D and trains f_θ. It flips b. For $b = 0$ it draws a fresh point from the distribution with $(x, y) \notin D$, and for $b = 1$ a point of D. The adversary has query access to the distribution and to f_θ. The paper says the game is inspired by Yeom et al. and Jayaraman et al. Against Yeom's Experiment 1 the bit is renamed and the OUT draw is conditioned on missing D, so the two games agree when that event has probability one, for instance for an atomless distribution.

Proof. Borrowed (a definition, read in full). The agreement is proven by the conditioning remark.

The other regime. Query access to the distribution is the resource HSU25 prices in samples. LiRA spends it on shadow training sets and reports runs.

B.20 reconstruction robustness and its oblivious baseline

`fact-balle-reconstruction-robustness-definition`, from `lab/theory/spaces/statistical-limits/round-01/builder.md`.

status: borrowed

uses: none

Statement. Key BCH22. Definition 1 is the informed adversary, holding D^-, the released θ, the training algorithm and optional side knowledge. Definition 2 calls M

(η, γ)-ReRo for a prior π and an error ℓ when every deterministic $R : \Theta \to \mathcal{Z}$ and every D^- give $\Pr_{Z\sim\pi,\theta\sim M(D^-\cup\{Z\})}[\ell(Z, R(\theta)) \le \eta] \le \gamma$. The adversary knows π and D^-, and its uncertainty is the target Z. The baseline $\kappa_{\pi,\ell}(\eta) = \sup_{z_0} \Pr_{Z\sim\pi}[\ell(Z, z_0) \le \eta]$ is defined in §VII-B without a number.

Proof. Borrowed (definitions, read). Randomized attacks give the same supremum by averaging over their coins. Attainment by a measurable posterior maximizer is not claimed.

The other regime. The definition is statistical and prices neither learning the prior nor training a reconstructor.

B.21 Rényi and pure differential privacy bound reconstruction by the baseline

`fact-balle-dp-bounds-reconstruction`, from `lab/theory/spaces/statistical-limits/round-01/builder.md`.

status: borrowed

uses: fact-balle-reconstruction-robustness-definition

Statement. Key BCH22, §VII. Write $\kappa = \kappa_{\pi,\ell}(\eta)$ with $\eta > 0$.

1. Theorem 2. For $\alpha > 1$ and $\varepsilon \ge 0$, an (α, ε)-RDP mechanism is (η, γ)-ReRo with $\gamma = (\kappa e^{\varepsilon})^{(\alpha-1)/\alpha}$.
2. Corollary 3. An ε-DP mechanism is $(\eta, \kappa e^{\varepsilon})$-ReRo.
3. Corollary 4. A ρ-zCDP mechanism with $\rho < \log(1/\kappa)$ is ReRo with $\gamma = \exp(-(\sqrt{\log(1/\kappa)} - \sqrt{\rho})^2)$.
4. Proposition 6. For a constant $\eta \in (0, 1)$, the uniform prior on the unit ball of $\mathbb{R}^d$ and ℓ_2 error, $\varepsilon = o(d)$ or $\rho = o(d)$ gives $\gamma = e^{-\Omega(d)}$. Proposition 7. For a constant $\eta > 0$, the prior $\mathcal{N}(w, \sigma^2 I_d)$ with $\sigma \ge 2\eta/\sqrt{d}$ and ℓ_2 error, the same.
5. Theorem 5. For $\varepsilon \ge 0$, $\eta \in (0, 1)$, $\gamma \in [0, 1]$ and the 0/1 error, ReRo against every prior on two points with mass $1/(e^{\varepsilon} + 1)$ on one of them gives $(\varepsilon, \max\{0, (e^{\varepsilon} + 1)\gamma - e^{\varepsilon}\})$-DP.

Proof. Borrowed (proven in BCH22, appendix proofs of Theorems 2 and 5, Corollary 4 and Proposition 6 read).

The other regime. All bounds hold against unbounded attackers and need a known privacy parameter. A coupled certificate of the learning-oracle theory bounds the divergence between two fixed inputs, which controls reconstruction for the two-point priors of Theorem 5 and not for a general prior.

B.22 how many of the trainer's outputs a reconstruction attacker must see

`fact-reconstruction-query-lower-bounds`, from `lab/theory/spaces/statistical-limits/round-01/builder.md` and `lab/theory/spaces/statistical-limits/round-02/builder.md`.

From `lab/theory/spaces/statistical-limits/round-01/builder.md`.

status: borrowed

uses: fact-balle-reconstruction-robustness-definition

Statement. Key ML24, arXiv 2303.16372v6. In the game of §3 the adversary receives n independent draws $h_1, \dots, h_n \sim M(D^- \cup \{z\})$ with D^- known, and the value is $V(M, n) = \inf_{\mathcal{A}} \sup_z \mathbb{E}[\rho(\mathcal{A}(h_{1:n}), z)^2]$. Theorem 9 states, for a compact metric space, $\varepsilon \ge 0$, $\delta \in [0, 1]$ and every (ε, δ)-DP learner, $V(M, n) \gtrsim \operatorname{diam}(\mathcal{Z})^2 (2(1 - \delta)/(e^{\varepsilon} + 1))^n$.

The query form needs nondegenerate parameters and a target error below the diameter. Theorem 10, the claimed matching upper bound, and Corollary 12, the Rényi extension, are stated in ML24 and not established. The adversary of Appendix B.5 uses the true target, and the proof of Lemma 16 in Appendix B.3 bounds the total variation from the wrong side.

Proof. Borrowed for Theorem 9 (the two-point proof of Appendix B.2, checked by the discussant of round 00 and not read by the builder). The two gaps were read by the builder.

The other regime. The unit counts outputs of the trainer that the attacker observes, not runs it makes, and the bound needs DP inside training.

From `lab/theory/spaces/statistical-limits/round-02/builder.md`.

status: borrowed

uses: fact-balle-reconstruction-robustness-definition

Statement. Key ML24, arXiv 2303.16372v6. The game and Theorem 9 are as stated in `round-01/builder.md`. Theorem 10, the claimed matching upper bound for every $\varepsilon \geq 0$ and $\delta \in [0, 1]$, is not established, since the adversary of the proof in Appendix B.5 uses the true target. Corollary 12 is not established, since the proof of Lemma 16 in Appendix B.3 bounds the total variation from the wrong side. The same Appendix B.5 holds, per the round-01 discussant, an unnumbered result on a target-independent majority adversary for privacy parameter at least 0.042 (pages 26 and 27). The builder has not read that passage. It does not give the uniform rate of Theorem 10.

Proof. Borrowed for Theorem 9. The two gaps were read by the builder.

The other regime. The unit counts the trainer's outputs that the attacker observes, and the bound needs differential privacy inside training.

B.23 reconstruction thresholds for subset sums

`fact-dinur-nissim-reconstruction-thresholds`, from `lab/theory/spaces/statistical-limits/round-01/builder.md`.

status: borrowed

uses: none

Statement. Key DN03. Definitions 1 to 3 and Theorems 2 and 3 as in round 00. Theorem 2 uses all 2^n answers and exponential time, Theorem 3 uses $n(\log n)^2$ random answers and a linear program. Theorem 5 states $(T(n), \delta)$-privacy in the two-phase game of Definition 4, where the adversary names an index after querying and then receives every other bit, for the uniform prior, $T(n) > \text{polylog}(n)$ and perturbation $\tilde{O}(\sqrt{T(n)})$.

Proof. Borrowed (proven in DN03, statements and the proof of Theorem 2 read).

The other regime. Theorems 2 and 3 change both the number of answers and the running time, so they are not a separation of statistical from computational reconstruction at a fixed observation budget.

B.24 mutual information bounds the gain of a reconstruction that beats the prior guess

`fact-pac-privacy-mutual-information-bound`, from `lab/theory/spaces/statistical-limits/round-01/builder.md`.

status: borrowed

uses: fact-balle-reconstruction-robustness-definition

Statement. Key XD23. For a mechanism M, a data law $\mathcal{D}$ known to the adversary and a criterion ρ, let $1 - \delta_o^\rho$ be the optimal prior success (Definition 3). Theorem 1 states that M is $(\delta, \rho, \mathcal{D})$ PAC private, meaning no adversary succeeds with probability at least $1 - \delta$, for every δ with $\delta \leq \delta_o^\rho$ and $\mathrm{kl}(\delta \| \delta_o^\rho) > I(X; M(X))$. Equivalently, an adversary whose failure δ satisfies $\delta \leq \delta_o^\rho$, the optimal adversary included, has $\mathrm{kl}(\delta \| \delta_o^\rho) \leq I(X; M(X))$. Corollary 1, $\delta_o^\rho - \delta \leq \sqrt{I(X; M(X))/2}$, holds for every adversary. Remark 1 conditions both sides on extra side knowledge.

Proof. Borrowed (proven in XD23, Appendix A not read). The restriction to adversaries at least as good as the prior guess is the discussant's correction of round 00, with the constant-release counterexample.

The other regime. Statistical, and it needs the data law known. §2 of XD23 states that the framework cannot quantify the hardness added by ignorance of the law.

B.25 a Gaussian release has mutual information set by its covariance

`fact-pac-privacy-gaussian-release-bound`, from `lab/theory/spaces/statistical-limits/round-01/builder.md`.

status: borrowed

uses: fact-pac-privacy-mutual-information-bound

Statement. Key XD23. Theorem 3. For a deterministic M with $\mathbb{E}\|M(X)\|_2^2 < \infty$ and independent $B \sim \mathcal{N}(0, \Sigma_B)$ with Σ_B positive definite, $I(X; M(X) + B) \leq \frac{1}{2} \log \det(I_d + \Sigma_{M(X)} \Sigma_B^{-1})$. Theorem 4. If $\|M(X)\|_2 \leq r$, Algorithm 1 evaluates M on m independent draws of X, estimates $\Sigma_{M(X)}$, and returns Σ_B with $I(X; M(X) + B) \leq v + \beta$ with confidence $1 - \gamma$ once the parameter c satisfies display (6).

Proof. Borrowed (proven in XD23, appendices C and D not read).

The other regime. One run of Algorithm 1 is one evaluation of M, a training run when M is training and a forward pass when M is a representation. It certifies the release, not an attack.

B.26 an accurate representation must leak a correlated attribute

`fact-zhao-accuracy-forces-attribute-leakage`, from `lab/theory/spaces/statistical-limits/round-01/builder.md`.

status: borrowed

uses: none

Statement. Unchanged from round 00, with both attribute groups of positive probability. Key ZCTG20, Theorems 3.2 and 3.3, for $\mathcal{H}_A$ all measurable binary classifiers on $\mathcal{Z}$.

Proof. Borrowed (proven in ZCTG20, appendix proofs read).

The other regime. Unchanged. Statistical, no class restriction.

B.27 reconstruction by a learned reconstructor costs thousands of shadow runs and needs the initialization

`fact-balle-reconnn-priced-in-shadow-runs`, from `lab/theory/spaces/statistical-limits/round-01/builder.md`.

status: borrowed

uses: fact-balle-reconstruction-robustness-definition

Statement. Key BCH22, §IV-B and §VI-B. The attack trains k shadow models $\bar{\theta}_i = A(D^- \cup \{\bar{z}_i\})$ and fits a reconstructor network to the pairs $(\bar{\theta}_i, \bar{z}_i)$. With full-batch training from an initialization known to the adversary, reconstructions beat the nearest-neighbour oracle on average from 8K shadow models on MNIST and from 10K on CIFAR-10, and high-fidelity MNIST reconstructions appear from 1K (Fig. 5). With a fresh random initialization per model the attack fails on MNIST. Table III gives MSE between 0.0691 and 0.0695 then, against 0.0041 to 0.0089 with the initialization known, and a pool of 5M shadow pairs over 500 initializations per target did not help.

Proof. Borrowed, empirical on MNIST and CIFAR-10 with the models of §V.

The other regime. This is the reconstructor-as-learner of Halil's example, trained on weights rather than on an intermediate representation, priced in shadow training runs. The dependence on the initialization is the seed acting as a resource, which the top oracle models and which part (d) of `thm-shadow-run-price-finite-views` prices in the distinguishing game.

B.28 the uniform regret profile of an unbounded adversary in learning units

`def-uniform-security-profile`, from `lab/theory/spaces/statistical-limits/round-01/builder.md` and `lab/theory/spaces/statistical-limits/round-02/builder.md`.

From `lab/theory/spaces/statistical-limits/round-01/builder.md`.

status: open

uses: thm-coupling-certificate, def-degenerate-oracles, fact-hsu-membership-sample-gap

Statement. Fix a procedure, a coalition, a channel interface with its Eval allowance on served names, a game and a policy $\mathsf{P} \in \{\text{bottom}, \text{top}\}$. An instance ι fixes the learning function, the laws of seeds and hardware randomness, the procedure's data and a reference law $\mathcal{D}_\iota$ from which reference samples are drawn. A budget is $\beta = (m, N) \in \mathbb{N}^2$, ordered coordinatewise, of m reference samples from $\mathcal{D}_\iota$ and at most N Train calls. $\mathcal{A}_\mathsf{P}(\beta)$ is the set of adversaries under P with the given interface and budget, which stop with probability one and are chosen before the instance. For distinguishing, Succ_ι is the advantage. For inverting it is the success probability on the game's event.

1. The informed optimum $\mathrm{Opt}_\mathsf{P}(\iota)$ is the supremum of $\mathrm{Succ}_\iota(\mathcal{A})$ over adversaries under P with the same interface and any finite budget, which may depend on ι.
2. The uniform regret at a budget, over a class $\mathfrak{J}$, is

$$g_{\mathfrak{J},\mathsf{P}}(\beta) = \inf_{\mathcal{A} \in \mathcal{A}_\mathsf{P}(\beta)} \sup_{\iota \in \mathfrak{J}} \big(\mathrm{Opt}_\mathsf{P}(\iota) - \mathrm{Succ}_\iota(\mathcal{A})\big).$$

   It is nonincreasing in β.
3. The attained price at accuracy $a \geq 0$ is the set of minimal elements of $F(a) = \{\beta : \exists \mathcal{A} \in \mathcal{A}_\mathsf{P}(\beta)\ \forall \iota \in \mathfrak{J},\ \mathrm{Opt}_\mathsf{P}(\iota) - \mathrm{Succ}_\iota(\mathcal{A}) \leq a\}$. It is finite by Dickson's lemma, and empty when $F(a)$ is. The approximate price uses $\{\beta : g_{\mathfrak{J},\mathsf{P}}(\beta) \leq a\}$ instead.
4. The threshold price of a criterion C, a finite list of conditions on the run's outcome laws such as HSU25's two conditional probabilities, is the set of minimal β for which one adversary in $\mathcal{A}_\mathsf{P}(\beta)$ meets C on every $\iota \in \mathfrak{J}$, possibly empty.

Properties, with labels.

1. Singleton classes. For one learning node seen through a channel, an adversary that depends on ι attains $\mathrm{Opt}_{\mathsf{P}}(\iota)$ with no sample and no Train call, by hard-wiring the test of (F1) under the bottom oracle and the test on (ρ, Y) under the top oracle (proven, step 3 of the proof of `thm-coupling-certificate` (b), whose bottom-oracle optimum is $\mathrm{TV}(P_0, P_1)$ and top-oracle optimum $\mathrm{D}_1^{\top}$). So $g_{\{\iota\},\mathsf{P}} \equiv 0$, and every positive regret is a price of not knowing which member of the class is the instance.
2. Finite views. For one learning node through a deterministic channel onto k views, $g_{\mathfrak{J}_{\mathrm{all}},\mathrm{bottom}}(0, N) \le 2\sqrt{k/\lfloor N/2 \rfloor}$, so it tends to zero along finite budgets (proven, `thm-shadow-run-price-finite-views` (a)).
3. HSU25 at $N = 0$. Let $\mathfrak{J}$ hold the populations $\mathcal{N}(\mu, \Sigma)$ of $\mathcal{P}_d$ with the estimator $\mathcal{A}_{n,\rho}$, and let the attacker have no Train access, which is $N = 0$. For $d > c_1 d^{\star}$, every budget $(m, 0)$ with $m \le c_2 d^{\star}$ is outside the threshold price of HSU25's criterion, and its regret form is at least $\mathrm{Opt} - 0.02$ on some instance (proven, by HSU25 Theorem 3.1 and its proof as cited in `fact-hsu-membership-sample-gap`). With Train access the question is `prob-train-access-collapses-sample-price`.

Sketch of well-posedness. Success is defined for every adversary that stops with probability one, since its run is a measurable function of countably many independent draws (`def-learning-oracle-model`). The infimum and supremum are over sets of reals in $[-1, 1]$. Monotonicity holds because a budget-β adversary is a budget-β' adversary for $\beta' \ge \beta$. Dickson's lemma gives finitely many minimal elements of any subset of $\mathbb{N}^2$.

What would settle it. Acceptance as well posed. The definition is notation and is a result only together with the theorem that computes it on one axis.

Regime. Statistical. Budgets count samples and training runs, and adversaries are unbounded.

Experiment. Run a fixed attack at a grid of budgets on instances sampled from the class. Its measured worst regret over the sampled instances, after estimation error, is a lower bound on that attack's worst regret over the class and an upper bound on the uniform regret of the sampled subclass. Lower bounds on the price come from theorems. The space `oracle-and-measurement` owns the measurement.

The other regime. A bounded adversary cannot hard-wire an arbitrary test, so its profile is not degenerate on a singleton, and a comparison there needs a bound on advice or description size. That object belongs to the space `computational-limits`.

From `lab/theory/spaces/statistical-limits/round-02/builder.md`.

status: open

uses: thm-coupling-certificate, thm-black-box-certifiability, def-degenerate-oracles, fact-hsu-membership-sample-gap

Statement. Items 1 to 4 (informed optimum, uniform regret, attained price, threshold price) are unchanged from `round-01/builder.md`.

Properties, with labels.

1. Singleton classes. (i) For any interface, an adversary that knows ι and has budget β simulates every finite-budget informed adversary, drawing reference samples from $\mathcal{D}_\iota$ and computing L on its own randomness, also at the honest seed it opens under the top oracle. So $g_{\{\iota\},\mathsf{P}}(\beta) = 0$ as an infimum at every budget (proven, by this simulation, since a simulated answer has the law of the oracle's answer by fidelity). (ii) For one learning node whose whole value reaches the adversary through a channel K on the plain channel, the optimum is $\mathrm{TV}(P_0, P_1)$ under the bottom oracle and $\mathrm{D}_1^{\top}(K)$ under the top oracle, and an adversary that hard-wires the optimal test attains it with no sample and no call (proven,

step 3 of `thm-coupling-certificate` (b)). (iii) For the query channel with a finite query set, finitely many outputs and allowance q_E, the bottom-oracle optimum is $\mathrm{Adv}^{\mathrm{bb}}(q_E)$, a maximum over finitely many strategy trees, attained by hard-wiring a best tree (proven, `thm-black-box-certifiability` (o)).
2. Finite views. Unchanged from round 01.
3. HSU25 at $N = 0$. Unchanged from round 01, with $\rho > 0$.

Sketch of well-posedness, what would settle it, regime and experiment are unchanged from round 01.

The other regime. Unchanged from round 01.

B.29 a random scramble of seed cells turns chosen and opened seeds into fresh draws

`lem-scrambled-seed-urn`, from `lab/theory/spaces/statistical-limits/round-02/builder.md`.

status: sketch

uses: def-learning-oracle

Statement. Let $\mathcal{V}$ be finite and M an integer. Partition $[0,1)$ into cells $c(t) = \lfloor Mt \rfloor \in [M] = \{0, \dots, M-1\}$. A learning setting Λ fixes a seed law μ on $[0,1)^2$ with uniform second coordinate, an evaluation map, an injection $s : \mathcal{V} \to \mathcal{M}$, a training input x_0, and a response at every input other than x_0, a measurable map of the seed and the input that the setting fixes. For a law p on $\mathcal{V}$ with $Mp(v) \in \mathbb{N}$, a labelling λ of $[M]$ with $|\lambda^{-1}(v)| = Mp(v)$ and a bijection π of $[M]$, the learning function $\mathsf{L}_\Lambda^{p,\pi}$ has deterministic kernels, returns $s(\lambda(\pi(c(\rho_b))))$ at x_0 for the seed $\rho = (\rho_a, \rho_b)$, and gives the setting's response elsewhere. Fix an adversary with at most $N < M$ Train calls, any Eval and Open calls, adaptive choices of inputs, seed values, seed names and draws, run against a fixed challenge value v. Let T_v be the law of its transcript under $\mathsf{L}_\Lambda^{p,\pi}$ with π uniform and independent of everything else. Let $\tilde{T}_v$ be the law of its transcript under the lazy oracle, which answers a Train call at x_0 whose seed lies in a cell not met by an earlier Train call at x_0 with $s(V)$ for a fresh $V \sim p$, answers a call at x_0 in an earlier cell with the earlier model, and answers every other call as $\mathsf{L}_\Lambda^{p,\pi}$ does. Then:

1. $\mathrm{TV}(T_v, \tilde{T}_v) \le N^2/(2(M-N))$.
2. For a second law p' and the lazy oracle $\tilde{T}'_v$ built from p' with the same setting Λ and the same adversary, $\mathrm{KL}(\tilde{T}_v \| \tilde{T}'_v) \le N\,\mathrm{KL}(p \| p')$.

Proof. Represent the run as a sequence of steps. Step j is either an action of the adversary, a kernel of the augmented history, or an answer of the oracle, a kernel of the augmented history and the instance. The augmented history h_j holds every action and answer so far and the seed value of every Train call made so far, also when the adversary holds it only by name. All spaces are standard Borel, so these kernels exist. Pad the run after it stops with a fixed symbol, so that it is an infinite sequence.

Step 1, the conditional laws. Fix a history h before a Train call at x_0. Its augmented part determines the cells $c_1, \dots, c_{t-1}$ met by earlier Train calls at x_0, their labels $u_1, \dots, u_{t-1}$, and the cell c of the present seed. Every other answer in h is a function of the seeds and the inputs that the setting fixes, or of a stored model $s(u_j)$ and fresh evaluation randomness, so it does not depend on π. Bayes' rule over the finite set of bijections therefore gives, under the uniform π given h, the uniform law on bijections with $\lambda(\pi(c_j)) = u_j$ for $j < t$. If c is a new cell, $\pi(c)$ is uniform on the $M - t + 1$ cells outside $\pi(\{c_1, \dots, c_{t-1}\})$ under that law, so the label has

law $q_h(v) = (Mp(v) - r_v)/(M - t + 1)$, with r_v the number of u_j equal to v, and $\mathrm{TV}(q_h, p) \le (t-1)/(M-t+1)$, as computed in round 01. If c is an earlier cell, both oracles return the earlier model. Every step that is not a Train call at x_0 has the same kernel under both oracles.

Step 2, the hybrid bound. Let $P_{1:j} = P_{1:j-1} \otimes K_j$ be the law of the first j steps under the scrambled oracle and $Q_{1:j} = Q_{1:j-1} \otimes K'_j$ under the lazy one. The triangle inequality through $P_{1:j-1} \otimes K'_j$ and data processing under K'_j give

$$\mathrm{TV}(P_{1:j}, Q_{1:j}) \le \mathrm{TV}(P_{1:j-1}, Q_{1:j-1}) + \int \mathrm{TV}(K_j(\cdot \mid h), K'_j(\cdot \mid h))\, P_{1:j-1}(dh)$$

(proven, standard). Summing over $j \le L$, the prefix of length L is within $\mathbb{E}_P\big[\sum_{j \le L} \mathrm{TV}(K_j(\cdot \mid h_{j-1}), K'_j(\cdot \mid h_{j-1}))\big]$, the expectation along paths of the scrambled run. On each path the summand is zero except at new-cell Train calls at x_0, of which there are at most N, and at the t-th of them it is at most $(t-1)/(M-N)$ by Step 1. So every finite prefix of the two runs is within $\sum_{t \le N}(t-1)/(M-N) \le N^2/(2(M-N))$. A bound with a supremum over histories at each fixed step index would not suffice, since the index of the t-th new-cell call varies with the path. The cylinder sets form an algebra that generates the product σ-algebra, so the total variation of the infinite sequences is the supremum over prefixes, and the transcript is a measurable function of the sequence. This gives claim 1 by data processing.

Step 3, the KL bound. Under the two lazy oracles the kernels differ only at new-cell Train calls at x_0, where they are s_*p and s_*p' whatever the history. The chain rule gives, for every finite prefix, the sum over steps of the expected conditional KL, which is at most $N\,\mathrm{KL}(p\|p')$. The KL of the infinite sequences is the supremum over the increasing prefix σ-algebras, and data processing passes it to the transcript. □

Status. Sketch until a round accepts it. No gap is known to the builder.

The other regime. The lemma makes the oracle's seed interface no stronger than a stream of fresh draws against an unknown scramble. The round-01 discussant points to the permutation against random function switching argument, of which this is the statistical form, with no hardness assumption.

B.30 the training-run price of distinguishing two training inputs on k views

`thm-shadow-run-price-finite-views`, from `lab/theory/spaces/statistical-limits/round-01/builder.md` and `lab/theory/spaces/statistical-limits/round-02/builder.md`.

From `lab/theory/spaces/statistical-limits/round-01/builder.md`.

status: sketch

uses: lem-scrambled-seed-urn, thm-coupling-certificate, thm-black-box-certifiability, thm-reference-access, def-degenerate-oracles, def-uniform-security-profile

Setting. One learning node trains x_b under H_b, with $x_0 \ne x_1$ known to the adversary. Its model W reaches the adversary only as $y = \kappa(W)$ through the plain channel, for a measurable map κ from the model space onto a finite set $\mathcal{V}$ with $k \ge 2$ elements, and a fixed injection $s : \mathcal{V} \to \mathcal{M}$ with $\kappa \circ s = \mathrm{id}$. An adversary makes at most N Train calls, any Eval and Open calls, chooses training inputs, seed values, seed names and draws adaptively, and may call before and after seeing y. $\mathfrak{J}_{\mathrm{all}}$ is the class of all learning functions with this model space, and $\mathfrak{J}_{\mathrm{det}}$ its members with deterministic kernels.

Statement.

(a) Upper bound. For every learning function, the adversary that makes $n = \lfloor N/2 \rfloor$ drawn-seed Train calls at each of x_0, x_1, opens the models, forms the empirical view laws $\hat{p}_0, \hat{p}_1$ and outputs 1 iff $\hat{p}_1(y) > \hat{p}_0(y)$ has $\mathrm{TV}(P_0, P_1) - \mathrm{Adv} \le 2\sqrt{k/n}$, in expectation over its shadow draws.

(b) Lower bound, general kernels, both oracles. Let $k \ge 4$. For every $N \ge 1$ and every adversary with at most N Train calls, some $\mathsf{L} \in \mathfrak{J}_{\mathrm{all}}$ has gap at least

$$\min\Big\{\frac{1}{8},\ \frac{1}{32}\sqrt{\frac{3k'}{N}}\Big\}$$

under the bottom oracle, and the same function has the same gap under the top oracle.

(c) Lower bound, deterministic kernels, bottom oracle. For every adversary with at most N Train calls and every $\zeta > 0$, some $\mathsf{L} \in \mathfrak{J}_{\mathrm{det}}$ has bottom-oracle gap at least $\min\{\frac{1}{8}, \frac{1}{32}\sqrt{3k'/N}\} - \zeta$.

(d) Top oracle, deterministic kernels. The adversary that calls $\mathsf{Train}(x_0, s^*)$ on the honest seed name s^*, opens the model m_0 and outputs 1 iff $y \ne \kappa(m_0)$ has advantage equal to the top optimum $\mathrm{D}_1^{\top}(\kappa) = \Pr_\rho[\kappa(\mathsf{L}(x_0;\rho)) \ne \kappa(\mathsf{L}(x_1;\rho))]$, which is at least $\mathrm{TV}(P_0, P_1)$, for every $\mathsf{L} \in \mathfrak{J}_{\mathrm{det}}$. It makes one Train call.

(e) Price. Let $0 < a < \frac{1}{8}$ and $k \ge 4$. Under the bottom oracle, over $\mathfrak{J}_{\mathrm{all}}$ or $\mathfrak{J}_{\mathrm{det}}$, an adversary whose gap is at most a at every member makes at least $3k'/(1024a^2)$ Train calls, and $2\lceil 4k/a^2 \rceil$ drawn-seed calls suffice in expectation. Under the top oracle over $\mathfrak{J}_{\mathrm{det}}$, one Train call reaches the top optimum. Under the top oracle over $\mathfrak{J}_{\mathrm{all}}$ the lower bound $3k'/(1024a^2)$ remains.

(f) Label-only views. Let U be a finite query set with $|U| = q$, $C \ge 2$ outputs, and deterministic evaluation $\mathsf{ev}(t, u) = t(u)$ on models that include the tables $t \in [C]^U$. Serve the model through the query channel with allowance $Q \ge q$. Then the label-only optimum is $\mathrm{Adv}^{\mathrm{bb}}(q) = \mathrm{TV}$ of the laws of the table, and the bounds of (a) to (e) hold with $k = C^q$. For a larger query set the lower bound of (b) and (c) still holds with $k = C^q$, and the upper bound of `thm-black-box-certifiability` (b) exceeds it by a factor of order $\ln|U|/(C-1) + 1$.

Proof of (a). For every set $A \subseteq \mathcal{V}$ and laws $P, \hat{p}$, $|P(A) - \hat{p}(A)| \le \frac{1}{2}\|P - \hat{p}\|_1$. Let $A^* = \{P_1 > P_0\}$ and $\hat{A} = \{\hat{p}_1 > \hat{p}_0\}$. Then

$$P_1(\hat{A}) - P_0(\hat{A}) \ge \hat{p}_1(\hat{A}) - \hat{p}_0(\hat{A}) - \tfrac{1}{2}\|\hat{p}_1 - P_1\|_1 - \tfrac{1}{2}\|\hat{p}_0 - P_0\|_1 \ge \mathrm{TV}(P_0, P_1) - \|\hat{p}_1 - P_1\|_1 - \|\hat{p}_0 - P_0\|_1,$$

since $\hat{p}_1(\hat{A}) - \hat{p}_0(\hat{A}) \ge \hat{p}_1(A^*) - \hat{p}_0(A^*)$. Drawn seeds give independent views at each input by fidelity, and $\mathbb{E}\|\hat{p}_b - P_b\|_1 \le \sum_v \sqrt{P_b(v)/n} \le \sqrt{k/n}$ by Jensen's and Cauchy and Schwarz's inequalities. The challenge is independent of the shadow draws, so the advantage is the expectation of $P_1(\hat{A}) - P_0(\hat{A})$. □

Proof of (b). Step 1, the decision rule. Fix a learning function. Under the bottom oracle the randomness of the adversary's own calls and its coins, together ω, is independent of the honest seed and hardware by fidelity, and the honest node reaches the adversary only through y. So the output is $F(\omega, y)$ for a measurable F, and with $\varphi(v) = \Pr_\omega[F(\omega, v) = 1]$, $\mathrm{Adv} = \sum_v \varphi(v)(P_1(v) - P_0(v))$. Under the top oracle the same holds for a learning function that ignores the seed, with the honest seed added to ω, since then the seed is independent of y and of every answer.

Step 2, the family. Pair the first k' views as $(2i-1, 2i)$ for $i \le r = k'/2$, and give any remaining view mass zero. For $\sigma \in \{\pm 1\}^r$ and $\delta \in (0, \frac{1}{2}]$ let

$$P_0^\sigma(2i-1) = \frac{1+\sigma_i\delta}{k'}, \qquad P_0^\sigma(2i) = \frac{1-\sigma_i\delta}{k'}, \qquad P_1(v) = \frac{1}{k'} \text{ for } v \le k'.$$

Let $\Xi = [0,1)$ with Lebesgue law and let $\mathsf{L}^\sigma(x_0;\rho,\xi) = s(Q^\sigma(\xi))$ and $\mathsf{L}^\sigma(x_1;\rho,\xi) = s(Q_1(\xi))$, with Q^σ and Q_1 the quantile maps of P_0^σ and P_1 in the order of $\mathcal{V}$, and let every other input train to one fixed model. Each Train call at x_0 returns s of a fresh draw of P_0^σ whatever its seed, and every other answer has a law free of σ. Then $\mathrm{TV}(P_0^\sigma, P_1) = \frac{1}{2}\sum_v |P_0^\sigma(v) - P_1(v)| = \frac{\delta}{2}$ and, by Step 1,

$$\mathrm{Adv}^\sigma = \frac{\delta}{k'}\sum_{i\le r}\sigma_i s_i^\sigma, \qquad s_i^\sigma = \varphi^\sigma(2i) - \varphi^\sigma(2i-1), \qquad \mathrm{gap}^\sigma = \frac{\delta}{k'}\sum_{i\le r}(1-\sigma_i s_i^\sigma).$$

Step 3, two runs. For $i \le r$ run the adversary twice, independently, against L^σ, with challenge $2i$ in the first run and $2i-1$ in the second, and let $Z_i \in \{-1,0,1\}$ be the first output minus the second. Then $\mathbb{E}^\sigma Z_i = s_i^\sigma$. Let $\sigma^\pm$ agree off coordinate i and have $\sigma_i = \pm 1$. Only the at most $2N$ Train calls at x_0 have laws that depend on σ, so the chain rule gives $\mathrm{KL} \le 2N\,\mathrm{KL}(P_0^{\sigma^+} \| P_0^{\sigma^-})$ for the two pair transcripts. Here $\mathrm{KL}(P_0^{\sigma^+} \| P_0^{\sigma^-}) = \frac{2\delta}{k'}\ln\frac{1+\delta}{1-\delta} \le \frac{16\delta^2}{3k'}$, since $\ln\frac{1+\delta}{1-\delta} \le 2\ln 3 \cdot \delta \le \frac{8}{3}\delta$ on $(0, \frac{1}{2}]$ by convexity. Pinsker's inequality bounds the total variation of the pair transcripts by $4\delta\sqrt{N/(3k')}$, and $|s_i^{\sigma^+} - s_i^{\sigma^-}|$ by twice that.

Step 4, averaging. For fixed σ_{-i}, $\frac{1}{2}[(1 - s_i^{\sigma^+}) + (1 + s_i^{\sigma^-})] \ge 1 - 4\delta\sqrt{N/(3k')}$. Averaging over σ uniform and summing over i,

$$\mathbb{E}_\sigma\,\mathrm{gap}^\sigma \ge \frac{\delta}{2}\Big(1 - 4\delta\sqrt{\frac{N}{3k'}}\Big).$$

If $N \ge 3k'/16$ take $\delta = \frac{1}{8}\sqrt{3k'/N} \le \frac{1}{2}$, which gives $\mathbb{E}_\sigma\mathrm{gap}^\sigma \ge \frac{\delta}{4} = \frac{1}{32}\sqrt{3k'/N}$. If $N < 3k'/16$ take $\delta = \frac{1}{2}$, which gives more than $\frac{1}{8}$. Some σ attains the average. Under the top oracle the family ignores the seed, so the top optimum equals TV and Step 1 applies with the honest seed in ω. □

Proof of (c). Replace the family of (b) by the scrambled family of `lem-scrambled-seed-urn`, with Ξ a point, seeds (ρ_a, ρ_b) uniform on $[0,1)^2$, $\mathsf{L}^{\sigma,\pi}(x_0;\rho) = s(\lambda_\sigma(\pi(c(\rho_b))))$ with $|\lambda_\sigma^{-1}(v)| = MP_0^\sigma(v)$, $\mathsf{L}(x_1;\rho) = s(Q_1(\rho_a))$, and a fixed model elsewhere. Take δ rational and M a multiple of k' times its denominator, which costs at most $\zeta/2$ in the bound by continuity in δ. The view laws are P_0^σ and P_1 for every π, so Step 1 holds for each instance (σ,π) and $\mathrm{gap}^{\sigma,\pi} = \frac{\delta}{k'}\sum_i(1-\sigma_i s_i^{\sigma,\pi})$. Average over π uniform. The average of $s_i^{\sigma,\pi}$ is the difference of the output probabilities under the mixture laws T_{2i}^σ and T_{2i-1}^σ of the lemma. Run Step 3 with these mixtures, independent across the two runs. By the lemma each mixture is within $N^2/(2(M-N))$ of its lazy process, whose KL to the lazy process of σ' is at most $N\,\mathrm{KL}(P_0^\sigma \| P_0^{\sigma'})$. So the pair total variation is at most $4\delta\sqrt{N/(3k')} + 2N^2/(M-N)$, and $M \ge N + 8N^2/\zeta$ makes the average gap at least the bound of (b) minus ζ. Some instance attains the average. □

Proof of (d). Under deterministic kernels the honest model is $\mathsf{L}(x_b;\rho^*)$, and the call returns $m_0 = \mathsf{L}(x_0;\rho^*)$. Under H_0, $y = \kappa(m_0)$ and the output is 0. Under H_1 the output is 1 exactly when $\kappa(\mathsf{L}(x_1;\rho^*)) \ne \kappa(\mathsf{L}(x_0;\rho^*))$. So the advantage is the divergence probability through κ, which equals the coupled divergence $\beta_1(\kappa)$ for a deterministic channel and hence the top divergence under deterministic kernels, and is at least $\mathrm{TV}(P_0, P_1)$, both by `thm-coupling-certificate` (b). By the same theorem it is the best top-oracle advantage. □

Proof of (e). If every member has gap at most $a < \frac{1}{8}$, then (b) or (c) gives $a \geq \frac{1}{32}\sqrt{3k'/N} - \zeta$ for every $\zeta > 0$, since the bound $\frac{1}{8}$ is excluded, and so $N \geq 3k'/(1024a^2)$. Part (a) with $n = \lceil 4k/a^2 \rceil$ gives gap at most a. Parts (d) and (b) give the top-oracle statements. □

Proof of (f). Let the served model be a table t with law P_b on $[C]^U$. An adversary with allowance $Q \geq q$ reads t by querying each point once, and a strategy tree of depth q that queries the q points along every path and outputs $\mathbf{1}[t \in A^*]$ attains TV, while no tree exceeds it by data processing. So $\mathrm{Adv}^{\mathrm{bb}}(q) = \mathrm{TV}(P_0, P_1)$, and every adversary's output is a function of t and its own transcript, which is the setting of (a) to (e) with $\mathcal{V} = [C]^U$ and κ the table. For $|U| > q$, let the family vary the table on q fixed points and keep the other values fixed, which gives the same lower bound. Finally $\ln|\mathfrak{T}_q| \leq \frac{C^q-1}{C-1}\ln|U| + C^q \ln 2$ from the count of `thm-black-box-certifiability`, which gives the stated factor. □

Status. Sketch, because (c) inherits the measurability gap of `lem-scrambled-seed-urn`. Parts (a), (b), (d), (e) for $\mathfrak{J}_{\mathrm{all}}$ and (f) are proofs.

Prior work and novelty. The passive version, where the attacker receives n independent views at each input and nothing else, is agnostic learning of the class of all subsets of $\mathcal{V}$, whose dimension is k. The rate $\Theta(k/a^2)$ for that problem is classical, and the round-00 discussant points to Daniely, Sabato, Ben-David and Shalev-Shwartz, JMLR 2015, Theorem 5, which this session has not read. The content here is the oracle part. Parts (b) and (c) show that chosen, opened and reused seeds, inputs chosen after the challenge, and coupled calls do not beat the passive rate, and (c) needs the scramble for that. Part (d) shows that the honest seed collapses the price to one call under deterministic kernels (proven above). No published statement of the oracle part is known to the space (open as a novelty claim, no search run).

Meaning in the trainer's units. The trainer spends one training run. Under the bottom oracle an attacker that must come within a of the best advantage on every learning function with k views spends order k/a^2 runs. LiRA's 64 to 256 runs per benchmark (`fact-lira-priced-in-training-runs`) sit below this worst case because its Gaussian model restricts the class, and BCH22's thousands of shadow runs and its failure without the initialization (`fact-balle-reconnn-priced-in-shadow-runs`) match the direction of part (d) in a reconstruction game (heuristic, the games differ).

The other regime. Without the oracle the attacker samples the learning function, and `thm-perfect-substitution` makes the bottom oracle the real procedure, so (b) and (c) are lower bounds on the shadow training runs of every attack that treats training as a black box. A computationally bounded attacker pays at least this price. The space `computational-limits` asks when it pays more. Degenerate oracles. The lower bound of (c) fails under the top oracle by (d), which meets the non-triviality bar of `def-non-triviality-bar` for a statement about adversaries.

From `lab/theory/spaces/statistical-limits/round-02/builder.md`.

status: sketch

uses: lem-scrambled-seed-urn, thm-coupling-certificate, thm-black-box-certifiability, thm-reference-access, def-degenerate-oracles, def-uniform-security-profile

Setting. As in `round-01/builder.md`.

Statement.

(a) Upper bound. Let $N \geq 2$. For every learning function, the plug-in adversary with $n = \lfloor N/2 \rfloor$ drawn-seed Train calls at each input has $\mathrm{TV}(P_0, P_1) - \mathrm{Adv} \leq 2\sqrt{k/n}$, in expectation over its shadow draws. For $N \leq 1$ the constant output 0 has gap $\mathrm{TV}(P_0, P_1)$.

(b) Lower bound, general kernels. Let $k \geq 4$ and $N \geq 1$. Under the bottom oracle, every adversary with at most N Train calls has, at some L^σ of the family of round

01, gap at least $g_N = \min\{\frac{1}{8}, \frac{1}{32}\sqrt{3k'/N}\}$. Under the top oracle, every adversary of the top oracle with at most N Train calls has, at some L^σ of the same family, a gap of at least g_N against the top optimum, which equals $\mathrm{TV}(P_0, P_1)$ on this family.

(c) Lower bound, deterministic kernels, bottom oracle. Every adversary with at most N Train calls and every $\zeta > 0$ have, at some $\mathsf{L}^{\sigma,\pi}$ of the scrambled family of round 01, gap at least $g_N - \zeta$.

(d) Top oracle, deterministic kernels. Unchanged. One Train call at the honest seed reaches the top optimum $\mathrm{D}_1^\top(\kappa)$, which is at least $\mathrm{TV}(P_0, P_1)$.

(e) Price. Unchanged. For $0 < a < \frac{1}{8}$ and $k \geq 4$, under the bottom oracle over $\mathfrak{J}_{\mathrm{all}}$ or $\mathfrak{J}_{\mathrm{det}}$, regret at most a everywhere needs at least $3k'/(1024a^2)$ Train calls, and $2\lceil 4k/a^2 \rceil$ drawn-seed calls suffice in expectation. Under the top oracle, one call suffices over $\mathfrak{J}_{\mathrm{det}}$, and $3k'/(1024a^2)$ calls remain necessary over $\mathfrak{J}_{\mathrm{all}}$.

(f) Label-only views. Let U be a finite query set, $C \geq 2$ outputs, deterministic evaluation $\mathsf{ev}(t, u) = t(u)$ on models that include the tables $t \in [C]^U$, and the query channel with allowance $q_E \leq |U|$. Under the bottom oracle the optimum is $\mathrm{Adv}^{\mathrm{bb}}(q_E)$, and over learning functions that vary the table only on a fixed set $U_0 \subseteq U$ of q_E points, it equals the total variation of the laws of the table on U_0. On such functions the bounds of (a), (b), (c) and (e) hold with $k = C^{q_E}$. Under the top oracle the optimum is the top divergence of the channel that reads the table on U_0, which can exceed the total variation of the marginal table laws, and part (d) applies to it with κ the restriction to U_0. The upper bound of `thm-black-box-certifiability` (b) at allowance q_E is $2\sqrt{\ln(2|\mathfrak{T}_{q_E}|)/n}$ with $\ln|\mathfrak{T}_{q_E}| \leq \frac{C^{q_E}-1}{C-1}\ln|U| + C^{q_E}\ln 2$, so the known upper and lower bounds on the bottom-oracle price differ by a factor of order $\ln|U|/(C-1) + 1$. Whether that factor is needed is open.

Proofs. Parts (a), (b), (d) and (e) as in `round-01/builder.md`, section `thm-shadow-run-price-finite-views` with two changes. In (a) the empirical laws exist because $n \geq 1$. In (b) the quantifiers are per policy. Step 1 is the decision-rule reduction for adversaries of one policy, and Steps 2 to 4 give each such adversary a hard sign vector. Under the top oracle the family ignores the seed, so the honest seed is one more auxiliary variable independent of the challenge, as the round-01 discussant noted. Part (c) as in round 01, now with `lem-scrambled-seed-urn` in its revised form. Its hypothesis holds, since the scrambled family shares the seed law, the evaluation map, the injection and the responses at x_1 and at other inputs across σ. Part (f). Let the family vary the table only on U_0 and keep fixed values elsewhere. A query outside U_0 then returns a value free of the instance, so the best strategy tree of depth q_E reads U_0 and outputs $\mathbf{1}[t|_{U_0} \in A^*]$, and $\mathrm{Adv}^{\mathrm{bb}}(q_E)$ is the total variation of the restricted table laws by data processing. Every adversary's output is a function of $t|_{U_0}$ and its own transcript, which is the setting of (a) to (e) with $\mathcal{V} = [C]^{U_0}$. Under the top oracle the discussant's example, two fair seed bits under x_0 and their complements under x_1, has marginal total variation zero and top divergence one, so the top optimum must be taken on the pair of seed and table. Part (d) gives one call that reaches it under deterministic kernels. □

Status. Sketch until a round accepts it. No gap is known to the builder.

Prior work and novelty. The passive rate is classical (Daniely, Sabato, Ben-David and Shalev-Shwartz, JMLR 2015, Theorem 5, as the round-00 and round-01 discussants cite, not read by the builder). The scrambling step is the statistical form of the switch from a random permutation to a random function (Bellare and Rogaway, EUROCRYPT 2006, as the round-01 discussant cites, not read). The search row

`search-shadow-run-lower-bound` of `round-01/summary.md` found no lower bound on shadow training runs. The oracle part, that chosen, opened and reused seeds do not beat the passive rate under the bottom oracle, while the honest seed collapses it to one call under deterministic kernels, has no known published statement (open as a novelty claim, one query run).

Meaning in the trainer's units and the other regime. Unchanged from round 01.

B.31 the trainer's error and the representation's residual information bracket an unbounded reconstructor

`prop-reconstruction-label-sandwich`, from `lab/theory/spaces/statistical-limits/round-01/builder.md` and `lab/theory/spaces/statistical-limits/round-02/builder.md`.

From `lab/theory/spaces/statistical-limits/round-01/builder.md`.

status: sketch

uses: fact-balle-reconstruction-robustness-definition, fact-pac-privacy-mutual-information-bound, fact-pac-privacy-gaussian-release-bound, fact-zhao-accuracy-forces-attribute-leakage

Statement. Let $(\mathcal{X}, d)$ be a Polish metric space, $\mathcal{Y}$ finite and $\mathcal{Z}$ standard Borel, and let (X, Y, Z) have any joint law on $\mathcal{X} \times \mathcal{Y} \times \mathcal{Z}$. Let $\mathcal{Y}^+$ be the labels of positive probability. For $\eta > 0$ and $y \in \mathcal{Y}^+$ let $\kappa_y(\eta) = \sup_{x_0 \in \mathcal{X}} \Pr[d(X, x_0) \le \eta \mid Y = y]$ and $\bar{\kappa}(\eta) = \sum_{y \in \mathcal{Y}^+} \Pr[Y = y] \kappa_y(\eta)$. Let $S^\star(\eta) = \sup_R \Pr[d(X, R(Z)) \le \eta]$ over measurable $R : \mathcal{Z} \to \mathcal{X}$, let $\varepsilon^\star = \inf_{\hat{y}} \Pr[\hat{y}(Z) \ne Y]$ over measurable $\hat{y} : \mathcal{Z} \to \mathcal{Y}$, and let $I(X; Z \mid Y) = \sum_{y \in \mathcal{Y}^+} \Pr[Y = y] \operatorname{KL}(P_{XZ|y} \| P_{X|y} \otimes P_{Z|y})$, possibly infinite.

1. Floor. $S^\star(\eta) \ge \bar{\kappa}(\eta) - \varepsilon^\star$. For every measurable head h with error ε_h and every $\zeta > 0$ the decoder $R_h(z) = x_{h(z)}$, with x_y within ζ of the supremum in κ_y, succeeds with probability at least $\bar{\kappa}(\eta) - \varepsilon_h - \zeta$.
2. Ceiling. If $S^\star(\eta) \ge \bar{\kappa}(\eta)$ then $\operatorname{kl}(S^\star(\eta) \| \bar{\kappa}(\eta)) \le I(X; Z \mid Y)$. Hence always $S^\star \le \bar{\kappa} + \sqrt{I(X; Z \mid Y)/2}$, and, if $0 < \bar{\kappa} < 1$, $S^\star \le (I(X; Z \mid Y) + \ln 2)/\ln(1/\bar{\kappa})$.
3. Gaussian release. If $Z = f(X) + B$ with $f : \mathcal{X} \to \mathbb{R}^D$ measurable, $\mathbb{E}\|f(X)\|^2 < \infty$, $B \sim \mathcal{N}(0, \Sigma_B)$ with Σ_B positive definite and B independent of (X, Y), then $I(X; Z \mid Y) \le \sum_{y \in \mathcal{Y}^+} \Pr[Y = y] \frac{1}{2} \ln\det(I + \Sigma_{f|y} \Sigma_B^{-1})$, with $\Sigma_{f|y}$ the covariance of $f(X)$ given $Y = y$.
4. Tightness. If $Z = Y$ then $\varepsilon^\star = 0$, $I(X; Z \mid Y) = 0$ and $S^\star = \bar{\kappa}$. If $Z = f(X)$ for an injective Borel f, then $S^\star(\eta) = 1$ for every $\eta > 0$.

Proof. Regular conditional laws exist on these spaces, and every success event $\{d(X, R(Z)) \le \eta\}$ is measurable, since d is continuous and R measurable.

Floor. The decoder R_h takes finitely many values, so it is measurable. Its success is at least $\Pr[d(X, x_Y) \le \eta,\ h(Z) = Y] \ge \Pr[d(X, x_Y) \le \eta] - \Pr[h(Z) \ne Y] \ge \bar{\kappa} - \zeta - \varepsilon_h$. A Bayes classifier attains $\varepsilon^\star$, and $\zeta \to 0$ gives the first claim.

Ceiling. Let $Q(dx, dy, dz) = \Pr[Y \in dy] P_{X|y}(dx) P_{Z|y}(dz)$, so $\operatorname{KL}(P_{XYZ} \| Q) = I(X; Z \mid Y)$. For every R, the event E_R has $Q(E_R) = \sum_y \Pr[Y = y] \int P_{Z|y}(dz) \Pr[d(X, R(z)) \le \eta \mid Y = y] \le \bar{\kappa}$. Data processing for the indicator of E_R gives $\operatorname{kl}(P(E_R) \| Q(E_R)) \le I(X; Z \mid Y)$. If $P(E_R) \ge \bar{\kappa}$, then $\operatorname{kl}(P(E_R) \| \bar{\kappa}) \le \operatorname{kl}(P(E_R) \| Q(E_R))$, since $q \mapsto \operatorname{kl}(p \| q)$ is nonincreasing on $[0, p]$. Take R_j with $P(E_{R_j}) \to S^\star$, and use lower semicontinuity of $\operatorname{kl}(\cdot \| \bar{\kappa})$ on $[\bar{\kappa}, 1]$. Pinsker's inequality $\operatorname{kl}(p \| q) \ge 2(p - q)^2$ gives the second form, and the case $S^\star < \bar{\kappa}$ is trivial. For the third form, $\operatorname{kl}(p \| q) \ge p \ln(1/q) - \ln 2$, which bounds $S^\star$ when $S^\star \ge \bar{\kappa}$. When $S^\star < \bar{\kappa}$ the bound holds too, since $\bar{\kappa} \le \ln 2/\ln(1/\bar{\kappa})$ on $(0, 1)$, as $u \ln(1/u) \le 1/e < \ln 2$.

Gaussian release. Fix $y \in \mathcal{Y}^+$. Given $Y = y$, Z has a density and covariance $\Sigma_{f|y} + \Sigma_B$, and given (X, Y) it is Gaussian with covariance Σ_B. So $I(X; Z \mid Y = y) = h(Z \mid Y = y) - h(B)$ with both entropies finite, and the Gaussian maximizes entropy at fixed covariance, which gives $\frac{1}{2} \ln \det(\Sigma_{f|y} + \Sigma_B) - \frac{1}{2} \ln \det \Sigma_B$.

Tightness. For $Z = Y$, the ceiling gives $S^\star \le \bar{\kappa}$ and the floor gives $S^\star \ge \bar{\kappa}$. For injective Borel f on a Polish space the inverse on the image is Borel (Lusin and Souslin), so $R = f^{-1}$ recovers X. $\square$

Status. Sketch, pending a discussant's check of the measurability steps. The proof is elementary. The ceiling is Theorem 1 of XD23 with side knowledge Y (its Remark 1), applied to an adversary that does not hold Y.

Reading in the trainer's terms. The floor is Halil's lower bound in the statistical regime, in the exact form an unbounded reconstructor permits. It recovers at least the class-conditional prior of the label the trained head predicts, less the head's error. The ceiling says the gain over that floor is paid for only by what the representation carries beyond the label. Part 4 shows that for a deterministic injective representation the statistical baseline is total, so the content of the reconstruction example lies in the computational regime, or in noisy, quantized or many-to-one representations. The proposition does not show that learning a reconstructor needs the trainer's samples, capacity or training cost.

Experiment. Train a split network on MNIST with Gaussian noise of positive definite covariance at the cut. Measure the head's test error. Fix a finite candidate set $\mathcal{C}$ of centres in advance, such as held-out training points, and estimate the class-conditional ball masses on $\mathcal{C}$ from m_y samples per class. Hoeffding's inequality and a union bound give a deviation of $\sqrt{\ln(2|\mathcal{C}||\mathcal{Y}|/\alpha)/(2m_y)}$ for all of them with probability $1 - \alpha$ (proven, standard), and the floor with $\mathcal{C}$ in place of $\mathcal{X}$ is a valid floor, attained by the class-centre decoder. Every learned decoder obeys the ceiling. The ceiling needs $\bar{\kappa}$ over all balls and the class-conditional covariances, and the space claims no estimation rate for the supremum over balls (open).

The other regime. A bounded reconstructor may fall far below $S^\star$, and whether it must, when the representation is injective, is the question of the space `computational-limits`. This proposition is the statistical floor and ceiling it is compared with.

From `lab/theory/spaces/statistical-limits/round-02/builder.md`.

status: sketch

uses: fact-balle-reconstruction-robustness-definition, fact-pac-privacy-mutual-information-bound, fact-pac-privacy-gaussian-release-bound, fact-zhao-accuracy-forces-attribute-leakage

Statement unchanged from round 01. Two changes. In the ceiling proof, if $S^\star = \bar{\kappa}$ the divergence is zero. If $S^\star > \bar{\kappa}$, a tail of any sequence R_j with $P(E_{R_j}) \to S^\star$ lies above $\bar{\kappa}$, where the bound for each R_j holds, and lower semicontinuity passes it to the limit. In the reading, the gain of an unbounded reconstructor above the floor $\bar{\kappa} - \varepsilon^\star$ can come from removing the head's error as well as from residual information, and only the gain above $\bar{\kappa}$ needs residual information $I(X; Z \mid Y)$.

B.32 the proof note's main theorem

`thm-fs-sq-black-box-reduction`, from `lab/theory/spaces/computational-limits/round-00/builder.md`.

status: refuted as stated (counterexample proven below) uses: fact-fgrvx17-planted-clique-sq-bound

Statement (theirs, Theorem 2 of the note). G is drawn from $G(n, 1/2)$ or from $G(n, 1/2)$ with a planted k-clique. The vertex-row encoding D_G draws $v \in [n]$ uniformly

and outputs the adjacency row $G[v, \cdot]$ with a label computed from it. A deterministic learner L asks q_L statistical queries of tolerance τ_L to D_G, and a black-box adversary A queries the model. The note claims $\mathrm{Adv}(A) \le \mathrm{Adv}_{SQ}(n, k, q_L, \tau_L)$, and in particular that for $q_L = \mathrm{poly}(n)$ at constant tolerance no black-box adversary has constant advantage, whatever its own query count.

Counterexample (proven). Take $\tau_L = 1/3$ and the exact oracle, which returns the vertex average itself and is a valid $\mathrm{STAT}(\tau)$ oracle for every τ. The same works with an honest oracle that averages each query over $B = \lceil 32n^2 \ln n \rceil$ fresh draws from D_G: by Hoeffding every answer is within $1/(4n)$ of the vertex average with probability $1 - 2n^{-4}$, and rounding to a multiple of $1/n$ makes it exact for all $O(n^2)$ queries with probability $1 - O(n^{-2})$. Let $T = [\ell]$ with $\ell = \lceil 3 \log_2 n \rceil$.

1. The learner builds the trie of row prefixes on T. For each length $j \le \ell$ and each surviving prefix b it asks $f_b(x) = \mathbf{1}[x_{1..j} = b]$, whose exact answer is the number of rows with that prefix over n. This takes at most $2n\ell$ queries.
2. For each full prefix p and each column $c \in [n]$ it asks $g_{p,c}(x) = x_c \mathbf{1}[x_T = p]$. If every prefix has count one, the answers are the rows themselves. The model M stores the multiset of rows as a lookup function. In all $q_L \le 2n\ell + n^2$ queries, every answer in $[0, 1]$.
3. Distinct prefixes. Under the null, two rows $u \ne v$ agree on T only if $G[u, j] = G[v, j]$ for every $j \in T \setminus \{u, v\}$, independent fair bits, so with probability at most $2^{2-\ell}$. A union bound over pairs gives failure at most $4/n$. Under the planted law the same holds on the event $T \cap S = \emptyset$, of probability at least $1 - \ell k/n$, since then no edge with an endpoint in T is planted.
4. The adversary reads the multiset of rows from M and outputs 1 iff some $C \subseteq [n]$ with $|C| = k$ has, for every $c \in C$, a row r with $r_C = \mathbf{1}_C - e_c$. Under the planted law $C = S$ passes, because a clique vertex's row is 1 on S except at itself (no loops). Under the null, a passing C needs k distinct rows (the patterns differ) whose $k \times k$ restriction contains at least $(k^2 - k)/2$ distinct edges with prescribed values. A union bound over C and the row assignment gives probability at most $n^{2k} 2^{-(k^2-k)/2}$, which tends to 0 for $k > 4 \log_2 n + 1$.

So $\mathrm{Adv}(A) \ge 1 - o(1)$ at $k = n^{1/2-\delta}$, with $q_L = O(n^2)$ queries of constant tolerance. The particular clause is false. The inequality is false for every reading of Adv_{SQ} that is $o(1)$ at $q = O(n^2)$ and $\tau = 1/3$, which is the reading the note needs, and it is vacuous under the only other reading, the best case over oracles.

Where the argument fails, with the remaining defects.

- Step 4 applies FGRVX17 to "vertex-SQ access to G", the empirical law of the n rows of one symmetric matrix. FGRVX17 bounds statistical access to D_S, a law of independent rows, and its Appendix A concerns n samples, not statistical access to them.
- Step 3 equates the two advantages for each fixed oracle, while the SQ lower bound holds for the worst oracle only. The two cannot be chained, and a real learner meets one particular oracle.
- The quoted form $\mathrm{Adv}_{SQ} \le O(q\tau^2 n/k^2)$ grows with τ, backwards, since a coarser tolerance cannot help. The calibration remark inherits the inversion. In the distributional problem at $n = 10^4$ and $k = n^{0.4} \approx 40$, tolerance $k/(4n) \approx 9.95 \cdot 10^{-4}$ recovers S with n queries (FGRVX17 Section 2.4), while the remark calls $\tau_L \in [10^{-5}, 10^{-3}]$ a regime of meaningful security.
- The remark that white-box adversaries escape the proof is wrong. Step 2(3) hands B the whole model, so the argument, where valid, covers white-box and unbounded adversaries alike.

- The membership corollary is ill-posed, since every row is in D_G by construction. The extraction corollary's $I(G;M) \le q_L \log(1/\tau_L)$ needs answers discretized to $\log(1/\tau_L)$ bits, and exact answers break it.
- The abstract says "under the planted-clique conjecture", but the argument would be unconditional, and the Hirahara and Shimizu equivalences (HS24 Theorem 1.3) play no role in it.
- The Brennan and Bresler pointer: BB20 carries its SQ bounds for the secret leakage variant in Section 12.3 (Corollaries 12.17 and 12.18, for specific leakage laws), not in a Theorem 5.1.

Other regime. The counterexample's adversary is unbounded. A polynomial-time adversary on the same model would face the planted clique conjecture after reading the graph, which is the only place computation enters, and the note does not use it.

B.33 what survives of the proof note

`lem-sq-learner-post-processing`, from `lab/theory/spaces/computational-limits/round-00/builder.md`.

status: proven uses: fact-fgrvx17-planted-clique-sq-bound, fact-sq-simulates-minibatch-sgd

Statement. Let $k \le n^{1/2-\delta}$ for a constant $\delta > 0$. Let L be a learner whose only access to its training law $P \in \{U\} \cup \{D_S : |S| = k\}$ is m calls to 1-STAT, with internal coins independent of P. Let A be any randomized map from the output of L, weights included, to $\{0,1\}$, with no bound on its computation. If $A \circ L$ outputs 0 on U and 1 on every D_S, each with probability at least $2/3$, then $m = \Omega(n^2/k^2)$. If instead L asks q queries to VSTAT$(n^2/(rk^2))$ and success is required against every valid oracle, FGRVX17 Theorem 5.4 bounds $A \circ L$ with the same parameters.

Proof. The queries of $A \circ L$ are those of L, chosen from earlier answers and from coins independent of P, and A makes no further call. So $A \circ L$ is a 1-bit sampling algorithm with m calls, or an SQ algorithm with the same q queries, and FGRVX17 Theorem 2.10, or Theorem 5.4, applies. By AKMSS21 Theorem 1c a bSGD learner with $b\rho^2 > C \log(Tp/\delta)$ is simulated by an SQ method with Tp queries of tolerance $\rho/8$, so the VSTAT clause covers it in the worst-case-rounding sense. □

The content is that a learner which cannot detect the planted structure cannot hand it to anyone. The bound is priced in the trainer's units (its fresh samples) and holds against unbounded white-box attackers, so it is a statistical statement about the trainer, and it is non-trivial under Halil's constraints only as a pricing convention. The VSTAT clause speaks about some valid answer sequence, not about the run that happened. A mini-batch SGD learner that reuses samples or reads real-valued per-sample gradients is not a 1-bit sampling algorithm, so the honest clause does not cover it.

Other regime. No oracle is used. The lemma belongs with the statistical space, which owns unbounded attackers, and is recorded here because it fixes the attacker-class question: an SQ bound on the learner binds every attacker, an SQ bound on the attacker binds only SQ attackers.

B.34 the reconstruction games with and without the seed

`def-seed-blind-reconstruction-game`, from `lab/theory/spaces/computational-limits/round-01/builder.md`.

status: proposed, revised uses: fact-cohen-narcissus-equals-one-wayness, fact-orthogonal-equivariance-of-training

Protocol class $\mathcal{P}_H$, for a closed subgroup $H \le O(d)$.

1. The client applies a pre-map $\pi : \mathbb{R}^d \to \mathbb{R}^d$ with $\pi(Qx) = Q\pi(x)$ for $Q \in H$ (the identity, radial normalization), then $a = W\pi(x) + b$ with $W \in \mathbb{R}^{m \times d}$, then any map $z = \psi_\theta(a)$. Every client computation depends on the record only through a.
2. The initialization $\omega = (W_0, b_0, \theta_0, \text{other states}, \text{server coins}, \text{schedule})$ is independent of the records and satisfies $(W_0 Q, \text{rest}) \overset{d}{=} (W_0, \text{rest})$ for $Q \in H$. The optimizer state of W starts at 0.
3. The update of (W, s), with G the gradient in W and s its optimizer state, satisfies $\mathcal{U}_t(WQ, GQ, sQ; c) = \mathcal{U}_t(W, G, s; c)\, Q$ for $Q \in H$ and all coins c. Other parameters update from quantities that depend on the record only through a.
4. The server may follow any strategy, honest or not. Its view $\mathcal{V}$ is every representation it receives, every message it sends, its coins and state, and the side data. The view never contains W_t, any gradient in W, any record, or any pair (record, representation).
5. Records $x_1, \dots, x_N$ are i.i.d. from D. Side data are $y_i = Y(x_i, \xi_i)$ with independent ξ_i and $Y(Qx, \xi) = Y(x, \xi)$ for $Q \in H$. The attacker knows D, Y and the protocol.

A target is a measurable $g : \mathbb{R}^d \to \mathbb{R}^s$ with $0 < \|g\|_{L^2}^2 = E\|g(x)\|^2 < \infty$, with a finite group $K \le O(s)$ of symmetries under which success is measured ($K = \{\pm 1\}$ for a symmetric hidden law). The game reconstructs the observed records, not fresh ones.

Sample game $\Gamma_{\text{samp}}(N, T)$. The attacker receives $\mathcal{V}$ of one run on N records, runs in time T, and outputs $t_1, \dots, t_N$. For a scale $\epsilon > 0$ let F_ϵ be the event that some $\kappa \in K$ has $\|t_i - \kappa g(x_i)\| \le \epsilon$ for every i. Fine success is $S_{\text{fine}}(\epsilon; N, T) = \sup_A \left(\Pr[F_\epsilon] - \Pr[F'_\epsilon] \right)$, where F'_ϵ is F_ϵ with every x_i replaced by an independent $x'_i \sim D$ (the Narcissus baseline, CKMMNST25 Definition 1.6 with $\varepsilon = 0$), the supremum over attackers of time T.

Population SQ game $\Gamma_{\text{SQ}}(q, \tau)$, frozen clients only (no update). The attacker has q queries to STAT(τ) for the law of $(z, y) = (\Phi_\omega(x), Y(x, \xi))$ given the realized ω, where $\Phi_\omega = \psi_{\theta_0}(W_0 \pi(\cdot) + b_0)$, and no other access. It outputs a measurable $R : \mathcal{Z} \times \mathcal{Y} \to [-1, 1]^s$. Its weak success is $\max_{\kappa \in K} E\langle R(z, y), \kappa g(x) \rangle / \|g\|_{L^2}$. It reaches c_0 if its weak success is at least c_0 with probability at least $2/3$ over ω and its coins, for every valid oracle.

Seed-known variants give the attacker ω as well. The price of the seed at a success level is the pair of budget sets at which the seed-known and the seed-blind attackers reach it. A theorem bounds a set from one side. An experiment measures the success of named attacks at given budgets, which bounds the seed-blind set from the other side only.

What would settle it. Well posed as stated. Useful if the theorems below hold in it, which this round claims.

Regime. The sample game against unbounded attackers is statistical and belongs to `statistical-limits`. Polynomial-time and SQ attackers are this space's.

Experiment. Named attacks' success at budgets, by Monte Carlo over records, initialization and coins.

Without the oracle. No learning oracle is needed.

B.35 the server's view is a function of rotated records

`lem-equivariant-view-coupling`, from `lab/theory/spaces/computational-limits/round-01/builder.md`.

status: proven uses: def-seed-blind-reconstruction-game, fact-orthogonal-equivariance-of-training

Statement. Fix a protocol in $\mathcal{P}_H$ and $Q \in H$. Write $\mathcal{V}(\omega; x_{1:N}, y_{1:N})$ for the view of the run with coins ω on records x and side data y, and ω_Q for ω with W_0 replaced by $W_0 Q^\top$. Then pathwise $\mathcal{V}(\omega; x, y) = \mathcal{V}(\omega_Q; Qx, y)$. Consequently, for any joint law of (x, y) independent of ω, $(\mathcal{V}(\omega; x, y), x, y) \stackrel{d}{=} (\mathcal{V}(\omega; Qx, y), x, y)$.

Proof. Induction on the step t, with hypothesis: in the run on (Qx, y) from ω_Q the first-layer weight and its state are $W_t Q^\top$ and $s_t Q^\top$, and every other quantity equals its value in the run on (x, y) from ω. At $t = 0$ this is the definition of ω_Q and $s_0 = 0$. At step t, the pre-activation is $W_t Q^\top \pi(Qx_i) + b_t = W_t Q^\top Q \pi(x_i) + b_t = a_i$, so every representation, every server message (a function of the view so far and the server's coins) and every back-propagated signal $\delta_i = \partial \ell / \partial a_i$ agree. The weight gradient is $\sum_i \delta_i \pi(Qx_i)^\top = G_t Q^\top$, and item 3 of the protocol class with $Q^\top \in H$ gives $\mathcal{U}_t(W_t Q^\top, G_t Q^\top, s_t Q^\top; c) = \mathcal{U}_t(W_t, G_t, s_t; c) Q^\top$. Other parameters see only a and the messages, so they agree. The view contains no first-layer quantity, so the two views are equal. For the second claim, $\omega_Q \stackrel{d}{=} \omega$ by item 2 applied to $Q^\top$, and ω_Q is a function of ω, hence independent of (x, y). □

The pathwise identity keeps the side data y fixed and does not use item 5 of the protocol class. Item 5 is needed only when a simulator must compute the side data from rotated records, as in the corollary.

The induction is that of LZA21 Theorem C.1, applied to the whole transcript, to an adaptive server, and to side data, and stated as a pathwise coupling of the view with the records rather than as equivariance of the output.

Corollary (the reduction). Let Q be Haar on H, independent of everything, and $u_i = Qx_i$. The simulator $\mathrm{Sim}(u, \xi)$ runs the protocol from fresh coins on $(u_i, Y(u_i, \xi_i))$. For every attacker A and target g, the joint law of $(A(\mathcal{V}), g(x_{1:N}))$ equals that of $(A(\mathrm{Sim}(u, \xi)), g(Q^\top u_{1:N}))$, because $Y(u_i, \xi_i) = y_i$ by H-invariance. The u_i are i.i.d. from $Q_\# D$ given Q. The overhead is one run of the protocol, the trainer's own cost. If $D = D_{\mathrm{lab}} \otimes P_V^A$ with V in the orthogonal complement of the label subspace U_y and $H = \{Q : Q|_{U_y} = \mathrm{id}\}$, then QV is a uniformly random frame in $U_y^\perp$ and $g(Q^\top u) = (QV)^\top u$ for $g(x) = V^\top x$: the attacker's task becomes NGCA coordinate recovery in dimension $d - \dim U_y$.

What breaks it, each by a hypothesis of $\mathcal{P}_H$: Adam, RMSProp, AdaGrad, sign updates, coordinate clipping and ℓ_1 penalties on W (item 3, LZA21 Table 1), pixel-wise augmentation, coordinate-wise normalization and convolutional first layers (item 1), publication of W_t or its gradients, and a server that knows some records or pairs (item 4), labels that depend on the hidden part (item 5).

Other regime. The lemma is distributional and holds for unbounded attackers. It is the common first step of the statistical and the computational statements.

B.36 a frozen linear client hands over Haar-rotated records

`lem-whitening-gives-rotated-records`, from `lab/theory/spaces/computational-limits/round-01/builder.md`.

status: proven uses: def-seed-blind-reconstruction-game

Statement. Let the client be frozen with $\pi = \mathrm{id}$ and $\psi = \mathrm{id}$, so $z = Wx + b$, with $W \in \mathbb{R}^{m \times d}$ of full column rank almost surely and $(WQ, b) \stackrel{d}{=} (W, b)$ for all $Q \in O(d)$. Let $E[x] = 0$ and $E[xx^\top] = I_d$. Let $\Sigma_z = WW^\top$ (the covariance of z), B an orthonormal basis of $\mathrm{range}(\Sigma_z)$ chosen by a fixed measurable rule from that subspace, and $T = B^\top \Sigma_z^{+1/2}$. Then $T(z - E[z]) = Rx$ with $R = B^\top W (W^\top W)^{-1/2} \in O(d)$, R is a function of W alone, and R is Haar-distributed and independent of the records. For a LeakyReLU first layer the coordinate-wise inverse is applied first.

Proof. $E[z] = b$. With the polar decomposition $W = OS$, $S = (W^\top W)^{1/2}$,

$O = WS^{-1}$ has orthonormal columns and $WW^\top = OS^2O^\top$, so $\Sigma_z^{+1/2} = OS^{-1}O^\top$ and $\Sigma_z^{+1/2}W = O$. B and O are orthonormal bases of the same d-dimensional subspace, so $R = B^\top O \in O(d)$ and $T(z-b) = Rx$. For $Q \in O(d)$, range(WQ) = range(W), so B is unchanged, and $O(WQ) = WQ(Q^\top S^2 Q)^{-1/2} = WQQ^\top S^{-1}Q = OQ$. Hence $R(WQ) = R(W)Q$, and $WQ \overset{d}{=} W$ gives $RQ \overset{d}{=} R$ for all Q. A probability measure on $O(d)$ invariant under right translations is the Haar measure. R is a function of W, which is independent of the records. $\square$

With N samples the server estimates Σ_z and b, and the error of the estimated T is a separate, standard covariance-estimation term (heuristic here, not used below in any proven bound).

Other regime. Statistical and computational alike: it is the upper-bound half of the seed price for frozen linear clients.

B.37 SQ price of blind reconstruction from a frozen client

`prop-sq-blind-reconstruction-bound`, from `lab/theory/spaces/computational-limits/round-02/builder.md`.

status: proven for part (a), proven conditional on DKRS23 Theorem 1.5 for part (b) uses: lem-equivariant-view-coupling, fact-clwe-hardness, fact-hclwe-sq-correlation-bound, fact-ngca-sq-lower-bound, fact-fgrvx17-planted-clique-sq-bound

Setting as in round 01: Γ_{SQ} with $H = O(d)$, a frozen client of $\mathcal{P}_{O(d)}$, no side data, $r = 1$, target $g(x) = \langle v, x\rangle$, $K = \{\pm 1\}$.

(a) Pancake records. Let $D = H_{v,\beta,\gamma}$ with $\gamma \geq 2$, $\beta \in (0,1)$ and $c_0 \in (0,1]$. Let $\rho_0 = c_0/3$, $\alpha_0 = \rho_0^2/2$, $\gamma_c = 8e^{-\pi\gamma^2(1-\alpha_0^2)}$, $\beta_c = 2(\gamma/\beta)^2$, $M = \lfloor e^{d\alpha_0^2/4}/2 \rfloor$, $L = \sqrt{\log(48\gamma d/(\beta c_0))}$ and $\tau_1 = c_0/(9\sqrt{2\pi}L\sqrt{d})$. Assume $M \geq 48/\rho_0^2$ and $\tau_*^2 \geq 6\gamma_c$, where $\tau_* = \min(\tau, \tau_1)$. Then an attacker that reaches weak success c_0 with q queries of STAT(τ) satisfies

$$q \;\geq\; \frac{11}{288}\, M\, \frac{\tau_*^2 \beta^2}{\gamma^2} \;-\; d.$$

Both assumptions hold for $d \geq C\log(1/c_0)/c_0^4$ and $\gamma^2 \geq C\log(d/c_0)$, in particular for $\gamma \geq 2\sqrt{d}$ and d large. So the price is $\exp(\Omega(c_0^4 d))\tau_*^2\beta^2/\gamma^2$ queries.

(b) Moment-matched records. Let $D = P_v^A$ with γ_d on $v^\perp$ and A of mean 0, variance 1 and $\Pr_A[|a| > t] \leq 2e^{-t^2/(2\sigma^2)}$. Let A satisfy the hypothesis of DKRS23 Theorem 1.5 in ambient dimension d with an even moment degree k, constants λ, c and slack ν, and let $|E_A[h_{j^*}]| \geq \kappa$ for a witness degree $j^* \geq 1$. Let $\sigma, j^*, \kappa, c_0, \lambda, c, k$ be fixed, independent of d. If $((1-\lambda)/4 - c)k > 1/2$ and $\nu \leq d^{-((1-\lambda)/4-c)k}$, then for d large every attacker reaching weak success c_0 with q queries of STAT(τ) has $q \geq 2^{d^{\Omega(c)}} - d - 1$ or $\tau \leq O_k(d^{-((1-\lambda)/4-c)k})$.

Proof. The common part is that of round 01 with the corrected covariance. $\sigma_{\max} = (2\pi)^{-1/2}$ and $\sigma_a \geq (2\pi)^{-1/2}(1 - 2\pi\Delta)^{1/2}$. For $\gamma \geq 2$, $2\pi\Delta \leq 6\pi \cdot 5e^{-4\pi} < 4\cdot 10^{-4}$, so $\sigma_a \geq 0.9998(2\pi)^{-1/2}$. The reducer's d queries $R(\Phi_{\omega'}(u))\mathrm{clip}(u_j, L)/L$ at tolerance τ_1 give each coordinate of $\bar{w} = E[R(\Phi_{\omega'}(u))u]$ within $L\tau_1 + E[|u_j|\mathbf{1}\{|u_j| > L\}]$. Here $L\tau_1 = c_0/(9\sqrt{2\pi}\sqrt{d}) \leq c_0\sigma_a/(8\sqrt{d})$, and by the density bound $H_w \leq (3\gamma/\beta)D_0$ the truncation term is at most $(3\gamma/\beta)e^{-\pi L^2}/\pi \leq c_0/(16\pi d) \leq c_0\sigma_a/(8\sqrt{d})$ for the stated L, since $E_{D_0}[|g|\mathbf{1}\{|g| > L\}] = e^{-\pi L^2}/\pi$ for g of density ρ and $e^{-\pi L^2} \leq \beta c_0/(48\gamma d)$. So $\|\hat{w} - \bar{w}\| \leq c_0\sigma_a/4$, $|\langle \hat{w}, w\rangle| \geq 3c_0\sigma_a/4$, $\|\hat{w}\| \leq \sigma_{\max} + c_0\sigma_a/4 \leq 2\sigma_{\max}$, and $|\langle \hat{s}, w\rangle| \geq 3c_0\sigma_a/(8\sigma_{\max}) \geq 0.37c_0 \geq \rho_0$.

Part (a). The search problem, the packing U of M unit vectors with pairwise overlap at most α_0, the Gram count of at most $4/\rho_0^2$ vectors that one output solves, BRST21 Proposition 8.3, FGRVX17 Lemma 3.10 and Theorem 3.2 are as in round 01. With $M \geq 48/\rho_0^2$, the solution-set fraction is $\eta = 4/(\rho_0^2 M) \leq 1/12$, so $(\alpha-\eta)/(1-\eta) \geq 2/3 - 1/12 \geq 1/2$ for $\alpha = 2/3$, and $|D_{\hat{s}}| \geq (11/12)M$. With $\bar{\gamma} = \tau_*^2/3$ and $\tau_*^2 \geq 6\gamma_c$, $\gamma' = \bar{\gamma} - \gamma_c \geq \tau_*^2/6$, and $\beta_c - \gamma_c \leq \beta_c = 2(\gamma/\beta)^2$. FGRVX17 Theorem 3.2 gives $q + d \geq \frac{1}{2} \cdot \frac{11}{12} M \cdot \frac{\tau_*^2/6}{2(\gamma/\beta)^2} = \frac{11}{288} M \tau_*^2 \beta^2/\gamma^2$. The two assumptions: $M \geq 48/\rho_0^2$ is $e^{dc_0^4/1296} \geq 864/c_0^2 + 2$, and $\tau_1^2 \geq 6\gamma_c$ is $c_0^2/(162\pi L^2 d) \geq 48e^{-\pi\gamma^2(1-\alpha_0^2)}$, which holds once $\gamma^2(1-\alpha_0^2) \geq \pi^{-1}\log(7776\pi L^2 d/c_0^2)$.

Part (b). As in round 01, with $\rho_1 = 3c_0/(8\sigma_{\max}) = 3c_0/8$ for the isotropic A. With j^*, κ, c_0, σ fixed, a fixed M_h meets the truncation condition, so $\tau_2 = \rho_1^{j^*}\kappa/(4M_h)$ is a positive constant. Since $|E_A[h_j]| \leq \nu$ for $1 \leq j \leq k$ and $\nu \to 0$, a fixed κ forces $j^* > k$ for d large. The composite tester uses $q + d + 1$ queries of tolerance $\min(\tau, \tau_1, \tau_2)$, and DKRS23 Theorem 1.5 bounds it. Since $\tau_1 = \tilde{\Theta}(d^{-1/2})$ and $\tau_2 = \Theta(1)$ exceed $O_k(d^{-((1-\lambda)/4-c)k}) + (1 + o(1))\nu$ for d large under the two stated conditions, the minimum is τ. □

Other regime. SQ only. Lattice methods are outside these bounds (`fact-ngca-sq-lower-bound`).

B.38 polynomial-time price of fine blind reconstruction

`prop-fine-reconstruction-to-hclwe`, from `lab/theory/spaces/computational-limits/round-02/builder.md`.

status: proven uses: lem-equivariant-view-coupling, fact-clwe-hardness

Statement. As in round 01: a protocol of $\mathcal{P}_{O(d)}$ (frozen or trained) on N records from $H_{v,\beta,\gamma}$, $\beta \in (0,1)$, $\gamma \geq 2\sqrt{d}$, γ/β polynomial, no side data, $c_1 = 1/(16\pi\sqrt{2\pi})$. An attacker of time T that, with probability p, outputs $t_{1:N}$ with $|t_i - \kappa\langle v, x_i\rangle| \leq c_1/\gamma$ for all i and some $\kappa \in \{\pm 1\}$ gives, for $N = N' = Cd\log^2(\gamma/\beta)$, an algorithm for $\mathrm{hCLWE}_{\beta,\gamma}$ of time $T + T_{\mathrm{run}} + \mathrm{poly}(d)$ and advantage at least $p - e^{-\Omega(d)}$, where T_{run} is the cost of one protocol run. If the protocol runs in polynomial time and arithmetic carries $\mathrm{poly}(d)$ bits, BRST21 Corollary 4.2 makes p negligible for every polynomial-time attacker unless $\mathrm{DGS}_{2\sqrt{2}d\eta_\varepsilon(L)/\beta}$ has a polynomial-time quantum algorithm.

Proof. Steps 1 and 3 are those of round 01, which astra accepted line by line. Step 2 with the corrected covariance. $\Sigma \succeq (1 - 2\pi\Delta)I/(2\pi) \succeq 0.99I/(2\pi)$ for $\gamma \geq 2$. BRST21 Lemma 7.4 applied to the isotropic rows $\Sigma^{-1/2}u_j$, of sub-Gaussian norm $O(\sqrt{\log(\gamma/\beta)})$, gives $\|\frac{1}{N}\Sigma^{-1/2}U^\top U\Sigma^{-1/2} - I\| \leq 1/2$ with probability $1 - e^{-\Omega(d)}$ for $N = Cd\log^2(\gamma/\beta)$, so $\sigma_{\min}(U) \geq (N \cdot 0.99/(4\pi))^{1/2} \geq \sqrt{N}/(2\sqrt{2\pi})$, the bound round 01 used. The least-squares error, the angle bound $\theta \leq 1/(8\gamma)$ and the test then follow as before. In step 3, $E|a| \leq \sigma_a \leq (2\pi)^{-1/2}$. With $\mathrm{poly}(d)$-bit arithmetic, rounding moves each $\langle\hat{s}, u_j\rangle/s_\ell$ by $2^{-\mathrm{poly}(d)}$, which changes S by less than the margin $\kappa_\beta/6$. □

Other regime. Conditional on worst-case lattice hardness, against polynomial-time attackers with the record-level view.

B.39 the seed separates attackers for pancake records

`thm-blind-split-seed-separation`, from `lab/theory/spaces/computational-limits/round-02/builder.md`.

status: proven uses: fact-lei-random-expansive-layers-invertible, prop-sq-blind-reconstruction-bound, prop-fine-reconstruction-to-hclwe, lem-equivariant-view-coupling

Statement. The client's first layer is $z = \mathrm{ReLU}(Wx)$, $W \in \mathbb{R}^{m\times d}$ with i.i.d. Gaussian entries, $m = c_0' d$, $c_0' \geq 2.1$, further layers optional and each satisfying

LJDD19 Assumption 1, the update satisfies item 3 of $\mathcal{P}_{O(d)}$, and the records are i.i.d. from $H_{v,\beta,\gamma}$, $\beta \in (0,1)$, $\gamma \geq 2\sqrt{d}$, γ/β polynomial, with no side data.

(i) Seed-known. For a frozen client, every record x_i is recovered exactly in polynomial time from z_i and W, with probability $1 - Ne^{-\Omega(d)}$ (LJDD19 Theorem 3 per record, union bound). For a trained client, the same holds for the records whose representation the server receives under W_0: the first processed batch, and every record when the protocol makes a forward pass over all records before the first update.

(ii) Seed-blind, population SQ, frozen client. Weak success c_0 costs at least $\frac{11}{288} M\tau_*^2\beta^2/\gamma^2 - d$ queries of STAT(τ), with $\tau_* = \min(\tau, \tau_1)$ and M, τ_1 as in part (a) of `prop-sq-blind-reconstruction-bou` for d large.

(iii) Seed-blind, record-level view, polynomial time, frozen or trained client of polynomial cost. Fine success at scale c_1/γ has negligible probability unless $\mathrm{DGS}_{2\sqrt{2d}\eta_\varepsilon(L)/\beta}$ is quantum polynomial-time (`prop-fine-reconstruction-to-hclwe`).

Proof. (i) LJDD19 Theorem 3 needs Assumption 1 for each layer and an input fixed independently of the weights, and each record is independent of W. A representation computed with W_0 is exactly the object Theorem 3 inverts. (ii) and (iii) are the cited propositions, whose hypotheses the client meets. □

Remark (heuristic). A seed-known server could track a trained client's weights: invert the first batch, recompute the client's update from the back-propagated signals it sent and the recovered records, and repeat. Each step needs the current weights to keep the injectivity of LJDD19 Lemma 1, which Theorem 3 does not give for trained weights.

The price of the seed: one linear solve per record for the seed-known attacker, against exponentially many statistical queries, or a quantum algorithm for a worst-case lattice problem, for the seed-blind one, plus one training run for the reduction. Astra judges this an artificial-distribution reduction: the Gaussian layer is no plug-in, but the hardness enters through the pancake record law and no learning task enters the separation. That judgement is a decision for Halil, named below.

Other regime. The gap is computational. An unbounded blind attacker is not bounded here.

B.40 a good conditional-mean predictor hands over the label direction

`thm-reconstruction-from-conditional-mean`, from `lab/theory/spaces/computational-limits/round-01/builder.md`.

status: proven uses: lem-whitening-gives-rotated-records, def-seed-blind-reconstruction-game

Statement. Let the client be frozen and linear of full column rank (or LeakyReLU, inverted coordinate-wise), records $x \sim \gamma_d$, side data $y = Y(\langle u, x\rangle, \xi)$ for an unknown unit u, $m(t) = E[y \mid \langle u, x\rangle = t]$ with $m \in L^2(\gamma_1)$ and $\beta_1 = E[m(g)g] \neq 0$. The server holds a top model f with $|f| \leq B$, the predictor $F(x) = f(Wx+b)$, its excess square risk $\varepsilon = E(F(x) - m(\langle u, x\rangle))^2$, the population whitening T, and the representations of N records independent of F (held-out batches or inference inputs). Let $\eta_N = B\sqrt{d/(N\delta)}$. If $\sqrt{\varepsilon} + \eta_N < |\beta_1|/2$, then with probability at least $1 - \delta$ the server's unit vector $\hat{s} = \hat{w}_R/\|\hat{w}_R\|$, $\hat{w}_R = \frac{1}{N}\sum_j f(z_j)T(z_j - b)$, satisfies $\|R^\top \hat{s} - \mathrm{sign}(\beta_1)u\| \leq 2(\sqrt{\varepsilon} + \eta_N)/|\beta_1|$, and the reconstructions $t(x) = \langle \hat{s}, T(z-b)\rangle$ satisfy $E_x(t(x) - \mathrm{sign}(\beta_1)\langle u, x\rangle)^2 \leq 4(\sqrt{\varepsilon} +$

$\eta_N)^2/\beta_1^2$ for a record independent of $\hat{s}$, and, for any n records $x_{1:n}$, also those the model was trained on, $\frac{1}{n}\sum_i (t(x_i) - \mathrm{sign}(\beta_1)\langle u, x_i\rangle)^2 \le \|R^\top \hat{s} - \mathrm{sign}(\beta_1)u\|^2 \, \lambda_{\max}(\frac{1}{n}\sum_i x_i x_i^\top)$.

Proof. By `lem-whitening-gives-rotated-records`, $T(z-b) = Rx$, so $R^\top \hat{w}_R = \hat{w} := \frac{1}{N}\sum_j F(x_j)x_j$ and $\|\hat{w}_R\| = \|\hat{w}\|$. Its mean is $E[F(x)x] = E[m(\langle u, x\rangle)x] + E[(F - m)x]$. The first term is $\beta_1 u$, since for unit $e \perp u$ the Gaussian $\langle e, x\rangle$ is independent of $\langle u, x\rangle$ with mean zero. For unit e, $E[(F-m)\langle e, x\rangle] \le \sqrt{\varepsilon}$ by Cauchy-Schwarz and $E\langle e, x\rangle^2 = 1$. By Chebyshev, $\Pr[\|\hat{w} - E\hat{w}\| > \eta_N] \le E[F^2\|x\|^2]/(N\eta_N^2) \le B^2 d/(N\eta_N^2) = \delta$. So $\hat{w} = \beta_1 u + e'$ with $\|e'\| \le \sqrt{\varepsilon} + \eta_N < |\beta_1|/2$, and $\|\hat{w}/\|\hat{w}\| - \mathrm{sign}(\beta_1)u\| \le 2\|e'\|/|\beta_1|$ by the triangle inequality and $|\|\hat{w}\| - |\beta_1|| \le \|e'\|$. Then $t(x) = \langle \hat{s}, Rx\rangle = \langle R^\top \hat{s}, x\rangle$, and $t(x) - \mathrm{sign}(\beta_1)\langle u, x\rangle = \langle R^\top \hat{s} - \mathrm{sign}(\beta_1)u, x\rangle$. For an independent record its second moment is the squared norm. For any n records the average is a quadratic form in the empirical second moment matrix, bounded by its top eigenvalue whatever the dependence on the records. □

For $x_i \sim \gamma_d$, $\lambda_{\max}(\frac{1}{n}\sum x_i x_i^\top) \le (1 + C\sqrt{d/n} + C\sqrt{\log(1/\delta)/n})^2$ with probability $1 - \delta$ (BRST21 Lemma 7.4), so the training records are reconstructed as well as fresh ones once $n \gtrsim d$. This is Halil's lower bound in its exact form: the server's reconstruction error is at most $4(\sqrt{\varepsilon} + \eta_N)^2/\beta_1^2$, set by the performance of the attacked model. CDGJMMSTT21 Theorems 4 and 5 give a distinguishing analogue with an encoder oracle. This theorem needs no encoder oracle and prices reconstruction of a real-valued coordinate.

Other regime. The bound holds for any server, efficient or not, and costs it one pass over N representations. It is an upper bound on attack cost.

B.41 what the labels cost the server

`prop-sq-label-relevant-reconstruction-bound`, from `lab/theory/spaces/computational-limits/round-02/builder.md`.

status: proven uses: lem-equivariant-view-coupling, lem-whitening-gives-rotated-records, fact-generative-exponent-sq-recovery-bound, fact-fgrvx17-planted-clique-sq-bound

Statement. Let the client be frozen and linear, $z = Wx$, with W right rotation invariant and of full column rank, records $x \sim \gamma_d$, labels with $(\langle u, x\rangle, y) \sim P \in \mathcal{G}$ (DPLB24 Definition 1.1), generative exponent $k^* \ge 3$ and $I_{\chi^2}[P] = \sum_k \lambda_k^2 < \infty$, target $g(x) = \langle u, x\rangle$, $K = \{\pm 1\}$. Fix P, $c_0 \in (0, 1]$ and $r \ge 1$ independently of d.

(a) Lower bound. In Γ_{SQ} with the attacker's queries answered by VSTAT(n) on the law of (z, y), an attacker of weak success c_0 with $q \le d^r - d$ queries needs $n \ge c(P, c_0, r)\, d^{k^*/2}$ for d large.

(b) Upper bound, sample-based. With population whitening, the server runs DPLB24's Algorithm 1 on $(Tz_i, T_P(y_i)) = (Rx_i, T_P(y_i))$ with the denoiser T_P of Theorem 4.1. With $n \ge C_{k^*}[d^{k^*/2}/\tilde{\lambda}_{k^*}^2 + d/(\tilde{\lambda}_{k^*}^2\epsilon^2)]$ labelled representations it obtains $\hat{s}$ with $(\hat{s} \cdot Ru)^2 \ge 1 - \epsilon^2$ with probability $1 - 2e^{-d^c}$, and reconstructs $\pm\langle u, x_i\rangle$ by $\langle \hat{s}, Tz_i\rangle$ with average error at most $2\epsilon^2 \lambda_{\max}(\frac{1}{n}\sum_i x_i x_i^\top)$.

Proof of (a). Reduction, as accepted in round 01: the reducer answers the attacker's queries $f(z, y)$ by $f(W'x', y)$ on the single-index law with a uniform direction w^*, and the d truncated queries at VSTAT(n_1), $n_1 = \lceil 64L^2 d/c_0^2\rceil$, $L = \sqrt{2\ln(16\sqrt{d}/c_0)}$, return $\hat{s}$ with $|\hat{s} \cdot w^*| \ge \rho = 3c_0/5$, with probability $2/3$, using $Q \le d^r$ queries to VSTAT(n'), $n' = \max(n, n_1)$.

Lower bound for that search problem, written out. Take the code of DPLB24 Lemma E.3: $m \le d^{d^{1/4}}$ unit vectors with pairwise overlap at most $\epsilon = \sqrt{C}\log_d(m)/\sqrt{d}$. By the series in the proof of Lemma E.2, $\chi_0(P_{w_i}, P_{w_j}) = \sum_{k \ge k^*} \lambda_k^2 (w_i \cdot w_j)^k$, so off

the diagonal $|\chi_0| \le \lambda_{k^*}^2 \epsilon^{k^*} + \epsilon^{k^*+1}/(1-\epsilon) \le 2\lambda_{k^*}^2 \epsilon^{k^*} =: \gamma_c$ once $\lambda_{k^*}^2 \ge 2\epsilon$ and $\epsilon \le 1/2$, and on the diagonal $\chi_0(P_w, P_w) = I_{\chi^2}[P] =: \beta_c$ by Lemma 2.3. If $\epsilon \le \rho^2/2$, the Gram argument lets one output solve at most $4/\rho^2$ code members, so the solution-set fraction is $\eta = 4/(\rho^2 m)$. FGRVX17 Lemma 3.10 and Theorem 3.2 with $\bar{\gamma} = 1/(3n')$, $\gamma' = \bar{\gamma} - \gamma_c$, give, for success $\alpha = 2/3$ and $\alpha' = (\alpha - \eta)/(1-\eta)$,

$$Q \ge \alpha'(m - 4/\rho^2)\frac{1/(3n') - \gamma_c}{\beta_c - \gamma_c}.$$

Choose $m = \lceil 12Qn'\beta_c/\alpha' \rceil + \lceil 4/\rho^2 \rceil$, so the inequality forces $1/(3n') - \gamma_c \le 1/(12n')$, that is $n' \ge 1/(4\gamma_c) = d^{k^*/2}/(8\lambda_{k^*}^2 C^{k^*/2} \log_d(m)^{k^*})$. Suppose $n' < d^{k^*/2}$, or the claim holds. Then $\log_d(m) \le r + k^*/2 + \log_d(12\beta_c/\alpha') + \log_d(1 + 4/\rho^2) + 1$, a constant $\ell(P, c_0, r)$ for d large, and $m \le d^{d^{1/4}}$. So $\epsilon = \sqrt{C}\ell/\sqrt{d}$, which meets $\epsilon \le \rho^2/2$, $\epsilon \le 1/2$ and $\lambda_{k^*}^2 \ge 2\epsilon$ for d large, and $\eta \to 0$, so $\alpha' \ge 1/2$. Hence $n' \ge c\, d^{k^*/2}$ with $c = 1/(8\lambda_{k^*}^2 C^{k^*/2} \ell^{k^*})$. For $k^* \ge 3$, $n_1 = \tilde{O}(d) < c\, d^{k^*/2}$ for d large, so $n' = n$. □

Proof of (b). By `lem-whitening-gives-rotated-records`, $(Tz_i, y_i) = (Rx_i, y_i)$ is a single-index sample with direction Ru, and DPLB24 Theorem 4.1 applies to $(Rx_i, T_P(y_i))$. The reconstruction error is as in round 01. □

The server's price for the label-relevant coordinate is the generative exponent of the task, whatever the trainer's model learned. The lower bound measures detection first (DPLB24 Remark 3.3) and the upper bound is sample-based, so the two sides are not one SQ price.

Other regime. SQ for (a), sample-based for (b). No oracle.

B.42 withdrawn as stated

`conj-reconstruction-at-training-cost`, from `lab/theory/spaces/computational-limits/round-01/builder.md`.

status: refuted as stated (part b), superseded uses: thm-reconstruction-from-conditional-mean, prop-sq-label-relevant-reconstruction-bound

Part (b) of round 00 claimed that blind reconstruction of the label-relevant coordinate is equivalent to learning the task. Astra's counterexample refutes it (proven): $y = \langle u, x \rangle \xi$ with an independent sign ξ has Bayes square-loss predictor 0, so learning the task needs nothing, while the label law reveals $\langle u, x \rangle$ up to sign at generative exponent 2. Part (a) is now `thm-reconstruction-from-conditional-mean`, with the sample complexity astra asked for, and the label side is `prop-sq-label-relevant-reconstruction-bound`. Nothing of the aim remains open under this slug.

Other regime. Not applicable.

B.43 the seed is worth the NGCA price of the record law

`conj-seed-price-is-ngca-price`, from `lab/theory/spaces/computational-limits/round-02/builder.md`.

status: open, restated uses: lem-equivariant-view-coupling, lem-whitening-gives-rotated-records, prop-sq-blind-reconstruction-bound, prop-fine-reconstruction-to-hclwe

Statement. Consider frozen clients of $\mathcal{P}_{O(d)}$ that are recoverable: a polynomial-time seed-known attacker recovers every record exactly from (ω, z_i) with probability $1 - o(1)$. Let the records be i.i.d. from P_V^A with A of mean 0 and covariance I_r, $r = O(1)$, and no side data. Then the budget sets of seed-blind attackers in Γ_{SQ} (weak success) and in Γ_{samp} (fine success) for the target $V^\top x$ modulo the symmetry group of A equal, up to $\mathrm{poly}(d)$ in each resource and constant factors in the success level, those of NGCA coordinate recovery for A with a uniformly random hidden frame.

Settled. Lower inclusion: for pancakes in both games (`prop-sq-blind-reconstruction-bound` part a, `prop-fine-reconstruction-to-hclwe`), and for moment-matched A with $k \geq 4$ and a fixed witness in the SQ game (part b). Upper inclusion: for frozen linear clients of full column rank given the population whitening map. Since A is isotropic, P_V^A is isotropic, `lem-whitening-gives-rotated-records` gives $u_i = Rx_i \sim P_{RV}^A$ with R Haar, so RV is a uniformly random frame and the NGCA output $(RV)^\top u_i = V^\top x_i$ is the target itself (proven). Linear clients are recoverable.

Open. The upper inclusion for ReLU and deeper recoverable clients, where the blind server must learn the client map without the seed. The cost of estimating the whitening map, in samples or statistical queries, and the robustness of NGCA algorithms to that estimate. Coarse reconstruction in polynomial time (`prob-coarse-reconstruction-lattice-hardness`).

What would settle it. A blind inverter for expansive Gaussian ReLU clients that learns the client up to rotation from representations of an isotropic record law, at polynomial cost. A refutation: a recoverable client for which blind reconstruction is easier or harder than NGCA by more than polynomial factors.

Regime and experiment. Computational. The experiment of round 00, with a lattice-reduction attacker added (`fact-ngca-sq-lower-bound`).

B.44 one learning oracle with two seed-access modes

`def-seed-access-modes`, from `lab/theory/spaces/oracle-and-measurement/round-02/builder.md`.

status: open uses: def-learning-oracle, def-learning-oracle-model, def-certifier, def-degenerate-oracles, def-passive-learning-oracle, def-generic-training-oracle

Statement. Fix a learning oracle $\mathcal{O}$ for (L, μ, π). A game assigns each caller four things.

1. A *seed-access mode*, named or drawn. In the *named mode* the caller's Train calls are those of def-learning-oracle. In the *drawn mode* the oracle accepts $\mathsf{Train}(x, s)$ only with $s = \perp$, returns the model's name only, keeps the seed under no name the caller holds, and refuses Open on any seed. A grant to a caller in the drawn mode carries model names only. In both modes the caller's k-th received name is k.
2. A *return label*, open or eval, carried by the model names that the caller's own Train calls return. With label eval the caller can query its own models through Eval but cannot open or retrain them.
3. A set $\Lambda_A \subseteq \Lambda$ of learner names, and a set of starting models, that its Train calls may use.
4. A *budget*, a bound for each operation (Train, Eval at each query type, Open). The budget (q_T, q_E) of def-learning-oracle-model is the case that bounds Train and Eval on names with label eval and leaves the rest free.

A *cost meter* is a measurable map $\mathsf{cost} = (\mathsf{work}, \mathsf{size}) : \mathcal{X} \to [0, \infty)^2$, where $\mathsf{work}(x)$ is the work of one run at x in a fixed machine model and $\mathsf{size}(x)$ the number of examples in the dataset of x. A caller's *oracle cost* is the coordinatewise sum of cost over its Train calls. The bottom oracle of def-degenerate-oracles is the named mode with return label open and no grant, the top oracle adds grants of the honest seeds, and an experimenter in the drawn mode follows the drawn-seed discipline of def-certifier.

Remark, the needle bound in oracle cost (proven). In the setting of lem-needle-lower-bound let N_X count the certifier's Train calls with input in $X = \bigcup_k X_k$. Its proof gives $\Pr_{\mathsf{L}^\circ}[\hat{t} \geq \vartheta] \geq 1 - \alpha - \mathbb{E}_{\mathsf{L}^\circ}[N_X]/J$, since $\sum_k \mathbf{1}[\mathrm{Hit}_k] \leq N_X$. A certifier whose work

is at most C satisfies $N_X \inf_{x \in X} \mathsf{work}(x) \le C$, which is vacuous when the infimum is zero. A run of thm-dataflow-composition (c) costs $\sum_{t \le R}(\mathsf{cost}(x_t^0) + \mathsf{cost}(x_t^1))$ plus the calls of the other nodes and of the strategy. A bound in oracle cost is a bound in machine compute only under FORMAL_SECURITY's no-skipping hypothesis.

Remark, the seed for an adversary. Under the top oracle with deterministic kernels and a deterministic channel, one Train call at the revealed seed attains $\mathrm{D}_1^{\top}$, which is thm-reference-access (a). Under hardware randomness the rerun-and-compare attack has advantage equal to a coupled tail minus the spread of the rerun, which can be negative. How many reruns an adversary fixed before L needs to approach $\mathrm{D}_1^{\top}$ is open.

The proposal for Halil. Both repositories use this one oracle, and each game fixes the four items per caller. The passive learning oracle and the generic training oracle of FORMAL_SECURITY are drawn-mode callers of it (prop-drawn-mode-is-fs-oracle). Drawn coins lose no soundness for any bottom-oracle target of the HE-ML theory, and lose the accuracy that its separations measure (thm-coupling-free-soundness). Named seeds are needed for the top-oracle threat model, for replay audits, and for the accuracy of coupled certificates. The decision left to Halil is whether to adopt this, to keep drawn coins only, which makes those three inapplicable, or to keep named seeds only, which needs the declared starting models of item 3 whenever the meter is work.

Without the oracle, or in the other regime. The definition adds no power. It names restrictions that a game imposes, per caller, and a meter. In the statistical regime the meter is a count of calls. In the computational regime it is the work and the examples of the runs, and a price in work transfers to real training only under the no-skipping hypothesis.

B.45 the two FORMAL_SECURITY oracles are drawn-mode callers

`prop-drawn-mode-is-fs-oracle`, from `lab/theory/spaces/oracle-and-measurement/round-02/builder.md`.

status: sketch uses: def-seed-access-modes, def-passive-learning-oracle, def-generic-training-oracle

Statement. (a) Let $\mathcal{M}$ be a standard Borel space of traces of length at most T, Λ_A the finite set of recipes a game names, and $m_\varnothing$ a fixed symbol, the only starting model allowed. Let $x(\ell, D) = (\ell, m_\varnothing, D, \mathcal{H})$ for a fixed $\mathcal{H}$, and let $\mathsf{L}(x(\ell, D); \rho, \xi)$ be the trace of recipe ℓ on D with coins (ρ, ξ), whose initial state the declared initialization rule of ℓ draws from the coins. A caller in the drawn mode with return label open, which opens each model name it receives, has, call by call, the transcript law of a caller of the passive learning oracle of `oracle.tex` (label `def:passive`) on the same sequence of recipes and corpora. Its oracle cost, the sums of work and of size, is the metered cost of `def:cost`, the work of the recipe on the corpus together with the size of the corpus. (b) Let $\mathsf{ev}(m, u)$ return $f_m(x)$, the input gradient of the loss at a labelled point z, or its parameter gradient at z, according as u is (out, x), (gx, z) or $(\mathrm{g}\theta, z)$. Fix the architecture, the training procedure and the utility condition α of def-generic-training-oracle, the last as a restriction on the admissible learning functions. A caller in the drawn mode, with return label eval and a budget on Train, on Eval at each query type and on Open, realizes an access class of that report. Train is train, Eval at the three query types is eval, grad_x and grad_θ, and reveal is Open, permitted exactly when the class contains reveal, in which case the return label is open.

Proof. (a) $\mathsf{Train}(x(\ell, D), \perp)$ draws $(\rho, \xi) \sim \mu \otimes \pi$, independent of all other randomness by fidelity, and stores the trace of ℓ on D with those coins. Open returns it. So each pair of calls returns a trace of ℓ on D with fresh coins, and two pairs on one corpus return independent traces, the two clauses of `def:passive`. The caller

can pass no seed and holds no seed name, so it cannot rerun at chosen coins, which `def:passive` forbids. The only starting model is $m_\varnothing$ and the initialization is the declared rule of ℓ, so no caller emits chosen weights as a trace, the laundering that the text after `def:cost` blocks. Learner names lie in Λ_A, so no caller authors a recipe. Summing $\mathsf{work}(x(\ell, D))$ and $\mathsf{size}(x(\ell, D)) = |D|$ gives the two parts of the metered cost. The coins of `def:passive` are one random object, and (ρ, ξ) represents it by (F9). (b) Each operation of the report either reads the stored parameter vector through a fixed public map, which is ev at the matching query, or returns it, which is Open. The report's train returns a fresh handle and never opens the randomness, which is the drawn mode. With return label eval the caller cannot open its own handles, as a class without reveal requires. The budgets per operation are item 4 of def-seed-access-modes. The utility condition restricts which learning functions the game admits and does not change the interface (proven).

Label. Proven, by construction.

Without the oracle, or in the other regime. The statement concerns oracles only. Its use is that every theorem of the HE-ML theory that holds for a drawn-mode caller holds for FORMAL_SECURITY's oracles under these identifications, and every theorem that needs the named mode is inapplicable to them as stated.

B.46 shared seeds buy accuracy, never soundness, for bottom-oracle targets

`thm-coupling-free-soundness`, from `lab/theory/spaces/oracle-and-measurement/round-02/builder.md`.

status: sketch uses: thm-coupling-certificate, thm-dataflow-composition, thm-bounded-deviation-transfer, def-isolated-client

Setting. A *twin rule* $\Theta = (\Theta_t)_{t \le R}$ is a sequence of laws on $(\mathcal{R} \times \Xi)^2$ whose two marginals are both $\mu \otimes \pi$. It *shares the seed* when its two seed coordinates agree almost surely. The identity coupling of the oracle theory is the twin rule with a common seed and independent hardware randomness. The *independent twin rule* is $(\mu \otimes \pi)^{\otimes 2}$. Both are implementable through Train, by one call with $s = \perp$ and a second with the returned seed name or with $s = \perp$. A general twin rule need not be implementable, and the theorem concerns an experimenter that can sample the rule it uses. In the setting of thm-dataflow-composition, the Θ-coupled execution Ex_0^Θ runs G under H_0 and, at each v_t that runs, draws $\zeta_t = ((\rho_t, \xi_t), (\rho'_t, \xi'_t)) \sim \Theta_t$ independently of ω, of the release noises $Z_1, \dots, Z_R$ and of the other ζ_s. The node uses (ρ_t, ξ_t), and the twin is $W'_t = \mathsf{L}(x(c_t, D_j^1, W'_{t-1}); \rho'_t, \xi'_t)$. The execution Ex_1^Θ runs G under H_1 with the two coordinates exchanged, the node using (ρ'_t, ξ'_t) and the twin, on D_j^0, using (ρ_t, ξ_t). Their exceedance probabilities are $\epsilon_0^\Theta(\Delta)$ and $\epsilon_1^\Theta(\Delta)$, quantities of the experiments, which differ from $\epsilon_0(\Delta)$ and $\epsilon_1(\Delta)$ in general. The *bottom-oracle targets* are $\mathrm{H}_\gamma(P_0 \| P_1)$ and $\mathrm{H}_\gamma(P_1 \| P_0)$ for the bottom-oracle view, and the *top-oracle targets* the same divergences for the top-oracle view, $\mathrm{D}_\gamma^\top(K)$ at one node.

Statement. (a) One node. For every coupling Γ' of Π_{x_0} and Π_{x_1}, every channel K and every $\gamma \ge 1$, $\mathrm{H}_\gamma(P_0 \| P_1) \le \mathbb{E}_{\Gamma'} \mathrm{H}_\gamma(K(\cdot \mid W_0) \| K(\cdot \mid W_1))$. If Γ' is the law of $(\mathsf{L}(x_0; \rho, \xi_0), \mathsf{L}(x_1; \rho, \xi_1))$ for a joint law of (ρ, ξ_0, ξ_1) in which each (ρ, ξ_b) has law $\mu \otimes \pi$, the same right side bounds $\mathrm{D}_\gamma^\top(K)$.

(b) Composition. Under the bottom oracle, for every twin rule Θ, every adversary, every $\varepsilon \ge 0$ and every $\Delta > 0$, $\mathrm{H}_\gamma(P_0 \| P_1) \le \delta_{\mathcal{N}}(\varepsilon; \sqrt{R}\Delta/\sigma) + \min\{\epsilon_0^\Theta(\Delta), \gamma\epsilon_1^\Theta(\Delta)\}$ and $\mathrm{H}_\gamma(P_1 \| P_0) \le \delta_{\mathcal{N}}(\varepsilon; \sqrt{R}\Delta/\sigma) + \min\{\epsilon_1^\Theta(\Delta), \gamma\epsilon_0^\Theta(\Delta)\}$, with thresholds per round as in thm-dataflow-composition (b). Under the top oracle the same holds

for every twin rule that shares the seed. The sum-of-tails bound of thm-dataflow-composition (b) is not claimed for general Θ.

(c) Certificates. Let an experimenter sample independent repetitions of a Θ-coupled experiment. The certificate of thm-coupling-certificate (b), computed on draws of Γ', is sound at level α for $\mathrm{H}_\gamma(P_0\|P_1)$, under its computability assumption on the channel divergence. With $\bar{p}^\Theta$ the Clopper and Pearson bound of thm-dataflow-composition (c) computed on Ex_0^Θ, $\min\{1, \delta_{\mathcal{N}}(\varepsilon; \sqrt{R}\Delta/\sigma) + \bar{p}^\Theta\}$ is an upper bound on $\mathrm{H}_\gamma(P_0\|P_1)$ with probability at least $1-\alpha$, for every Θ. The same holds for the top-oracle targets when Θ shares the seed. Under a rule other than the identity coupling these give, in general, no certificate of $\epsilon_0(\Delta)$.

(d) Necessity. The independence from the release noise and the shared seed for top targets cannot be dropped.

Proof. (a) (F4) with $\Theta = \Gamma'$ gives the first bound, since $\Gamma' K_b$ has b-th marginal P_b. Step 2 of the proof of thm-coupling-certificate (b) writes $\mathrm{D}_\gamma^\top(K) = \int \mathrm{H}_\gamma(P_0^\rho\|P_1^\rho)\,\mu(d\rho)$, with P_b^ρ the law of the view given the seed. For μ-almost every ρ, the conditional law of (ξ_0, ξ_1) given ρ has marginals π, so (F4) with it bounds the integrand by the conditional mean of $\mathrm{H}_\gamma(K(\cdot \mid \mathsf{L}(x_0; \rho, \xi_0))\|K(\cdot \mid \mathsf{L}(x_1; \rho, \xi_1)))$, and integrating over μ gives the second bound.

(b) Follow the proof of thm-dataflow-composition (b) with one change, which is which coordinates drive which chain. Write $\zeta_t = ((\rho_t^0, \xi_t^0), (\rho_t^1, \xi_t^1)) \sim \Theta_t$, independent of ω, of the Z_s and of the other ζ_s, and let v_t use (ρ_t^b, ξ_t^b) under H_b. That pair has law $\mu \otimes \pi$ and is independent of everything else in the H_b execution, so the law of that execution is unchanged. Step 1 is unchanged, with $\bar{\rho}_t = \rho_t^b$ under the top oracle. Step 2 drives the hybrid's chain of D_j^b by (ρ_t^b, ξ_t^b). Step 3 compares the hybrid with Ex_0^Θ, whose twin is driven by (ρ_t^1, ξ_t^1). Before the first exceedance the clip does nothing, so the two have the same choices, chains, releases and stops, and $\mathrm{TV}(\mathcal{L}_0, \mathcal{L}_{\mathrm{hyb}}) \le \epsilon_0^\Theta(\Delta)$. In step 4 the kernel draws $\zeta_t \sim \Theta_t$ and extends both chains. Given its past outputs, both chain states and ζ_t, when $\mathfrak{z}_t \sim \mathcal{N}(\Delta/\sigma, 1)$ the output y_t has law $\mathcal{N}(O_t + F_t^1, \sigma^2 I_d)$ whatever F_t^0 is, by (F6)(ii), and Z_t'' is fresh. That law does not depend on the chain of D_j^0, which can therefore be discarded. The chain of D_j^1 is driven by (ρ_t^1, ξ_t^1), of law $\mu \otimes \pi$ and independent of the past, so by induction over the rounds the output has law $\mathcal{L}_1$ in every coordinate but one. Under the top oracle the kernel must also output a revealed seed, which is ρ_t^1 in $\mathcal{L}_1$ and ρ_t^0 in the hybrid as step 3 needs. One kernel outputs one value, so step 4 needs $\rho_t^0 = \rho_t^1$ almost surely. Under the bottom oracle $\bar{\rho}_t = \star$ and the condition does not arise. Step 5 is unchanged. The bounds with ϵ_1^Θ repeat the construction with the roles of the hypotheses exchanged, the hybrid agreeing with Ex_1^Θ, whose node is driven by the second coordinate, which is why the coordinates of Ex_1^Θ are exchanged. A twin rule need not be exchangeable, so the two bounds use the two executions as defined.

(c) By fidelity across repetitions the indicators of $D_0^\star > \Delta$ in Ex_0^Θ are independent Bernoulli variables with parameter $\epsilon_0^\Theta(\Delta)$, so (F8) gives $\Pr[\epsilon_0^\Theta(\Delta) \le \bar{p}^\Theta] \ge 1-\alpha$, and (b) with $\mathrm{H}_\gamma \le 1$ gives the bound. For thm-coupling-certificate (b), the summands of $\hat{t}$ are independent, lie in $[0,1]$, and have mean the right side of (a), so Hoeffding's inequality (F8) gives soundness. For the last sentence, argon's example in (d) has $\epsilon_0^\Theta(0.5) = 1/2$ under the independent twin rule and $\epsilon_0(0.5) = 1$.

(d) Shared seed. Let $\mathcal{R} = [0,1)$, deterministic kernels, the plain channel, $\mathsf{L}(x_0; \rho) = \mathbf{1}[\rho < 1/2]$ and $\mathsf{L}(x_1; \rho) = \mathbf{1}[\rho \ge 1/2]$. Then $\mathrm{D}_1^\top = 1$, while the independent twin

rule gives $\Pr[W_0 \neq W_1] = 1/2$, so the certificate of thm-coupling-certificate (b) concentrates at $1/2$ and is unsound for $\mathrm{D}_1^{\top}$. As a release on $\mathbb{R}$ with $R = 1$ and $\Delta = 0.5$, the identity coupling has $|W_1 - W_0| = 1$ always and the independent rule half the time. Release noise. Take $R = 2$, $d = 1$, $\sigma = 1$, $O_t = 0$, $f_t(c, m) = m$, round-1 models of law $\mathcal{N}(0, 1)$ under both hypotheses, the choice $c_2 = \mathbf{1}[|y_1| > 1]$, and models in round 2 that differ by 10Δ between the hypotheses exactly when $c_2 = 1$. A round-1 coupling that sets both models to $-Z_1$ has the right marginals, yet makes $y_1 = 0$, $c_2 = 0$ and the measured exceedance zero. In the true execution $y_1 \sim \mathcal{N}(0, 2)$ under both hypotheses, so by (F5) $\mathrm{TV}(P_0, P_1) = \Pr[|y_1| > 1]\,(2\Phi(5\Delta) - 1)$, which at $\varepsilon = 0$ and $\Delta = 1/2$ is $0.4795 \times 0.9876 = 0.474$, above the claimed $\delta_{\mathcal{N}}(0; \sqrt{2}\Delta) = 0.276$ (computed). The example is excluded because ζ_1 must be independent of Z_1.

Label. Proven, (a) to (d). Part (b) modifies a proof that round 03 of the formal tuning checked, and astra re-checked the change line by line in round 01. Argon's review states a wider form with twin kernels that adapt to the past, driven by fresh randomness independent of ω and of $Z_{\geq t}$. That form is not re-derived here and stays open in this file.

Meaning. A real twin runs at a chosen seed on whatever hardware is available, so the realized pair need not have independent hardware randomness. For bottom-oracle targets that does not matter. Soundness needs each run, alone, to have the law of the deployed run at its input, and repetitions to be independent. The shared seed and the choice of coupling move only the size of ϵ_b^{Θ}, so only the accuracy, and the coupled tail $\epsilon_0(\Delta)$ itself is certified, in general, only under the identity coupling.

Without the oracle, or in the other regime. Without the oracle no twin can be sampled. In the drawn mode the independent twin rule is available, and its certificate is sound and, by thm-dataflow-composition (e)(ii), inaccurate where the separations place it. The statement is statistical. In the computational regime the work of a certificate is the work of its runs, whatever their coupling.

B.47 what a finite experiment certifies about a security profile

`prop-profile-run-coordinate`, from `lab/theory/spaces/oracle-and-measurement/round-02/builder.md`.

status: sketch uses: def-security-profile-curve, def-learning-oracle-model, thm-coupling-certificate, lem-atomless-two-point

Setting. Fix a procedure, a coalition and a distinguishing goal under the bottom oracle, and a learning function L. For a budget $b = (q_T, q_E)$ let $\epsilon_{\mathsf{L}}(b)$ be the best advantage of an adversary with that budget. A *declared family* is a sequence $\mathcal{A} = (\mathcal{A}_k)_{k \in \mathbb{N}}$ of adversaries, fixed before L, where $\mathcal{A}_k$ makes k Train calls of its own, its *shadow runs*, and its advantage $\mathrm{Adv}(\mathcal{A}_k)$ averages over them. The family is *two-phase* when each $\mathcal{A}_k$ first makes its k Train calls, with inputs and seeds that depend on no procedure value, and then outputs a bit from the procedure's channel outputs and its own models with no further Train call. Its profile is $\bar{\epsilon}_{\mathcal{A}}(k) = \max_{k' \leq k} \mathrm{Adv}(\mathcal{A}_{k'})$, nondecreasing in k.

Statement. (a) Flatness. For every learning function and every (q_T, q_E), $\epsilon_{\mathsf{L}}(q_T, q_E) = \epsilon_{\mathsf{L}}(0, q_E)$. In the setting of thm-coupling-certificate with no query channel, $\epsilon_{\mathsf{L}}(b) = \mathrm{TV}(P_0, P_1)$ for every b.

(b) Lower band of a declared family. Fix shadow budgets $k_1 < \cdots < k_r$ and trial counts $n_1, \ldots, n_r$ in advance. For a two-phase family a trial of $\mathcal{A}_{k_i}$ makes fresh

shadow runs and then one execution of the procedure under each hypothesis, both tested with the same shadow models. For a general family a trial makes two complete and independent executions of $\mathcal{A}_{k_i}$ against the procedure, one under each hypothesis, each with its own shadow runs. A trial records the difference $D \in \{-1, 0, 1\}$ of the output under H_1 and the output under H_0. Let $\bar{D}_i$ be the mean of n_i independent trials and $\underline{\ell}_i = \bar{D}_i - \sqrt{2\ln(r/\alpha)/n_i}$. Then, for every learning function, with probability at least $1-\alpha$, $\bar{\epsilon}_{\mathcal{A}}(k) \geq \max\{\underline{\ell}_i : k_i \leq k\}$ for every $k \geq k_1$. A trial costs k_i shadow runs and two executions of the procedure for a two-phase family, and $2k_i$ shadow runs and two executions for a general one.

(c) Upper band. In the setting of thm-coupling-certificate, the coupled certificate of its part (b) at $\gamma = 1$ is an upper band on $\epsilon_{\mathsf{L}}(b)$ for every b at once, sound at level α for every learning function. Through a deterministic channel with atomless view laws, no drawn-mode certificate sound for every learning function outputs an upper band below one with probability above α.

Proof. (a) Monotonicity gives $\geq$. For $\leq$, let $\mathcal{A}$ have budget (q_T, q_E). Under the bottom oracle every Train call of $\mathcal{A}$ has arguments it holds with label open, so its training inputs are built from values it received through the plain channel, its own choices, models of its own calls, and models and seeds of the nodes its coalition owns, and never from a name served through the query channel, which carries label eval. Let $\mathcal{A}'$ run $\mathcal{A}$ and keep a private table of virtual names in the order $\mathcal{A}$ expects them. It answers each Train call of $\mathcal{A}$ by computing $\mathsf{L}(x; \rho', \xi')$ itself, at the given L, with fresh hardware randomness and either a fresh seed from its own coins or the seed that a virtual or a coalition seed name stands for, which it reads by Open. It answers later Open and Eval calls on virtual names from the table, supplies the value of a virtual model when one enters a procedure choice, and passes calls on real query-channel names to the oracle, within their allowances. By fidelity the answers $\mathcal{A}$ receives have the same joint law with everything else as in the oracle, so $\mathcal{A}'$ has the same advantage, the same Eval calls, and no charged Train call. Such an $\mathcal{A}'$ exists for each L, since the adversary may depend on L when L is fixed first (remark of def-degenerate-oracles). This is a statement about measurable strategies, not an efficient simulation. With no query channel, (F1) and the hard-wired Neyman and Pearson set attain $\mathrm{TV}(P_0, P_1)$ at budget zero, and step 3 of the proof of thm-coupling-certificate (b) bounds every bottom-oracle adversary by it.

(b) In a two-phase trial the shadow models are independent of the two executions of the procedure and serve both, so $\mathbb{E}[D] = \mathbb{E}_S[\Pr[\mathcal{A}_{k_i} = 1 \mid H_1, S] - \Pr[\mathcal{A}_{k_i} = 1 \mid H_0, S]] = \mathrm{Adv}(\mathcal{A}_{k_i})$. In a general trial the two executions are independent complete runs of the attack, so $\mathbb{E}[D] = \mathrm{Adv}(\mathcal{A}_{k_i})$ directly. Trials are independent by fidelity. D takes values in $[-1, 1]$, so Hoeffding's inequality (F8) gives $\Pr[\underline{\ell}_i > \mathrm{Adv}(\mathcal{A}_{k_i})] \leq \alpha/r$. A union bound over i and $\mathrm{Adv}(\mathcal{A}_{k_i}) \leq \bar{\epsilon}_{\mathcal{A}}(k)$ for $k \geq k_i$ finish. Two remarks show why the two kinds of trial differ. A general adversary may choose its shadow inputs from the challenge, so one bank drawn before the challenge need not implement it under either hypothesis. One bank reused across trials does not give a band either. Let $P_0 = \delta_0$, $P_1 = \delta_1$ on $\mathcal{V} = \{0, 1\}$, and let $\mathcal{A}_0$ draw a fair bit S and output $\mathbf{1}[Y = S]$. Then $\mathrm{Adv}(\mathcal{A}_0) = 0$, while the test realized at $S = 1$ has advantage one, so a band built on one realized bank exceeds the profile with probability $1/2$.

(c) The first sentence is thm-coupling-certificate (b) with $\epsilon_{\mathsf{L}}(b) \leq \mathrm{TV}(P_0, P_1)$ from (a). The second is thm-coupling-certificate (a)(iii), since an upper band below one on $\epsilon_{\mathsf{L}}(0) = \mathrm{TV}(P_0, P_1)$ is a certificate of the total variation (proven).

Label. Proven, (a) to (c).

Reading. For one deployed pipeline the run axis of a profile carries information only for a declared family, which is what an experiment measures, or uniformly over a class of learning functions, which is the shadow-run price of the space statistical-limits, or for a computationally bounded class, which is the space computational-limits. The object proposed in FORMAL_SECURITY's terminology ledger, the curve from a budget to the success, is well defined as a measured object only with its family named. A finite experiment certifies a lower band for that family, with fresh shadow runs in every trial, and an upper band for the whole class only from coupled runs.

Remark, paired trials. Under any coupling of the two executions of a trial whose marginals are right, the mean of D is the advantage and $\mathrm{Var}(D) = p_0(1-p_0) + p_1(1-p_1) - 2\,\mathrm{Cov}(T_0, T_1)$, with T_b the output under H_b and p_b its mean. The variance is at most the discordance $\Pr[T_0 \neq T_1]$. A shared seed lowers the variance against independent executions exactly when it makes $\mathrm{Cov}(T_0, T_1) > 0$, and astra's complement example, $T_1 = 1 - T_0$ with $p_0 = p_1 = 1/2$, raises it from one half to one. The measured covariance therefore reports what a shared seed gives the measurement. With a positive covariance, an empirical Bernstein interval turns the smaller variance into a smaller trial count, which is common random numbers.

Without the oracle, or in the other regime. Without the oracle a declared family still has its band from real shadow runs. LiRA's evaluations use one bank for all targets (fact-lira-priced-in-training-runs), so what they report is the advantage conditional on that bank. In the computational regime the family's budget is its shadow work, and part (a) does not apply, since a bounded adversary cannot hard-wire an unbounded table.

B.48 certifying leakage at a zero-leakage function costs exact twin runs

`thm-exact-twin-price`, from `lab/theory/spaces/oracle-and-measurement/round-02/builder.md`.

status: sketch uses: thm-dataflow-composition, lem-needle-lower-bound, thm-coupling-free-soundness, def-seed-access-modes

Setting. As in thm-dataflow-composition (e)(ii), one release $y = W + Z$ with $Z \sim \mathcal{N}(0, \sigma^2 I_d)$, a fixed choice c, $\mathcal{M} = \mathbb{R}^d$ and $x_b = x(c, D_j^b)$, with $x_0 \neq x_1$. Three targets are considered. The coupled tail $\theta_{\mathrm{ex}} = \epsilon_0(\Delta)$ for a fixed $\Delta > 0$, the total variation $\theta_{\mathrm{tv}} = \mathrm{TV}(P_0, P_1)$ of the release, and its one-sided hockey-stick divergence $\theta_{\mathrm{hs}} = \mathrm{H}_{e^\varepsilon}(P_1 \| P_0)$ for a fixed $\varepsilon \geq 0$. Let L° have deterministic kernels and $\mathsf{L}^\circ(x_0; \rho) = \mathsf{L}^\circ(x_1; \rho)$ for every ρ, so all three vanish there. Fix $\epsilon_{\mathrm{acc}} \in (0,1)$, $\alpha > 0$ and $\alpha' \geq 0$ with $\alpha + \alpha' < 1$. A certifier is *accurate at* L° when $\Pr_{\mathsf{L}^\circ}[\hat{t} \leq \epsilon_{\mathrm{acc}}] \geq 1 - \alpha'$. Let N_b count its Train calls at x_b, $c_{\mathrm{acc}} = -\ln(1 - \epsilon_{\mathrm{acc}})$ and $\Lambda_{\alpha,\alpha'} = (1-\alpha')\ln((1-\alpha')/\alpha)$.

Statement. (a) One pair. Let a certifier, in either mode and passing seeds by value or not, be sound at level α over every learning function with deterministic kernels and accurate at L°. For each of the three targets, $\mathbb{E}_{\mathsf{L}^\circ}[N_1] \geq \Lambda_{\alpha,\alpha'}/c_{\mathrm{acc}}$. For θ_{ex} and θ_{tv}, also $\mathbb{E}_{\mathsf{L}^\circ}[N_0] \geq \Lambda_{\alpha,\alpha'}/c_{\mathrm{acc}}$. Calls at every other training input, which include fine-tunes from a trained model, warm starts and approximate-unlearning steps, do not enter these bounds. For θ_{ex} some certifier sound over every learning function is accurate at L° with $\mathbb{E}[N_0] = \mathbb{E}[N_1] = (1-\alpha')\lceil \ln((1-\alpha')/\alpha)/c_{\mathrm{acc}} \rceil$. For θ_{tv} and θ_{hs} the same holds with ϵ_{acc} replaced by $\epsilon_{\mathrm{acc}} - \eta$ in c_{acc}, for every $\eta \in (0, \epsilon_{\mathrm{acc}})$.

(b) Every record. Let D hold n_D distinct records $z_1, \ldots, z_{n_D}$ and let $D' \mapsto x(c, D')$ be injective on $\{D\} \cup \{D \setminus \{z_i\}\}_i$. Put $x^{(0)} = x(c, D)$ and $x^{(i)} = x(c, D \setminus \{z_i\})$, record i having $D_j^0 = D$ and $D_j^1 = D \setminus \{z_i\}$, take as target the maximum over i

of the target of record i, and let L° return one model at all $n_D + 1$ inputs for each seed. Under the hypotheses of (a), for each of the three targets the calls at $x^{(1)}, \ldots, x^{(n_D)}$ together have expectation at least $n_D \Lambda_{\alpha,\alpha'}/c_{\mathrm{acc}}$, and for θ_{ex} and θ_{tv} the calls at $x^{(0)}$ have expectation at least $\Lambda_{\alpha,\alpha'}/c_{\mathrm{acc}}$. With $N = \lceil \ln(1/\alpha)/c_{\mathrm{acc}} \rceil$ common seeds, one base run and n_D twin runs at each give a certificate of θ_{ex} sound at level α and accurate at L° with $\alpha' = 0$, at $N(n_D + 1)$ calls, and the same holds for θ_{tv} and θ_{hs} with $\epsilon_{\mathrm{acc}} - \eta$ in place of ϵ_{acc}.

(c) The base input for the one-sided target. Some certifier of θ_{hs} at $\varepsilon = 1$, sound at level 0.05 over every learning function and accurate at L° with $\epsilon_{\mathrm{acc}} = 0.01$ and $\alpha' = 0$, makes four calls at x_0, while $\Lambda_{0.05,0}/c_{0.01}$ is about 298. So no bound at the base input of the kind in (a) holds for θ_{hs}.

Proof of (a), lower bound at x_1 for θ_{ex}. The construction is argon's. By (F9) identify $(\mathcal{R}, \mu)$, outside a μ-null set $\mathcal{N}_0$, with $[0,1)$ and Lebesgue measure. Fix $p \in (\epsilon_{\mathrm{acc}}, 1)$, an integer M with $m = \lceil pM \rceil < M$, the cells $I_r = [(r-1)/M, r/M)$, and a unit vector $\mathbf{e}$. For an m-subset B of $\{1, \ldots, M\}$ let $S_B = \bigcup_{r \in B} I_r$, viewed in $\mathcal{R} \setminus \mathcal{N}_0$, and let L_B agree with L° except that $\mathsf{L}_B(x_1; \rho) = \mathsf{L}^\circ(x_1; \rho) + 2\Delta\mathbf{e}\,\mathbf{1}[\rho \in S_B]$. At seeds in $\mathcal{N}_0$, which a caller may pass by value, L_B equals L°. It has deterministic kernels, and the twin of x_0 at seed ρ differs from the node's model by $2\Delta\mathbf{1}[\rho \in S_B]$, so $\theta_{\mathrm{ex}}(\mathsf{L}_B) = m/M \geq p > \epsilon_{\mathrm{acc}}$.

Couple the certifier's runs against L° and L_B with common coins and a common triple per call, as in the proof of lem-needle-lower-bound. A call at an input other than x_1, a call at x_1 whose seed lies outside S_B, and every Eval and Open call on what these return get the same answer in both runs. A seed is drawn, named or passed by value, and in each case it is a function of the history and of fresh draws, and the run against L° does not depend on B. So the runs agree, and output the same $\hat{t}$, unless one of the at most N_1 cells that the seeds of the L° run's calls at x_1 visit lies in B. Let $E = \{\hat{t} \leq \epsilon_{\mathrm{acc}}\}$ in the L° run, with probability $q \geq 1 - \alpha'$. Soundness at L_B and $\epsilon_{\mathrm{acc}} < m/M$ give $\Pr_{\mathsf{L}_B}[\hat{t} < m/M] \leq \alpha$. For B uniform, averaging gives

$$\alpha \geq \mathbb{E}_{\mathsf{L}^\circ}\Big[\mathbf{1}_E\, \mathbf{1}[N_1 < M - m] \prod_{j=0}^{N_1 - 1} \Big(1 - \frac{m}{M - j}\Big)\Big],$$

since the probability that B avoids $K \leq N_1$ given cells is $\binom{M-K}{m}/\binom{M}{m}$, at least the product. If $\mathbb{E}_{\mathsf{L}^\circ}[N_1] = \infty$ there is nothing to prove, so $N_1 < \infty$ almost surely. The law of the L° run does not depend on M, so dominated convergence as $M \to \infty$, with $m/M \to p$, gives $\mathbb{E}[\mathbf{1}_E (1-p)^{N_1}] \leq \alpha$. Jensen's inequality for the convex map $u \mapsto (1-p)^u$ under the law given E gives $(1-p)^{\mathbb{E}[N_1 | E]} \leq \alpha/q$, so $\mathbb{E}[N_1] \geq q \ln(q/\alpha)/(-\ln(1-p))$. The map $q \mapsto q \ln(q/\alpha)$ increases on $q > \alpha/e$, and $q \geq 1 - \alpha' > \alpha$, so $\mathbb{E}[N_1] \geq \Lambda_{\alpha,\alpha'}/(-\ln(1-p))$. Letting $p \downarrow \epsilon_{\mathrm{acc}}$ gives the bound.

Proof of (a), the other lower bounds. For θ_{ex} at x_0, perturb x_0 instead, $\mathsf{L}_B(x_0; \rho) = \mathsf{L}^\circ(x_0; \rho) + 2\Delta\mathbf{e}\,\mathbf{1}[\rho \in S_B]$, which moves the node's model by the same amount, so $\theta_{\mathrm{ex}}(\mathsf{L}_B) = m/M$ and the proof is unchanged. For θ_{tv} and θ_{hs} at x_1, use the shift $s\mathbf{e}$ in place of $2\Delta\mathbf{e}$. Let Q_B be the law of $\mathsf{L}^\circ(x_0; \rho) + Z$ given $\rho \in S_B$. The laws P_0 and P_1 agree on the part carried by seeds outside S_B, so $\mathrm{TV}(P_0, P_1) = \mu(S_B)\, \mathrm{TV}(Q_B, Q_B(\cdot - s\mathbf{e}))$, and a probability law on $\mathbb{R}^d$ and its translate by $s\mathbf{e}$ have total variation tending to one as $s \to \infty$. For θ_{hs}, the half-space $\{y : \langle y, \mathbf{e} \rangle > s/2\}$ has P_1-probability tending to $\mu(S_B)$ and P_0-probability tending to zero as $s \to \infty$. There are finitely many B for each M, so some s_M makes both targets at least $\frac{1}{2}(1 + \epsilon_{\mathrm{acc}}/p)\, m/M \geq \frac{1}{2}(p + \epsilon_{\mathrm{acc}}) > \epsilon_{\mathrm{acc}}$ for every B, and the rest of the proof is unchanged. For θ_{tv} at x_0, perturb x_0 by $s\mathbf{e}$, and total variation is symmetric. This step fails for θ_{hs}, since perturbing the denominator law P_0 adds mass that the factor e^{ε} can cover, which is what (c) exploits.

Proof of the upper bounds in (a). For θ_{ex}, with probability α' output one at no call. Otherwise run the certificate of thm-dataflow-composition (c) at level $\alpha/(1-\alpha')$ with $n = \lceil \ln((1-\alpha')/\alpha)/c_{\mathrm{acc}} \rceil$ coupled pairs. It fails to cover with probability at most $(1-\alpha') \cdot \alpha/(1-\alpha') = \alpha$, since an output of one always covers. At L° every pair coincides, no run exceeds Δ, and the bound $1-(\alpha/(1-\alpha'))^{1/n}$ is at most ϵ_{acc} by the choice of n. For θ_{tv} and θ_{hs}, thm-dataflow-composition (b) with $R=1$ gives $\mathrm{TV}(P_0,P_1) \le \delta_{\mathcal{N}}(0;\Delta/\sigma) + \epsilon_0(\Delta)$ and $\mathrm{H}_{e^\varepsilon}(P_1 \| P_0) \le \delta_{\mathcal{N}}(\varepsilon;\Delta/\sigma) + \epsilon_1(\Delta)$. At $R=1$ with a fixed choice, the identity-coupled pairs of Ex_0 and of Ex_1 have one law up to the order of their coordinates, so $\epsilon_1(\Delta) = \epsilon_0(\Delta)$ and one set of pairs serves both. Choose Δ with $\delta_{\mathcal{N}}(\varepsilon;\Delta/\sigma) \le \eta$, possible since $\delta_{\mathcal{N}}(\varepsilon;r) \to 0$ as $r \to 0$, and output $\min\{1, \delta_{\mathcal{N}}(\varepsilon;\Delta/\sigma) + \bar{p}\}$, with $\varepsilon = 0$ for total variation, with $\bar{p}$ the bound above at accuracy $\epsilon_{\mathrm{acc}} - \eta$ (proven).

Proof of (b). Use the family $\mathsf{L}_{k,B}$ that perturbs only $x^{(k)}$ on S_B, with k uniform on $\{1,\dots,n_D\}$ and independent of B. Its target is that of record k, at least m/M for θ_{ex} and above ϵ_{acc} for the other two after the shift of (a). With $N^{(k)}$ the calls at $x^{(k)}$ in the L° run, the argument of (a) gives $\alpha \ge \mathbb{E}[\mathbf{1}_E \frac{1}{n_D}\sum_k (1-p)^{N^{(k)}}] \ge \mathbb{E}[\mathbf{1}_E (1-p)^{N_{\mathrm{tw}}/n_D}]$ in the limit, by convexity of $u \mapsto (1-p)^u$, with $N_{\mathrm{tw}} = \sum_k N^{(k)}$, and the rest of (a) gives the first bound. For θ_{ex} and θ_{tv}, perturbing $x^{(0)}$ moves every record's pair at once and gives the bound at $x^{(0)}$. For the upper bound, let $\bar{p}_k$ be the level-α Clopper and Pearson bound for record k from the N common seeds, and output $\max_k \bar{p}_k$ for θ_{ex} and $\min\{1, \delta_{\mathcal{N}}(\varepsilon;\Delta/\sigma) + \max_k \bar{p}_k\}$ for the other two. For a fixed $k^\star$ attaining the maximal target, the output undercovers only if $\bar{p}_{k^\star}$ undercovers the exceedance probability of record $k^\star$, which has probability at most α, since that record's indicators are independent across seeds. At L° no pair exceeds, and the output is at most $1-\alpha^{1/N}$ plus the allowance (proven, the certificate without a union bound is argon's).

Proof of (c). Astra's certifier, re-checked here. Let L° return the zero model at both inputs. The certifier draws and opens four models at x_0 and 368 at x_1, and outputs 0.01 if all are zero and one otherwise. Let a and b be the probabilities that x_0 and x_1 return exactly the zero model. If $a \ge 1/e$ and $b \ge 0.99$, then $P_1(E) - eP_0(E) \le b\mathcal{N}(0,\sigma^2 I_d)(E) - ea\mathcal{N}(0,\sigma^2 I_d)(E) + (1-b) \le 0.01$ for every event E, so $\theta_{\mathrm{hs}} \le 0.01$ and every output covers. If $a < 1/e$, four zeros at x_0 have probability below $e^{-4} < 0.0184$. If $b < 0.99$, 368 zeros at x_1 have probability below $0.99^{368} < 0.0248$. So whenever $\theta_{\mathrm{hs}} > 0.01$ the output 0.01 has probability below 0.05. At L° the output is 0.01 always, with four calls at x_0, while $\ln(1/0.05)/(-\ln 0.99) = 298.1$ (computed).

Label. Proven, (a) to (c).

Numbers. At $\alpha = \alpha' = 10^{-3}$ and $\epsilon_{\mathrm{acc}} = 10^{-2}$, (a) gives at least 686.5 expected calls at x_1, and at x_0 for the coupled tail and total variation, and 688 pairs suffice for the coupled tail. For 60,000 distinct records, (b) gives at least 4.12×10^7 expected twin calls, and $688 \times 60{,}001 \approx 4.13 \times 10^7$ calls suffice (computed). Each count multiplies the cost of one run, by the remark of def-seed-access-modes. thm-record-level-certificate certifies the fraction of records whose tail exceeds a threshold at a cost free of n_D, so the price of the every-record certificate is the price of its worst-case target.

Realization (heuristic). With the seed fixed, a Train call is one run of the pipeline. For θ_{tv} and θ_{hs}, thm-coupling-free-soundness makes the certificate sound under any implementable coupling with the right marginals, so realization needs each run to have the deployed law at its input and repetitions to be independent, which on GPUs is the fidelity assumption of def-learning-oracle. Accuracy at a zero-leakage function needs the pairs to coincide, which needs deterministic kernels. For θ_{ex} the certificate needs the identity coupling itself. fact-controlled-nondeterminism-exact-replication realizes deterministic kernels on one GPU type, and across types at 1.2 to 1.7 times the training time, for a pipeline modified in precision, rounding and dropout. The certified pipeline is then that modified one, and it must be the deployed one. fact-rerun-reproduction-error

shows that whole uncontrolled GPU reruns do not coincide.

Remark, scope. The bound holds because soundness ranges over every learning function, so the model at x_1 is unlinked from the model at any surrogate input. A class restricted by a known stability bound on $\|\mathsf{L}(x_1;\rho) - \mathsf{L}(x';\rho)\|$ for a surrogate x' would let surrogate calls count (argon). XD23 certifies a different target, a mutual-information bound averaged over a data law, from drawn runs of the mechanism, and this theorem says nothing about it. The proof is a rare-event change of measure. Kaufmann, Cappé and Garivier (JMLR 2016, Lemma 1), as astra points out, relate expected sample counts to discrimination under mutual absolute continuity, which the singular construction here does not satisfy, so the finite-partition argument is needed for adaptive seeds passed by value.

Without the oracle, or in the other regime. Without named seeds, thm-dataflow-composition (e) prices the twin certificate above the coupled one in the drawn mode. With the oracle, (a) says the oracle supplies no twin for free. Each twin is one call, a surrogate is a call at another input and counts for nothing, and what the oracle supplies is the named seed, which real training also offers. The statement is statistical. By the remark of def-seed-access-modes it becomes a bound in oracle cost, and a bound in compute under the no-skipping hypothesis.

Appendix C

The learning-oracle theory in full

The final document of the learning-oracle rounds of 16 and 17 October 2026, converted at build time from `sink/2026-10-16-formal-tuning/final/learning-oracle-theory.md`. Its slugs match the objects of the chapter on the learning oracle.

C.1 Definitions and prominent results of the learning oracle

C.1.1 What the theory shows

A privacy or security proof for a procedure that trains a model must say how the trained model depends on the data. For a transformer trained by stochastic gradient descent nobody can write that law down, and one sample of it costs one training run. Proofs about procedures with transformer-sized learning nodes therefore assume a mechanism with a known law, such as clipping and noise inside training.

The theory replaces each training run by a call, Train, to an idealized training oracle. It runs an arbitrary learning function on a training input and a seed, and names the seed. A caller can rerun training on another input with the same seed, which samples a coupling of two runs. Every bound below depends on the learning function only through quantities that such calls define. So each bound holds for every learning function, transformers included. Train calls certify those quantities where a result says so.

Two degenerate oracles test whether a result uses the oracle. The bottom oracle gives an adversary what a real training service gives, and the top oracle also hands it the honest seeds. For an experimenter or an auditor the two cases are seeds that the oracle draws and hides, and seeds that the caller passes. A result counts as non-trivial when four conditions hold. The oracle is in it, and removing the oracle leaves a problem that needs the law of the trained model. Its bound depends on the learning function only through oracle-measured quantities, and it fails, or becomes vacuous, under one degenerate case.

The main results follow. Each claim is labelled proven, with a proof here or in a round file, heuristic, with its assumption, or open. Here R is a number of rounds, σ a noise scale, d the model dimension, B_A a bound on a coalition's deviation and α the error probability of a certifier.

1. Coupling certificate. One training node leaks at most the coupled divergence. Coupled Train calls certify it for every learning function when the divergence of the channel at two models is computable. Through a deterministic channel with atomless view laws, drawn seeds certify nothing below one for the total variation (proven).

2. Black-box certifiability. Label-only leakage at a fixed number of queries, on a finite query set with finitely many outputs, is certifiable from both sides. With deterministic evaluation and atomless model laws, drawn seeds never certify that leakage below one equals white-box leakage (proven).
3. Dataflow composition. If client data reach a coalition only via R Gaussian releases, as def-isolated-client makes precise, leakage is at most a Gaussian term plus the exceedance probability. That is the probability that coupled updates differ by more than a threshold. Coupled runs of one strategy certify it, with a number of runs free of d (proven).
4. Bounded-deviation transfer. Over the same releases, let a coalition follow the protocol except for additive inputs within B_A of the prescribed ones. Its exceedance probability is then at most a fixed function of the protocol's, set by $\mathfrak{e} = (R-1)B_A^2/(2\sigma^2)$. Coupled runs of the protocol certify this bound at level α, free of d, and $e^{(1+o(1))\mathfrak{e}} \ln(1/\alpha)$ runs reach a fixed accuracy when none exceeds the threshold. In the clipped aggregation instance, let a certifier sound for every learning function be accurate where the leakage is zero, below a fixed fraction of one clipped release's leakage. At fixed error probabilities summing below one, it needs $e^{(1-o(1))\mathfrak{e}}$ Train calls when $\mathfrak{e}/d \to 0$ (proven).
5. Replay-audit steering. Let seeds come from a random oracle and the dataset be committed. A prover then passes a replay audit with a model in a target set with probability at most a multiple of the target's reach. Under deterministic kernels, grinding with at least one random-oracle query attains this bound within a factor 3.2 if the reach times the query count is at most one (proven).
6. Needle lower bound. A certifier sound for every learning function that makes N Train calls excludes leakage placed in one of J disjoint places with probability at most $\alpha + N/J$ (proven).

C.1.2 Definitions

Every space below is a standard Borel space. A remark after a definition is not part of it. `theory/CURRENT.md` adds the clauses that only the results stated with pointers use.

Learning oracle [def-learning-oracle]

Fix the data space $\mathcal{Z}$, the model space $\mathcal{M}$, the query space $\mathcal{U}$, the output space $\mathcal{Y}$, the set Λ of learner names and the space $\mathcal{X}$ of training inputs. A *dataset* is a finite multiset over $\mathcal{Z}$. A *training input* $x = (\ell, m, D, \mathcal{H})$ holds a learner name ℓ, a starting model m, a dataset D and a hypothesis class $\mathcal{H} \subseteq \mathcal{M}$ from a finite family. The *learning function* is a measurable map $\mathsf{L} : \mathcal{X} \times \mathcal{R} \times \Xi \to \mathcal{M}$, written $\mathsf{L}(x; \rho, \xi)$. The *seed* ρ has an atomless law μ on $\mathcal{R}$, and the *hardware randomness* ξ has a law π on Ξ. Π_x is the law of $\mathsf{L}(x; \rho, \xi)$ under $\mu \otimes \pi$. Training has *deterministic kernels* when Ξ is a point. The *evaluation map* $\mathsf{ev}(m, u; \upsilon) \in \mathcal{Y}$, with υ of law ν, is public.

A *learning oracle* $\mathcal{O}$ for (L, μ, π) stores models and seeds and names them per caller. The k-th name a caller receives is the integer k, and each name carries a label, open or eval. $\mathcal{O}$ answers three calls.

1. $\mathsf{Train}(x, s)$. The starting model in x is a model or a model name, the dataset is given by value, and s is a seed, a seed name or $\perp$. On $\perp$ the oracle draws $\rho \sim \mu$ and stores it under a new name. Otherwise the run uses the seed that s gives. The oracle draws $\xi \sim \pi$, stores $\mathsf{L}(x; \rho, \xi)$ under a new name and returns the new names.
2. $\mathsf{Eval}(h, u)$, for a model name h, returns $\mathsf{ev}(m, u; \upsilon)$ for the stored model m and a fresh $\upsilon \sim \nu$.

3. $\mathsf{Open}(h)$ returns the stored object.

A *grant* gives a caller a new name of a stored object, with a label, when a procedure or a policy directs it. A grant is not a call. Fidelity. Every seed drawn on $\perp$, every ξ and every υ is independent of all other randomness. Access. Train and Open require label open on every name among their arguments, and Eval requires label eval or open. The oracle refuses any other call. A caller holds label open on every name that its own call returns, and it holds a name only in this way or through a grant.

Remark. Label eval gives black-box access to a model, and label open white-box access. Fidelity of hardware randomness is a modelling assumption, heuristic for GPU training.

Learning-oracle model [def-learning-oracle-model]

A *procedure* is a dataflow graph G whose nodes run in a fixed topological order. Each node has an owner and one of four operations. An input node holds a dataset, a seed or a public object, and a sampler of fresh records is one. A learning node calls $\mathsf{Train}(x_v, s_v)$, with x_v formed from parent values and choices, and $s_v = \perp$ unless the procedure feeds the node a seed node. An evaluation node calls Eval. A general node computes a randomized function of its parents with randomness of its own, which the oracle does not hold. In $G^{\mathcal{O}}$ the calls go to $\mathcal{O}$, and the procedure grants each honest owner label open on the objects its node uses. In the real procedure G the nodes compute L and ev directly with fresh randomness.

Each node value reaches a *coalition* A through one of three channels. Through the plain channel A receives the value. Through the query channel the procedure grants A a name of the value with label eval, and it fixes a query set $U \subseteq \mathcal{U}$ and an allowance Q. The oracle then answers at most Q Eval calls of A on that name, each at a query in U. Through channel none A receives nothing.

The coalition acts through an *adversary* $\mathcal{A}$, an oracle algorithm that reaches L only through $\mathcal{O}$, computes ev itself and stops with probability one. It is unbounded, except in a statement with a computational assumption, which bounds its running time. Its *choices* are the node arguments that A sets, the values and learner names of nodes that A owns, and its queries. They also include the learner names of nodes whose training code A supplies. A strategy is *passive* when the choices that enter procedure nodes are the functions of procedure values that the protocol prescribes, and *active* otherwise.

For each node in order, the adversary receives the channel outputs of the nodes run so far, makes oracle calls and sets its choices for the node. Then the node runs. After the last node it makes further calls and stops. Its *view* is its transcript, made of the channel outputs, the answers to its calls and its coins. The *budget* (q_T, q_E) bounds its Train calls and its Eval calls on names with label eval. Calls that the procedure requires of an owner in A are not charged, and an Eval call on a name with label open is free.

To *distinguish*, the adversary faces hypotheses H_0, H_1 on the inputs and outputs a bit, with advantage $\Pr[1 \mid H_1] - \Pr[1 \mid H_0]$. A privacy statement bounds the hockey-stick divergence $\mathrm{H}_\gamma(P_0 \| P_1) = \sup_E (P_0(E) - \gamma P_1(E))$, for $\gamma = e^\varepsilon \geq 1$, over measurable sets E of views. Here P_b is the view's law under H_b. At $\gamma = 1$ the divergence is the total variation $\mathrm{TV}(P_0, P_1)$. To *steer*, the adversary faces a node, a target set $T \subseteq \mathcal{M}$ and an acceptance event, and succeeds when the node's value is in T and the event holds.

Remarks. A sum, a ciphertext or a noisy value is a general node released through the plain channel. To invert, an adversary wins on an event E of views, and $P_0(E) \leq e^\varepsilon P_1(E) + \mathrm{H}_{e^\varepsilon}(P_0 \| P_1)$ by the definition of H_γ (proven). So every privacy bound below also bounds inversion.

Certifier [def-certifier]

An *experimenter* is a caller outside every procedure that reaches L only through $\mathcal{O}$. A caller works in the *drawn-seed discipline* when every Train call it makes has $s = \perp$ and it opens no seed name, and in the *coupled discipline* otherwise. A *target* is a functional $\theta(\mathsf{L}) \in [0,1]$. A *certifier* is an experimenter that makes at most N Train calls and outputs $\hat{t} \in [0,1]$. It is *sound* at level α over a class $\mathfrak{L}$ of learning functions when $\Pr_{\mathsf{L}}[\theta(\mathsf{L}) \le \hat{t}] \ge 1 - \alpha$ for every $\mathsf{L} \in \mathfrak{L}$. A lower certifier outputs $\underline{t}$ with $\Pr_{\mathsf{L}}[\underline{t} \le \theta(\mathsf{L})] \ge 1 - \alpha$.

Remarks. The main targets have the channel form. For training inputs $x_0 \neq x_1$ and a channel K, a Markov kernel from $\mathcal{M}$ to a view space $\mathcal{V}$, the view laws are $P_b = \int K(\cdot \mid m)\, \Pi_{x_b}(dm)$, and θ is $\mathrm{TV}(P_0, P_1)$ or $\mathrm{H}_\gamma(P_0 \| P_1)$. A deterministic channel is a measurable map $\kappa : \mathcal{M} \to \mathcal{V}$. A certifier sound for a target $\theta' \ge \theta$ is sound for θ. So each lower bound below binds certifiers of every larger target (proven).

Degenerate oracles [def-degenerate-oracles]

Fix a procedure G and a coalition A. A *policy* directs grants to A beyond those of the procedure.

1. The *bottom oracle* is the realizable policy, which grants nothing, with no bound on q_T.
2. The *top oracle* is the seed-revealing policy, which grants A label open on the seed name of every honest learning node when the node runs.
3. For an experimenter, or an honest party whose Train calls a theorem designs such as an auditor, the bottom and the top case are the drawn-seed and the coupled discipline.

A statement about adversaries is tested under the bottom and the top oracle, and a statement about an experimenter or an auditor under the two disciplines.

Remark. The bottom oracle is the real procedure against an adversary that samples L (thm-perfect-substitution). An adversary can hard-wire one fixed learning function, so knowing L matters only when a statement chooses the adversary first. Drawn seeds match a training service that hides its randomness.

Isolated client [def-isolated-client]

Fix a client j, candidate datasets $D_j^0 \neq D_j^1$, a space $\mathcal{C}$ of choices, a number $R \ge 1$ of rounds, a dimension d and a noise scale $\sigma > 0$. Fix also an initial state $w_0 \in \mathcal{M}$, a measurable map $(c, D, w) \mapsto x(c, D, w) \in \mathcal{X}$ and measurable release maps $f_t : \mathcal{C} \times \mathcal{M} \to \mathbb{R}^d$. A choice holds the starting model and everything else that the node's input takes from outside the client, the learner name of the round included. Let G be a procedure in which client j owns the input node of D_j. Let its nodes include, for $t \le R$, a learning node v_t owned by j and a general node r_t, with $v_1 \prec r_1 \prec \cdots \prec v_R \prec r_R$. Then G *isolates client j behind Gaussian releases* when the following hold.

1. v_t calls $\mathsf{Train}(x(c_t, D_j, W_{t-1}), \perp)$, where W_t is the value of v_t and $W_0 = w_0$. Here c_t is the value of a parent of v_t other than v_{t-1} and the input node of D_j, or a choice of the coalition.
2. No node other than $v_1, \ldots, v_R$ reads D_j, and the children of v_t are among r_t and v_{t+1}.
3. r_t outputs the release $y_t = O_t + f_t(c_t, W_t) + Z_t$, where the noise $Z_t \sim \mathcal{N}(0, \sigma^2 I_d)$ is randomness of r_t. The contribution O_t is a measurable function of the other parents of r_t and of randomness of r_t independent of Z_t.

4. The coalition owns no v_t or r_t, and the input node of D_j and every v_t reach it through channel none.

The client is stateless when $x(c, D, w)$ does not depend on w. The procedure may stop after a release, on a value the coalition sets, and then no later node runs. Under H_b, $D_j = D_j^b$ and every other input is fixed, the same under both.

The *coupled execution* Ex_b runs G under H_b against a given strategy. At each v_t that runs it also computes the twin $W'_t = \mathsf{L}(x(c_t, D_j^{1-b}, W'_{t-1}); \rho_t, \xi'_t)$, with $W'_0 = w_0$, the seed ρ_t of v_t and fresh ξ'_t. No node reads a twin. With the maximum over the rounds that run, the *exceedance* and the *exceedance probability* at a threshold $\Delta > 0$ are $D_b^\star = \max_t \|f_t(c_t, W'_t) - f_t(c_t, W_t)\|$ and $\epsilon_b(\Delta) = \Pr[D_b^\star > \Delta]$.

For states w^0, w^1 the *coupled tail* is $\beta_t(c, w^0, w^1, \Delta) = \Pr[\|f_t(c, \mathsf{L}(x(c, D_j^1, w^1); \rho, \xi_1)) - f_t(c, \mathsf{L}(x(c, D_j^0, w^0); \rho, \xi_0))\| > \Delta]$, with $\rho \sim \mu$ and $\xi_0, \xi_1 \sim \pi$ independent. For a stateless client it is $\beta_t(c, \Delta)$.

Remark. An experimenter that holds every input and runs the strategy samples Ex_b. Per round it makes two Train calls for client j, the second with the first's seed name, plus those of the other nodes and of the strategy. A passive strategy also takes the protocol's stop. Its choices fix the law of $D_b^\star$, so every passive adversary has the same ϵ_b, and the experimenter need not make the strategy's own calls.

C.1.3 Standing facts

The proofs use the following facts. Each is standard, and a source or a one-line proof is given.

- (F1) Test fact. The best advantage of a test between laws P_0 and P_1 is $\mathrm{TV}(P_0, P_1)$, attained on a set where the density of P_1 exceeds that of P_0 (Neyman and Pearson).
- (F2) Data processing. For a Markov kernel K and $\gamma \geq 1$, $\mathrm{H}_\gamma(KP_0 \| KP_1) \leq \mathrm{H}_\gamma(P_0 \| P_1)$. A measurable function with values in $[0, 1]$ has integral against $P_0 - \gamma P_1$ at most $\mathrm{H}_\gamma(P_0 \| P_1)$.
- (F3) Splits. For laws $\mathcal{L}, \mathcal{L}', \mathcal{L}''$ on one space, $\mathrm{H}_\gamma(\mathcal{L} \| \mathcal{L}'') \leq \mathrm{TV}(\mathcal{L}, \mathcal{L}') + \mathrm{H}_\gamma(\mathcal{L}' \| \mathcal{L}'')$ and $\mathrm{H}_\gamma(\mathcal{L}'' \| \mathcal{L}) \leq \mathrm{H}_\gamma(\mathcal{L}'' \| \mathcal{L}') + \gamma\, \mathrm{TV}(\mathcal{L}', \mathcal{L})$. Add and subtract $\mathcal{L}'(E)$ inside the supremum.
- (F4) Mixtures. For a law Θ of a pair $(\mathfrak{w}_0, \mathfrak{w}_1)$ and kernels K_0, K_1 to $\mathcal{V}$, $\mathrm{H}_\gamma(\Theta K_0 \| \Theta K_1) \leq \int \mathrm{H}_\gamma(K_0(\cdot \mid \mathfrak{w}_0) \| K_1(\cdot \mid \mathfrak{w}_1))\, \Theta(d\mathfrak{w})$. Integrate $K_0(E \mid \mathfrak{w}_0) - \gamma K_1(E \mid \mathfrak{w}_1)$ against Θ.
- (F5) Averaging. For a law $\mathbb{P}_\mathfrak{w}$ of $\mathfrak{w}$ and kernels $\mathfrak{w} \mapsto P_0^\mathfrak{w}, P_1^\mathfrak{w}$, the laws $\mathbb{P}_\mathfrak{w}(d\mathfrak{w}) P_b^\mathfrak{w}(dy)$ have divergence $\int \mathrm{H}_\gamma(P_0^\mathfrak{w} \| P_1^\mathfrak{w})\, \mathbb{P}_\mathfrak{w}(d\mathfrak{w})$. Sections give the upper bound, and densities with Dynkin's theorem the lower one (round-01/builder.md, coupling certificate, step 3 of (b)).
- (F6) Gaussian facts. Φ, $\bar{\Phi} = 1 - \Phi$ and φ are the standard normal distribution function, tail and density, and φ_σ is the density of $\mathcal{N}(0, \sigma^2 I_d)$. (i) $\mathrm{H}_{e^\varepsilon}(\mathcal{N}(\mathbf{a}_0, \sigma^2 I_d) \| \mathcal{N}(\mathbf{a}_1, \sigma^2 I_d)) = \delta_\mathcal{N}(\varepsilon; \|\mathbf{a}_0 - \mathbf{a}_1\|/\sigma)$, where $\delta_\mathcal{N}(\varepsilon; r) = \mathrm{H}_{e^\varepsilon}(\mathcal{N}(0, 1) \| \mathcal{N}(r, 1))$ is nondecreasing in the shift $r \geq 0$ and, for $r > 0$, equals $\Phi(r/2 - \varepsilon/r) - e^\varepsilon \Phi(-r/2 - \varepsilon/r)$ (Balle and Wang, ICML 2018, from memory). (ii) For $\mathfrak{f} \in [0, 1]$ and $r > 0$, the kernel $\mathfrak{z} \mapsto \mathfrak{f}(\mathfrak{z} - r) + \sqrt{1 - \mathfrak{f}^2}\, Z'$ with $Z' \sim \mathcal{N}(0, 1)$ maps $\mathcal{N}(r, 1)$ to $\mathcal{N}(0, 1)$ and $\mathcal{N}(0, 1)$ to $\mathcal{N}(-\mathfrak{f}r, 1)$. (iii) $\bar{\Phi}(r) \leq \frac{1}{2} e^{-r^2/2}$ for $r \geq 0$. (iv) $\frac{r}{1+r^2} \varphi(r) \leq \bar{\Phi}(r) \leq \varphi(r)/r$ for $r > 0$. (v) $\mathbb{E} e^{\langle Z, \mathbf{a} \rangle} = e^{\sigma^2 \|\mathbf{a}\|^2/2}$ for $Z \sim \mathcal{N}(0, \sigma^2 I_d)$.
- (F7) Change of measure. For laws $\mathbb{P}' \ll \mathbb{P}$ and $\varpi > 1$, Hölder's inequality gives $\mathbb{P}'(\cdot) \leq \mathbb{P}(\cdot)^{1 - 1/\varpi} (\int (d\mathbb{P}'/d\mathbb{P})^\varpi d\mathbb{P})^{1/\varpi}$ on every measurable set. By (F6)(v), $\int (\varphi_\sigma(y - \mathbf{a})/\varphi_\sigma(y))^\varpi \varphi_\sigma(y)\, dy = \exp(\varpi(\varpi - 1)\|\mathbf{a}\|^2/(2\sigma^2))$, the Gaussian Rényi moment.

- (F8) Statistics. The one-sided Clopper and Pearson upper bound at level α covers a binomial parameter with probability at least $1-\alpha$ at every value. With no success in n trials it equals $1-\alpha^{1/n} \le \ln(1/\alpha)/n$. Hoeffding's inequality bounds the deviation of a mean of n independent variables in $[0,1]$ by λ except with probability $e^{-2n\lambda^2}$ per side. McDiarmid's inequality gives $e^{-2\lambda^2/\sum_k \mathfrak{b}_k^2}$ for a function of n independent arguments whose k-th argument moves it by at most $\mathfrak{b}_k$.
- (F9) Realization. A standard atomless probability space is isomorphic modulo null sets to $[0,1)$ with Lebesgue measure (Kechris, Classical Descriptive Set Theory, Theorem 17.41).

The round files number these facts differently, and `theory/CURRENT.md` maps their numbers to these.

C.1.4 Needle lower bound [lem-needle-lower-bound]

Statement. Let θ be a target and $\vartheta \in [0,1]$. Let L° be a learning function, the *reference function*, and $X_1, \dots, X_J$ pairwise disjoint measurable sets of training inputs. For each $k \le J$ let L_k agree with L° at every training input outside X_k and satisfy $\theta(\mathsf{L}_k) \ge \vartheta$. Every certifier of θ with at most N Train calls, in either discipline, sound at level α over a class that contains $\mathsf{L}_1, \dots, \mathsf{L}_J$, satisfies $\Pr_{\mathsf{L}^\circ}[\hat{t} \ge \vartheta] \ge 1 - \alpha - N/J$.

Label. Proven.

Without the oracle, that is without Train calls, a certifier sound at the needle functions behaves at the reference function as at each of them. So it outputs at least ϑ there with probability at least $1-\alpha$. With N Train calls it rules out at most N of the J places where leakage can hide, whatever seeds it uses.

Proof. Fix the certifier and k. Put its runs against L° and against L_k on one probability space. Both runs use the same coins, and the i-th oracle call of both uses the same triple $(\rho_i', \xi_i', \upsilon_i') \sim \mu \otimes \pi \otimes \nu$, independent across i. By fidelity each run has the law that its oracle prescribes. Names are integers fixed by the count of calls, so while the transcripts agree the two oracles store the same objects under the same names. A Train call outside X_k gets the same answer in both runs, and so does every Eval and Open call. By induction the runs agree up to the first Train call with training input in X_k. Let Hit_k be the event that the run against L° makes such a call. Off Hit_k the outputs agree, so $\Pr_{\mathsf{L}_k}[\hat{t} \ge \vartheta] \le \Pr_{\mathsf{L}^\circ}[\hat{t} \ge \vartheta] + \Pr_{\mathsf{L}^\circ}[\mathrm{Hit}_k]$. Soundness at L_k gives $\Pr_{\mathsf{L}_k}[\hat{t} \ge \vartheta] \ge \Pr_{\mathsf{L}_k}[\hat{t} \ge \theta(\mathsf{L}_k)] \ge 1-\alpha$. A Train call names one training input, lying in at most one X_k, so $\sum_k \mathbf{1}[\mathrm{Hit}_k] \le N$. Averaging over k gives the claim. $\square$

C.1.5 Sphere code [lem-sphere-code]

Statement. (a) For every integer $d \ge 1$ there are at least $2^{0.18d}$ points of the unit sphere of $\mathbb{R}^d$ with pairwise distances at least one. (b) Let $\mathbf{p}_1, \dots, \mathbf{p}_J \in \mathbb{R}^d$ have pairwise distances at least $\mathfrak{s} > 0$, and let V_k be the set of points strictly nearer to $\mathbf{p}_k$ than to every other $\mathbf{p}_l$. The cells V_k are open and pairwise disjoint, and $\Pr[\mathbf{p}_k + \mathbf{g} \notin V_k] \le (J-1)\bar{\Phi}(\mathfrak{s}/(2\sigma')) \le \frac{J}{2} e^{-\mathfrak{s}^2/(8\sigma'^2)}$ for $\mathbf{g} \sim \mathcal{N}(0, \sigma'^2 I_d)$.

Label. Proven.

Proof. (a) Let $\mathrm{h}_2(p) = -p\log_2 p - (1-p)\log_2(1-p)$. For $p \le 1/2$ and every integer $r \le pd$, $\sum_{i \le r} \binom{d}{i} \le 2^{d\mathrm{h}_2(p)}$, since $1 \ge \sum_{i \le r} \binom{d}{i}(1-p)^d (p/(1-p))^i$ and $(p/(1-p))^i \ge (p/(1-p))^{pd}$. Let $\mathcal{W} \subseteq \{\pm 1\}^d$ be maximal with pairwise Hamming distances at least $\lceil d/4 \rceil$. Every word lies within $\lceil d/4 \rceil - 1 \le d/4$ of $\mathcal{W}$, so $|\mathcal{W}|\, 2^{d\mathrm{h}_2(1/4)} \ge 2^d$. Since $1 - \mathrm{h}_2(1/4) = 0.1887$ (computed), $|\mathcal{W}| \ge 2^{0.18d}$. The points $\mathbf{w}/\sqrt{d}$ for $\mathbf{w} \in \mathcal{W}$ have squared distances $4\,\mathrm{Ham}(\mathbf{w}, \mathbf{w}')/d \ge 1$. (b) V_k is an intersection of open half-spaces.

$\mathbf{p}_k + \mathbf{g} \notin V_k$ needs $\|\mathbf{p}_k + \mathbf{g} - \mathbf{p}_l\| \le \|\mathbf{g}\|$ for some $l \neq k$, which is $\langle \mathbf{g}, \mathbf{e} \rangle \ge \|\mathbf{p}_l - \mathbf{p}_k\|/2$ for the unit vector $\mathbf{e}$ along $\mathbf{p}_l - \mathbf{p}_k$. A union bound and (F6)(iii) finish. □

C.1.6 Coupling certificate [thm-coupling-certificate]

Setting. One learning node trains x_b under H_b, with $x_0 \neq x_1$, and its model reaches the view through a channel K, with view laws P_b as in def-certifier. The *identity coupling* Γ is the law of $(W_0^\Gamma, W_1^\Gamma) = (\mathsf{L}(x_0; \rho, \xi_0), \mathsf{L}(x_1; \rho, \xi_1))$, with $\rho \sim \mu$ and $\xi_0, \xi_1 \sim \pi$ all independent. An experimenter samples it with $\mathsf{Train}(x_0, \perp)$, which returns a seed name s, and $\mathsf{Train}(x_1, s)$. For $\gamma \ge 1$ the *coupled divergence* is $\beta_\gamma(K) = \mathbb{E}_\Gamma \, \mathrm{H}_\gamma(K(\cdot \mid W_0^\Gamma) \| K(\cdot \mid W_1^\Gamma))$, and the *divergence probability* is $\beta = \Pr_\Gamma[W_0^\Gamma \neq W_1^\Gamma]$. Let $P_b^\top$ be the law of (ρ, Y) with $\rho \sim \mu$, $\xi \sim \pi$ and $Y \sim K(\cdot \mid \mathsf{L}(x_b; \rho, \xi))$, which is what the top oracle shows an adversary that sees the node through K. The *top divergence* is $\mathrm{D}_\gamma^\top(K) = \mathrm{H}_\gamma(P_0^\top \| P_1^\top)$.

Statement. (a) Drawn seeds. (i) Fix an event $E \subseteq \mathcal{V}$ in advance and make n drawn-seed calls per input. One-sided Clopper and Pearson bounds at level $\alpha/2$, below on $P_1(E)$ and above on $P_0(E)$, give a lower certifier of $\mathrm{TV}(P_0, P_1)$ at level α, sound for every learning function. (ii) For finite $\mathcal{V}$ and the empirical law $\hat{p}_b$ of n drawn-seed views at x_b, $\min\{1, \mathrm{TV}(\hat{p}_0, \hat{p}_1) + \sqrt{|\mathcal{V}|/n} + \sqrt{2\ln(2/\alpha)/n}\}$ is sound at level α over every learning function. (iii) For a deterministic channel and atomless P_0, P_1, no drawn-seed certifier of $\mathrm{TV}(P_0, P_1)$ sound over every learning function outputs $\hat{t} < 1$ with probability above α.

(b) Shared seeds. For every learning function, every channel K and every $\gamma \ge 1$, $\mathrm{H}_\gamma(P_0 \| P_1) \le \mathrm{D}_\gamma^\top(K) \le \beta_\gamma(K) \le \beta$, with $\mathrm{D}_\gamma^\top(K) = \beta_\gamma(K)$ under deterministic kernels. At $\gamma = 1$ the left end is the best advantage under the bottom oracle, and $\mathrm{D}_1^\top(K)$ the best under the top oracle, at every budget. If $(m, m') \mapsto \mathrm{H}_\gamma(K(\cdot \mid m) \| K(\cdot \mid m'))$ is computable, then from n independent draws of Γ, at $2n$ Train calls, $\hat{t} = \min\{1, \frac{1}{n}\sum_{i \le n} \mathrm{H}_\gamma(K(\cdot \mid W_0^{\Gamma,i}) \| K(\cdot \mid W_1^{\Gamma,i})) + \sqrt{\ln(1/\alpha)/(2n)}\}$ is sound at level α over every learning function, for $\mathrm{H}_\gamma(P_0 \| P_1)$ and for $\mathrm{D}_\gamma^\top(K)$. For the Gaussian channel $K(\cdot \mid m) = \mathcal{N}(f(m), \sigma^2 I_d)$ and every $\Delta > 0$, $\beta_{e^\varepsilon}(K) \le \delta_{\mathcal{N}}(\varepsilon; \Delta/\sigma) + \Pr_\Gamma[\|f(W_0^\Gamma) - f(W_1^\Gamma)\| > \Delta]$.

(c) Tightness. (i) Some learning function with deterministic kernels has $\mathrm{TV}(P_0, P_1) = 0$ and $\mathrm{D}_1^\top = \beta = 1$ for the plain channel. (ii) A coupling through a seed map ψ with $\psi_* \mu = \mu$ also bounds $\mathrm{H}_\gamma(P_0 \| P_1)$. Some ψ makes it exact for TV under deterministic kernels through a deterministic channel with a finite view.

Label. Proven, (a), (b), (c)(i), and (c)(ii) for finite views. Open, (c)(ii) for atomless views and the search for a seed map by oracle calls (open problem 8).

Without the oracle, let the view be a deterministic function of the model with atomless laws. Part (a)(iii) shows that then no number of drawn-seed runs certifies the total variation below one, soundly for every learning function. Analysis needs the law of the trained transformer, or a mechanism with a known law such as DP-SGD. With the oracle, part (b) certifies $\mathrm{H}_\gamma(P_0 \| P_1)$ for every learning function when the divergence of the channel at two models is computable. It uses the learning function only through $\beta_\gamma(K)$, the mean of that divergence over coupled pairs.

Degenerate oracles. Through a deterministic channel with atomless view laws, the certificate of (b) at $\gamma = 1$ has no drawn-seed counterpart, by (a)(iii). On the adversary side, (b) brackets the best advantages under the two oracles, and (c)(i) shows that their gap can be one.

Proof of (a)(i). By fidelity the number of views in E at x_b is binomial with parameters n and $P_b(E)$. By (F8) both bounds hold together with probability at least $1 - \alpha$, and then $\underline{t} = \underline{p}_1 - \bar{p}_0 \le P_1(E) - P_0(E) \le \mathrm{TV}(P_0, P_1)$. □

Proof of (a)(ii). The triangle inequality, $\mathbb{E}\|\hat{p}_b - P_b\|_1 \le \sqrt{|\mathcal{V}|/n}$ and McDiarmid's inequality give it (round-01/builder.md). □

Proof of (a)(iii). This is lem-atomless-two-point (c), proven in round-01/builder.md, corrected in rounds 02 and 03. Conditioning the model laws on complementary random halves of a fine partition separates the views. The law of the drawn-seed draws moves by at most $N(2N-1)$ over the cell count. □

Proof of (b). Step 1, measurability. H_γ is a supremum over a countable algebra that generates the Borel sets of $\mathcal{V}$, by approximation in the sum of the two laws. So the integrand of $\beta_\gamma(K)$ is measurable.

Step 2, the chain. Let $P_b^\rho = \int K(\cdot \mid \mathsf{L}(x_b;\rho,\xi))\,\pi(d\xi)$, a kernel in ρ by Fubini's theorem. Then $P_b = \int P_b^\rho\,\mu(d\rho)$ and $P_b^\top(d\rho,dy) = \mu(d\rho)P_b^\rho(dy)$, so (F5) gives $\mathrm{D}_\gamma^\top(K) = \int \mathrm{H}_\gamma(P_0^\rho\|P_1^\rho)\,\mu(d\rho)$. (F4) with Θ the law of (ρ,ρ) gives $\mathrm{H}_\gamma(P_0\|P_1) \le \mathrm{D}_\gamma^\top(K)$. For fixed ρ, (F4) with $\Theta = \pi\otimes\pi$ on (ξ_0,ξ_1) bounds $\mathrm{H}_\gamma(P_0^\rho\|P_1^\rho)$ by the mean of $\mathrm{H}_\gamma(K(\cdot \mid \mathsf{L}(x_0;\rho,\xi_0))\|K(\cdot \mid \mathsf{L}(x_1;\rho,\xi_1)))$, with equality when Ξ is a point. Integrating over μ gives $\mathrm{D}_\gamma^\top(K) \le \beta_\gamma(K)$. Finally $\mathrm{H}_\gamma(P\|P) = 0$ and $\mathrm{H}_\gamma \le 1$, so the integrand of $\beta_\gamma(K)$ is at most $\mathbf{1}[W_0^\Gamma \ne W_1^\Gamma]$. This event is measurable, since the diagonal of a standard Borel space is Borel.

Step 3, the advantages. Under the bottom oracle the oracle answers the adversary's own calls with randomness free of the hypothesis and of the honest seed. So its transcript is the image of the channel output under a kernel free of the hypothesis, and (F2) with (F1) bound its advantage by $\mathrm{TV}(P_0,P_1)$. For one fixed learning function the adversary that hard-wires the set of (F1) attains it. Under the top oracle every transcript is the image of (ρ,Y) under a kernel free of the hypothesis, and an adversary that opens the seed name holds (ρ,Y). So (F2) and (F1) make $\mathrm{D}_1^\top(K)$ the best advantage.

Step 4, statistics. The pairs $(W_0^{\Gamma,i},W_1^{\Gamma,i})$ are independent draws of Γ, by fidelity. The summands lie in $[0,1]$ and have mean $\beta_\gamma(K)$, so Hoeffding's inequality makes $\hat{t}$ sound for $\beta_\gamma(K)$, hence for the smaller two. For the Gaussian channel, (F6)(i) gives the summand $\delta_{\mathcal{N}}(\varepsilon;\|f(W_0^\Gamma)-f(W_1^\Gamma)\|/\sigma)$. Monotonicity and $\delta_{\mathcal{N}} \le 1$ bound it by $\delta_{\mathcal{N}}(\varepsilon;\Delta/\sigma) + \mathbf{1}[\|f(W_0^\Gamma)-f(W_1^\Gamma)\| > \Delta]$, and expectation gives the tail form. □

Proof of (c). For (i) let $\mathcal{R} = \mathcal{M} = [0,1)$, Ξ a point and $\mathsf{L}(x_b;\rho) = \rho + b/2 \bmod 1$. Both model laws are uniform, so $\mathrm{TV} = 0$, and $W_0^\Gamma \ne W_1^\Gamma$ at every seed, so β, $\beta_\gamma(K)$ at $\gamma = 1$ and $\mathrm{D}_1^\top$ all equal one. Part (ii) is proven in round-01/builder.md, section Coupling certificate, by gluing measure isomorphisms between the level sets of the two views. □

C.1.7 Black-box certifiability [thm-black-box-certifiability]

Setting. One learning node trains x_b under H_b, and its model reaches the adversary through the query channel, with a finite query set U and an allowance $Q \ge q_E$. The output space $\mathcal{Y}$ is finite with $C \ge 2$ elements, and ev may be randomized. A *strategy tree* $\mathfrak{t}$ with q queries is a complete C-ary tree of depth q whose internal nodes carry queries in U and whose leaves carry bits. $\mathfrak{T}_q$ is the set of these trees, so $|\mathfrak{T}_q| \le |U|^{(C^q-1)/(C-1)}2^{C^q}$. For a tree let $\mathrm{Adv}(\mathfrak{t}) = \Pr_{H_1}[\mathfrak{t}=1] - \Pr_{H_0}[\mathfrak{t}=1]$, with the model drawn from Π_{x_b} and fresh evaluation randomness per query. Let $\mathrm{Adv}^{\mathrm{bb}}(q) = \max_{\mathfrak{t}\in\mathfrak{T}_q}\mathrm{Adv}(\mathfrak{t})$ and $\mathrm{Adv}^{\mathrm{wb}} = \mathrm{TV}(\Pi_{x_0},\Pi_{x_1})$.

Statement. (o) Reduction. Under the bottom oracle every adversary with q_E Eval calls on the served name has advantage at most $\mathrm{Adv}^{\mathrm{bb}}(q_E)$, whatever its Train calls, also when it knows L.

(a) Certificate from both sides. An experimenter draws n models at each input with drawn seeds, opens them, and runs every tree once on each model with fresh evaluation randomness. Let $\widehat{\mathrm{Adv}}(\mathfrak{t})$ be the frequency of output one over the

models at x_1 minus that over the models at x_0. For every learning function, with probability at least $1-\alpha$, $\max_{\mathfrak{t}\in\mathfrak{T}_q} |\widehat{\mathrm{Adv}}(\mathfrak{t}) - \mathrm{Adv}(\mathfrak{t})| \le \sqrt{2\ln(4|\mathfrak{T}_q|/\alpha)/n}$. The same bound holds between $\max_{\mathfrak{t}} \widehat{\mathrm{Adv}}(\mathfrak{t})$ and $\mathrm{Adv}^{\mathrm{bb}}(q)$.

(b) to (d), stated. One bottom-oracle adversary with $2n$ shadow models comes within $2\sqrt{\ln(2|\mathfrak{T}_{q_E}|)/n}$ of $\mathrm{Adv}^{\mathrm{bb}}(q_E)$ for every learning function. With deterministic ev and atomless model laws, a drawn-seed procedure that rejects every false claim $\mathrm{Adv}^{\mathrm{wb}} \le \mathrm{Adv}^{\mathrm{bb}}(Q) + \lambda < 1$ except with probability α accepts it with probability at most α everywhere. Coupled pairs certify $\mathrm{Adv}^{\mathrm{wb}} \le \beta$.

Label. Proven, (o) to (d). Open, a matching lower bound for (b) (open problem 3). Parts (b) to (d) are proven in round-01/builder.md, section Black-box certifiability, and revised in round-02/builder.md.

Without the oracle, a bound on what q_E label queries reveal needs the law of the trained transformer's labels on U. With the oracle, part (a) certifies $\mathrm{Adv}^{\mathrm{bb}}$ from both sides for every learning function, at a cost fixed by C, $|U|$, the number of queries, α and the accuracy. Part (c) concerns deterministic evaluation and atomless model laws. There drawn seeds can refute, and never certify, a claim that label queries reveal all that the model reveals, when the claimed value is below one.

Degenerate oracles. Part (o) holds under the bottom oracle and fails under the top oracle. In the example of thm-coupling-certificate (c)(i), let $\mathsf{ev}(m,u)$ be the u-th binary digit of m. Both model laws are uniform, so $\mathrm{Adv}^{\mathrm{bb}}(q) = 0$ for every q. An adversary that opens the honest seed ρ reads its first digit. It queries that digit of the served model once and wins with advantage one, since the shift by one half flips it.

Proof of (o). Represent the randomness of the adversary's own calls in advance, and call it ω together with the coins. It is independent of the hypothesis and of the served model. Given ω the adversary is deterministic, and its next query on the served name is a function of the earlier answers on that name. Each answered query lies in U, and a refused call returns a fixed symbol that can be removed. Padded to q_E queries, the interaction with the served model is a tree $\mathfrak{t}_\omega \in \mathfrak{T}_{q_E}$, and the output is its leaf bit. The advantage is $\mathbb{E}_\omega \mathrm{Adv}(\mathfrak{t}_\omega) \le \mathrm{Adv}^{\mathrm{bb}}(q_E)$. Nothing restricts ω, so it may hold knowledge of L. □

Proof of (a). For a fixed tree the n outcomes at x_b are independent Bernoulli draws with mean $\Pr_{H_b}[\mathfrak{t}=1]$. Hoeffding's inequality bounds each frequency's deviation by $\lambda/2$ except with probability $2e^{-n\lambda^2/2}$, so $|\widehat{\mathrm{Adv}}(\mathfrak{t}) - \mathrm{Adv}(\mathfrak{t})| \le \lambda$ except with probability $4e^{-n\lambda^2/2}$. A union bound over $\mathfrak{T}_q$, with $4|\mathfrak{T}_q|e^{-n\lambda^2/2} = \alpha$, gives the bound, and $|\max \widehat{\mathrm{Adv}} - \max \mathrm{Adv}| \le \max |\widehat{\mathrm{Adv}} - \mathrm{Adv}|$. □

C.1.8 Dataflow composition [thm-dataflow-composition]

Setting. G isolates client j behind Gaussian releases, P_b is the law of the adversary's view under H_b, and $\gamma = e^{\varepsilon}$. $\mathrm{clip}_B(\mathbf{w}) = \mathbf{w}\min\{1, B/\|\mathbf{w}\|\}$, with $\mathrm{clip}_B(0) = 0$.

Statement. (b) Composition. For every adversary under the bottom or the top oracle, with any strategy, with or without a stop, every $\varepsilon \ge 0$ and every $\Delta > 0$,

$$\mathrm{H}_\gamma(P_0\|P_1) \le \delta_{\mathcal{N}}(\varepsilon; \sqrt{R}\,\Delta/\sigma) + \min\{\epsilon_0(\Delta), \gamma\,\epsilon_1(\Delta)\}, \qquad \mathrm{H}_\gamma(P_1\|P_0) \le \delta_{\mathcal{N}}(\varepsilon; \sqrt{R}\,\Delta/\sigma) + \min\{\epsilon_1(\Delta), \gamma\,\epsilon_0(\Delta)\},$$

with ϵ_b computed for the adversary's strategy. Moreover $\epsilon_0(\Delta) \le \sum_{t\le R} \mathbb{E}[\mathbf{1}[t \text{ runs}]\, \beta_t(c_t, W_{t-1}, W'_{t-1}, \Delta)]$ in Ex_0, and the same holds for ϵ_1 in Ex_1 with the states exchanged. With a threshold Δ_t per round, and ϵ_b the probability that some round t has $\|f_t(c_t, W'_t) - f_t(c_t, W_t)\| > \Delta_t$, $\sqrt{R}\Delta$ becomes $(\sum_t \Delta_t^2)^{1/2}$. The bounds hold for an adversary that knows L.

(c) Certificate. Fix a strategy, the protocol's for a passive coalition, and fix Δ in advance. An experimenter samples Ex_0 independently n times, and $\bar{p}$ is the one-sided Clopper and Pearson upper bound at level α from the number of runs with $D_0^\star > \Delta$. Then $\Pr[\epsilon_0(\Delta) \le \bar{p}] \ge 1-\alpha$ for every learning function, and $\bar{p} \le \ln(1/\alpha)/n$ when no run exceeds Δ. Each run costs at most $2R$ Train calls for client j, plus the calls of the other nodes and the strategy's own Train calls. The certificate holds for every passive adversary and every strategy that agrees with the protocol's until a stop. Runs of Ex_1 serve the reverse order, and the DKW inequality gives a form uniform in Δ (proven, round-02/builder.md).

(e)(ii) Drawn seeds pay. Take $R = 1$, $O_1 = 0$, a fixed choice c, $\mathcal{M} = \mathbb{R}^d$, $f_1(c,m) = m$ and $x_b = x(c, D_j^b)$, so $P_b = \Pi_{x_b} * \mathcal{N}(0, \sigma^2 I_d)$, and let $\theta = \mathrm{H}_\gamma(P_0 \| P_1)$. Let $\mathfrak{L}_B$ be the class of learning functions with $\|\mathsf{L}(x_b; \rho, \xi)\| \le B$ for $b \in \{0,1\}$. Let $J_{\mathrm{cd}} \ge 1$ with $2J_{\mathrm{cd}} \le \lceil 2^{0.18d} \rceil$. For every drawn-seed certifier of θ with N Train calls, sound at level α over $\mathfrak{L}_B$, some $\mathsf{L}^\circ \in \mathfrak{L}_B$ with deterministic kernels and $\mathsf{L}^\circ(x_0; \rho) = \mathsf{L}^\circ(x_1; \rho)$ for every ρ gives $\Pr_{\mathsf{L}^\circ}[\hat{t} \ge 1 - (1+\gamma)(2J_{\mathrm{cd}} - 1)\bar{\Phi}(B/(2\sigma))] \ge 1 - \alpha - N(2N-1)/J_{\mathrm{cd}}$. At L°, $\theta = 0$, every coupled pair coincides, and the certificate (c) outputs $\bar{p} \le \ln(1/\alpha)/n$ with certainty.

Reading of (e)(ii). For $d \ge 6$ and $J_{\mathrm{cd}} = \min\{\lfloor 2^{0.18d-1} \rfloor, \lfloor e^{B^2/(16\sigma^2)} \rfloor\}$, certifying below $1-(1+\gamma)e^{-B^2/(16\sigma^2)}$ at L° with probability above $\alpha+\eta$ needs $N(2N-1) > \eta J_{\mathrm{cd}}$, so about $\sqrt{\eta/2}\min\{2^{0.09d-1/2}, e^{B^2/(32\sigma^2)}\}$ drawn-seed Train calls. The certificate (c) needs $\lceil \ln(1/\alpha)/\epsilon_{\mathrm{acc}} \rceil$ coupled runs there for accuracy ϵ_{acc}, whatever B, σ and d. Each run needs only client j's two Train calls, since the choice is fixed (proven).

The other parts, stated. (a) The ciphertexts of an aggregation sub-protocol reduce to the releases up to its realization error, for a coalition that follows or deviates inside it (proven given that error, round-03/builder.md). For threshold CKKS with smudging that error is negligible under RLWE, against a deviating coalition only with six additions (heuristic, open problem 6). Unchecked decryption lets a public scalar a divide the protecting noise when the decryptors add it at decryption. Noise that client j adds inside its ciphertext scales with the update, and the scalar then removes at most the decryption error (round-04/builder.md and round-05/builder.md). These claims are proven given that decryption of a times a ciphertext returns a times its message plus an error of bounded norm and any noise the decryptors add. That form is heuristic for CKKS. (d) In some one-round procedure, take a certifier sound for every learning function against all active strategies. For every $\eta \in (0,1)$, at some function where that leakage is zero, it outputs at least $1-\eta$ with probability at least $1-\alpha$, whatever its number of Train calls (proven, round-03/builder.md). (e)(i) In the setting of (e)(ii) without clipping, a drawn-seed certifier sound for every learning function does the same with probability at least $1-\alpha-\eta$. (e)(iii) Over $\mathfrak{L}_B$ and for $\alpha \le 1/2$, $O(e^{B^2/\sigma^2}\epsilon_{\mathrm{acc}}^{-2}\ln(1/\alpha))$ drawn-seed Train calls reach accuracy ϵ_{acc} in every dimension, at fixed γ (proven, round-02/builder.md).

Label. Proven, (a) given the realization error, (b), (c), (d) and (e). Heuristic, the realization by threshold CKKS and the form of its decryption under a public scalar. Open, the sum channel, the drawn-seed price above d, and a multi-round drawn-seed certifier free of d (open problems 1, 2 and 4).

Without the oracle, a privacy bound for this procedure needs the law of the trained models, or a worst-case bound that an unclipped update lacks. Parts (e)(i) and (e)(ii) show that drawn-seed runs do not replace the oracle at a cost free of d and B. With the oracle, part (b) holds for every learning function, every state and every strategy. It uses the learning function only through $\epsilon_b(\Delta)$. Part (c) certifies it for a fixed strategy, with a number of runs free of d, of B and of the architecture.

Degenerate oracles. Part (b) holds under both adversary-side oracles, and an instance makes it an equality under the top oracle. The certificate (c) holds in the

coupled discipline and fails under drawn seeds, by (e). So the theorem meets the test through its certificate, quantitatively in the clipped case.

Proof of (b). Step 1, the execution as a recursion. Let $\bar\rho_t = \rho_t$ under the top oracle and $\bar\rho_t = \star$, a fixed symbol, under the bottom oracle. Let ω collect three things. They are the adversary's coins, an independent sequence of triples $(\rho', \xi', \upsilon') \sim \mu \otimes \pi \otimes \nu$ for the calls it makes beyond those the procedure requires, and the randomness of every node other than $v_1, \dots, v_R$. The sequence serves those calls in order. Names are per-caller counters that the adversary's own calls shift, so a node argument that is a name stands for the object it names. The last item includes the oracle's draws for the calls that the procedure requires of the coalition's nodes, and the randomness of each r_t other than Z_t. Its law does not depend on b. Let $\zeta_t = (\rho_t, \xi_t^0, \xi_t^1) \sim \mu \otimes \pi \otimes \pi$, with ω, the ζ_t and the Z_t independent. Under H_b the node v_t uses (ρ_t, ξ_t^b).

Claim. There are measurable maps χ_t, o_t, st_t and Ψ, the same under both, such that under H_b, in every round that runs, $c_t = \chi_t(\omega, y_{<t}, \bar\rho_{<t})$ and $O_t = o_t(\omega, y_{<t}, \bar\rho_{\le t})$. The procedure stops after round t exactly when $\mathrm{st}_t(\omega, y_{\le t}, \bar\rho_{\le t}) = 1$, and the transcript is $\Psi(\omega, y_{\le R}, \bar\rho_{\le R})$, with $y_t = \bar\rho_t = \star$ after a stop.

Run the procedure in its order and answer the adversary's calls from ω. Under the bottom oracle the adversary holds no name of a seed or a model of any v_t. Under the top oracle it holds the seed name of v_t from the time v_t runs, and a Train call with it returns $\mathsf{L}(x; \rho_t, \xi')$ with ξ' from ω. Consider every node other than the input node of D_j, the v_t and the r_t. By induction over the order, its value and the adversary's state are measurable functions of ω, of the releases made so far and of the revealed seeds. Three facts carry the step. By condition 2, the input node of D_j has children only among the v_t, and W_t enters only r_t and v_{t+1}. By condition 4, the adversary sees that node and the v_t only through the releases and the revealed seeds. Every other input node has a value whose law the hypotheses fix, carried by ω. The stop is a value that the coalition sets after a release. No map involves D_j, so none depends on b. By condition 1, c_t is neither D_j nor W_{t-1}. Since $r_{t-1} \prec v_t \prec r_t$, c_t and O_t have the stated arguments.

So under H_b the chain and the releases solve $W_t^b = \mathsf{L}(x(c_t, D_j^b, W_{t-1}^b); \rho_t, \xi_t^b)$, with $W_0^b = w_0$, and $y_t = O_t + f_t(c_t, W_t^b) + Z_t$. Let $\mathcal{L}_b$ be the law of $(\omega, y_{\le R}, \bar\rho_{\le R})$ under H_b. Then $P_b = \Psi_* \mathcal{L}_b$, and by (F2) it suffices to bound the divergences of $\mathcal{L}_0$ and $\mathcal{L}_1$.

Step 2, the hybrid. On this space define recursively, applying the maps of step 1 to the hybrid's past,

$$\tilde c_t = \chi_t(\omega, \tilde y_{<t}, \bar\rho_{<t}), \qquad \tilde W_t^b = \mathsf{L}(x(\tilde c_t, D_j^b, \tilde W_{t-1}^b); \rho_t, \xi_t^b), \qquad F_t^b = f_t(\tilde c_t, \tilde W_t^b),$$

$$\tilde y_t = o_t(\omega, \tilde y_{<t}, \bar\rho_{\le t}) + F_t^1 - \mathrm{clip}_\Delta(F_t^1 - F_t^0) + Z_t,$$

for $b \in \{0, 1\}$, with $\tilde W_0^b = w_0$ and the stop given by st_t. Let $\mathcal{L}_{\mathrm{hyb}}$ be the law of $(\omega, \tilde y_{\le R}, \bar\rho_{\le R})$.

Step 3, $\mathrm{TV}(\mathcal{L}_0, \mathcal{L}_{\mathrm{hyb}}) \le \epsilon_0(\Delta)$. Let y^0 be the solution under H_0 on this space, with chain W_t and twin $W_t' = \mathsf{L}(x(c_t, D_j^1, W_{t-1}'); \rho_t, \xi_t^1)$. The H_0 run does not use ξ_t^1, so the space carries Ex_0. Let $t^\star$ be the first round of the hybrid with $\|F_t^1 - F_t^0\| > \Delta$, or ∞. Before $t^\star$ the clip does nothing, so by induction the hybrid and Ex_0 have the same choices, chains, twins, releases and stops. At $t^\star$ they have the same choice and the same chains up to round $t^\star - 1$, so the same difference. The same induction from the side of Ex_0 shows that $t^\star$ is also its first exceedance. Hence the two agree off $\{D_0^\star > \Delta\}$, and the coupling inequality gives the claim. For the sum, let $\mathcal{G}_t$ be generated by ω, $\zeta_{<t}$ and $Z_{<t}$. The event that round t runs, c_t, W_{t-1} and W_{t-1}' are $\mathcal{G}_t$-measurable, and ζ_t is independent of $\mathcal{G}_t$. So Fubini's theorem gives the conditional probability of an exceedance in round t as $\beta_t(c_t, W_{t-1}, W_{t-1}', \Delta)$, and a union bound finishes.

Step 4, the kernel. We build one kernel from $\mathbb{R}^R$ to the space of $(\omega, y_{\le R}, \bar\rho_{\le R})$ that maps $\mathcal{N}(0, I_R)$ to $\mathcal{L}_{\mathrm{hyb}}$ and $\mathcal{N}((\Delta/\sigma, \dots, \Delta/\sigma), I_R)$ to $\mathcal{L}_1$. Given $\mathfrak{z} \in \mathbb{R}^R$ the kernel draws ω. In each round, unless it has stopped, it draws ζ_t, sets $\bar\rho_t$, computes c_t and O_t from its past outputs, extends both chains and computes F_t^0, F_t^1. Let $\mathbf{n}_t$ be the unit vector along $F_t^1 - F_t^0$ (a fixed unit vector if they are equal) and $\mathfrak{f}_t = \min\{1, \|F_t^1 - F_t^0\|/\Delta\}$, so that $\mathrm{clip}_\Delta(F_t^1 - F_t^0) = \Delta \mathfrak{f}_t \mathbf{n}_t$. Draw $Z_t' \sim \mathcal{N}(0,1)$ and $Z_t'' \sim \mathcal{N}(0, I_d)$. It outputs

$$\tilde g_t = \mathfrak{f}_t(\mathfrak{z}_t - \Delta/\sigma) + \sqrt{1 - \mathfrak{f}_t^2}\, Z_t', \qquad y_t = O_t + F_t^1 + \sigma \tilde g_t \mathbf{n}_t + \sigma(I_d - \mathbf{n}_t \mathbf{n}_t^\top) Z_t'',$$

and then applies st_t. The coordinate $\mathfrak{z}_t$ is independent of the kernel's past and of ζ_t. If $\mathfrak{z}_t \sim \mathcal{N}(\Delta/\sigma, 1)$, then $\tilde g_t \sim \mathcal{N}(0,1)$ by (F6)(ii), and $y_t \sim \mathcal{N}(O_t + F_t^1, \sigma^2 I_d)$ given the past. That is the conditional law of the release under H_1, and the D_j^1 chain grows as under H_1, so the outputs have law $\mathcal{L}_1$. If $\mathfrak{z}_t \sim \mathcal{N}(0,1)$, then $\tilde g_t \sim \mathcal{N}(-\mathfrak{f}_t \Delta/\sigma, 1)$ and $y_t \sim \mathcal{N}(O_t + F_t^1 - \Delta \mathfrak{f}_t \mathbf{n}_t, \sigma^2 I_d)$, the hybrid's conditional law, so the outputs have law $\mathcal{L}_{\mathrm{hyb}}$. (F2) in both orders and (F6)(i) give $\mathrm{H}_\gamma(\mathcal{L}_{\mathrm{hyb}} \| \mathcal{L}_1)$ and $\mathrm{H}_\gamma(\mathcal{L}_1 \| \mathcal{L}_{\mathrm{hyb}})$ at most $\delta_{\mathcal{N}}(\varepsilon; \sqrt{R}\Delta/\sigma)$.

Step 5, the four bounds. The first split of (F3) with $\mathcal{L}' = \mathcal{L}_{\mathrm{hyb}}$ gives $\mathrm{H}_\gamma(\mathcal{L}_0 \| \mathcal{L}_1) \le \epsilon_0(\Delta) + \delta_{\mathcal{N}}(\varepsilon; \sqrt{R}\Delta/\sigma)$. The second gives $\mathrm{H}_\gamma(\mathcal{L}_1 \| \mathcal{L}_0) \le \delta_{\mathcal{N}}(\varepsilon; \sqrt{R}\Delta/\sigma) + \gamma\epsilon_0(\Delta)$. Exchanging the roles of the hypotheses gives the two bounds with ϵ_1. Here the hybrid releases $O_t + F_t^0 + \mathrm{clip}_\Delta(F_t^1 - F_t^0) + Z_t$ and agrees with the H_1 run until the first exceedance in Ex_1. For thresholds per round, the hybrid clips round t at Δ_t, step 3 reads the exceedance of round t at Δ_t, and step 4 uses $\mathcal{N}((\Delta_t/\sigma)_t, I_R)$. The bounds hold for each fixed L and every adversary, hence for one that knows L. □

Proof of (c). By fidelity the n runs are independent, so the indicators of $D_0^\star > \Delta$ are independent Bernoulli variables with parameter $\epsilon_0(\Delta)$, and (F8) gives the coverage. A strategy that agrees with the protocol's until a stop has, on the space of step 1, the same rounds as the protocol's before the stop and none after. So its exceedance is a maximum over fewer rounds, and its $\epsilon_0(\Delta)$ is no larger. □

Proof of (e)(ii). Step 1, the pair code. By lem-sphere-code (a) there are $2J_{\mathrm{cd}}$ points of the sphere of radius B with pairwise distances at least B. Pair them as $\mathbf{p}_i^+, \mathbf{p}_i^-$ for $i \le J_{\mathrm{cd}}$. By (F9) identify $(\mathcal{R}, \mu)$ with $[0,1)$ under Lebesgue measure. Let $\mathrm{ix}(\rho) = \lfloor J_{\mathrm{cd}}\rho \rfloor + 1$, and let $\mathrm{sg}(\rho) = +1$ when the fractional part of $J_{\mathrm{cd}}\rho$ is below one half and -1 otherwise. Then $(\mathrm{ix}, \mathrm{sg})$ is uniform. Let $\mathsf{L}^\circ(x_0; \rho) = \mathsf{L}^\circ(x_1; \rho) = \mathbf{p}_{\mathrm{ix}(\rho)}^{\mathrm{sg}(\rho)}$. For signs $\varsigma \in \{\pm 1\}^{J_{\mathrm{cd}}}$ let $\mathsf{L}_\varsigma(x_0; \rho) = \mathbf{p}_i^{\varsigma_i}$ and $\mathsf{L}_\varsigma(x_1; \rho) = \mathbf{p}_i^{-\varsigma_i}$, with $i = \mathrm{ix}(\rho)$. All of them agree at every other training input, have deterministic kernels and lie in $\mathfrak{L}_B$.

Step 2, drawn seeds see nearly the same thing. In the drawn-seed discipline every Train call has $s = \perp$ and the certifier opens no seed name, so each answer at x_b is a fresh draw of Π_{x_b}. So the transcript is a fixed map of N independent draws at x_0, N at x_1, used in order, and ω'. Here ω' holds the coins and an independent triple (ρ', ξ', υ') per call, which serves every other input and every Eval call. It has the same law under every member of the family. Under L° the draws are $\mathbf{p}_{\mathrm{ix}}^{\mathrm{sg}}$ with independent uniform index and sign. Under the mixture over a uniform ς they are $\mathbf{p}_{\mathrm{ix}}^{\pm\varsigma_{\mathrm{ix}}}$ with independent uniform indices. Given $2N$ distinct indices, the signs are independent and uniform in both. So the two laws of the $2N$ draws differ in total variation by at most the chance that two of $2N$ indices coincide, which is at most $N(2N-1)/J_{\mathrm{cd}}$. By (F2) the same holds for every event of the transcript.

Step 3, every member leaks. Let $\mathbb{1}_\varsigma(y) = 1$ when the code point nearest to y lies in $\{\mathbf{p}_i^{\varsigma_i}\}$, the support of the model at x_0 under L_ς. Under H_0 the view is a code point of that support plus $\mathcal{N}(0, \sigma^2 I_d)$ noise, and lem-sphere-code (b) with $\mathfrak{s} = B$ bounds the decoding error by $(2J_{\mathrm{cd}} - 1)\bar\Phi(B/(2\sigma))$. Under H_1 the view is a point of the other half plus noise,

and $\not\vdash_\varsigma = 1$ needs the same error. So $\theta(\mathsf{L}_\varsigma) \geq 1 - (1+\gamma)(2J_{\mathrm{cd}} - 1)\bar{\Phi}(B/(2\sigma)) =: \vartheta$ for every ς.

Step 4, transfer. Soundness gives $\Pr_{\mathsf{L}_\varsigma}[\hat{t} \geq \vartheta] \geq 1 - \alpha$ for every ς, and step 2 gives the claim at L°. At L° the model does not depend on the dataset, so $\theta = 0$ and every coupled pair coincides. For the reading, (F6)(iii) gives $(2J_{\mathrm{cd}} - 1)\bar{\Phi}(B/(2\sigma)) \leq J_{\mathrm{cd}} e^{-B^2/(8\sigma^2)} \leq e^{-B^2/(16\sigma^2)}$. □

C.1.9 Bounded-deviation transfer [thm-bounded-deviation-transfer]

Setting. G isolates client j behind Gaussian releases, with or without state, under the bottom or the top oracle. For each $t \leq R$ the coalition may own input nodes whose only child is r_t. Let $\mathbf{u}_t \in \mathbb{R}^d$ be the sum of their values in round t, or zero when it owns none. Assume $O_t = \mathbf{u}_t + \bar{o}_t$, where $\bar{o}_t$ is a measurable function of the parents of r_t other than v_t and these input nodes, and of randomness of r_t independent of Z_t. The protocol prescribes the values of these input nodes, and $\mathbf{u}_t^{\mathrm{pr}}$ is the sum of the prescribed values. It is a measurable function of the values of the nodes that the coalition owns or receives through the plain channel before r_t runs, other than these input nodes.

Fix $B_A \geq 0$. A strategy is *B_A-deviating* when three conditions hold. Every argument and value that it sets for a node other than these input nodes is the protocol's. In every round that runs, $\|\mathbf{u}_t - \mathbf{u}_t^{\mathrm{pr}}\| \leq B_A$. It stops, if at all, after a release. Its queries and other oracle calls are unrestricted. The *protocol's strategy* is the one with $\mathbf{u}_t = \mathbf{u}_t^{\mathrm{pr}}$ that never stops, and $\epsilon_b^{\mathrm{pr}}(\Delta)$ is its exceedance probability along H_b. Let θ^{dev} and $\theta^{\mathrm{dev},\leftarrow}$ be the suprema of $\mathrm{H}_\gamma(P_0 \| P_1)$ and of $\mathrm{H}_\gamma(P_1 \| P_0)$ over adversaries with a B_A-deviating strategy. Let $\mathfrak{e} = (R-1)B_A^2/(2\sigma^2)$, $\Upsilon_\mathfrak{e}(p) = \exp(-(\sqrt{\ln(1/p)} - \sqrt{\mathfrak{e}})_+^2)$ for $p \in (0,1]$, $\Upsilon_\mathfrak{e}(0) = 0$ and $\Upsilon_\mathfrak{e}(p) = 1$ for $p > 1$.

The instance G_R^{agg} has $\mathcal{M} = \mathbb{R}^d$, an honest stateless client j, an honest server, and one coalition client j' whose prescribed input is zero. The choice of round t is (t, m_t), with training input $x((t,m_t), D) = (\ell_t, m_t, D, \mathcal{H})$ for distinct learner names ℓ_t. The release is $y_t = \mathbf{u}_t + \mathrm{clip}_B(W_t - m_t) + Z_t$ with $B \in (0, \infty]$, and the server sets $m_{t+1} = m_t + y_t$ from a fixed m_1. Its target is $\theta_R^{\mathrm{agg}} = \max\{\theta^{\mathrm{dev}}, \theta^{\mathrm{dev},\leftarrow}\}$.

Statement. (a) Transfer. For every learning function, every B_A-deviating strategy, every $b \in \{0,1\}$, every $\Delta > 0$ and every $\varpi > 1$, $\epsilon_b(\Delta) \leq \epsilon_b^{\mathrm{pr}}(\Delta)^{1-1/\varpi} e^{(\varpi-1)\mathfrak{e}}$, hence $\epsilon_b(\Delta) \leq \Upsilon_\mathfrak{e}(\epsilon_b^{\mathrm{pr}}(\Delta))$. So $\theta^{\mathrm{dev}} \leq \delta_{\mathcal{N}}(\varepsilon; \sqrt{R}\Delta/\sigma) + \Upsilon_\mathfrak{e}(\epsilon_0^{\mathrm{pr}}(\Delta))$ and $\theta^{\mathrm{dev},\leftarrow} \leq \delta_{\mathcal{N}}(\varepsilon; \sqrt{R}\Delta/\sigma) + \Upsilon_\mathfrak{e}(\epsilon_1^{\mathrm{pr}}(\Delta))$. At $B_A = 0$, $\Upsilon_0(p) = p$ for $p \leq 1$, and the statement is the passive case of thm-dataflow-composition with stops.

(b) Certificate. Let $\bar{\epsilon}_b$ be the one-sided Clopper and Pearson upper bound at level α from N' independent runs of Ex_b under the protocol's strategy, for Δ fixed in advance. Then $\hat{t}_b = \min\{1, \delta_{\mathcal{N}}(\varepsilon; \sqrt{R}\Delta/\sigma) + \Upsilon_\mathfrak{e}(\bar{\epsilon}_b)\}$ is sound at level α over every learning function, $\hat{t}_0$ for θ^{dev} and $\hat{t}_1$ for $\theta^{\mathrm{dev},\leftarrow}$. Each run costs $2R$ Train calls for client j plus the calls of the other nodes, and the protocol's strategy adds none. If no run exceeds Δ and $N' \geq \ln(1/\alpha)\exp((\sqrt{\mathfrak{e}} + \sqrt{\ln(1/\epsilon_{\mathrm{acc}})})^2)$ with $\epsilon_{\mathrm{acc}} \in (0,1]$, then $\hat{t}_b \leq \delta_{\mathcal{N}}(\varepsilon; \sqrt{R}\Delta/\sigma) + \epsilon_{\mathrm{acc}}$.

(c) Tightness of the transfer. In G_R^{agg} let $R \geq 2$, $B_A > 0$ and $B \geq 2\Delta$. For every $\iota > \sqrt{2\mathfrak{e}} + 1$ some learning function and B_A-deviating strategy have $\epsilon_0^{\mathrm{pr}}(\Delta) = \bar{\Phi}(\iota)$ and $\epsilon_0(\Delta) = \bar{\Phi}(\iota - \sqrt{2\mathfrak{e}})$, and $\ln(1/\epsilon_0(\Delta)) \leq \ln(1/\Upsilon_\mathfrak{e}(\epsilon_0^{\mathrm{pr}}(\Delta))) + \ln(\iota+1) + 1$. Let a function Υ' satisfy $\epsilon_0(\Delta) \leq \Upsilon'(\epsilon_0^{\mathrm{pr}}(\Delta))$ in every instance of G_R^{agg} with these R, B_A, σ, Δ and B. Then for every $p \in (0, \bar{\Phi}(\sqrt{2\mathfrak{e}} + 1))$, $\Upsilon'(p) \geq e^{-1}\Upsilon_\mathfrak{e}(p)/\sqrt{8\ln(1/p)}$, while $\Upsilon_\mathfrak{e}(p)/p = \exp(2\sqrt{\mathfrak{e}\ln(1/p)} - \mathfrak{e})$. So no such Υ' improves on $\Upsilon_\mathfrak{e}$ by more than a factor $e\sqrt{8\ln(1/p)}$, while $\Upsilon_\mathfrak{e}(p)/p$ is unbounded as $p \to 0$.

(d) Lower bound. In G_R^{agg} let $B < \infty$ and $B_A > 0$, and let L° return the starting model at every training input. Then $\theta_R^{\mathrm{agg}}(\mathsf{L}^\circ) = 0$, and every coupled pair at L° coincides. For $\mathfrak{c}, \eta \in (0,1)$ let $\mathfrak{q}_\eta$ solve $\bar{\Phi}(\mathfrak{q}_\eta) = \eta/4$, and let $J_3 = \min\{\lfloor e^{\mathfrak{c}^2 d/4} \rfloor, \lfloor \frac{\eta}{2} \exp(\frac{1}{2}((1-\mathfrak{c})\sqrt{2\mathfrak{e}} - \mathfrak{q}_\eta)_+^2) \rfloor\}$. If $J_3 \geq 1$, every certifier of θ_R^{agg} with N Train calls, in either discipline, sound at level α over every learning function, satisfies $\Pr_{\mathsf{L}^\circ}[\hat{t} \geq (1-\eta/2)\delta_{\mathcal{N}}(\varepsilon; B/\sigma)] \geq 1 - \alpha - N/J_3$. If $\mathfrak{e} < d/4$, $\mathfrak{c} = 2\sqrt{\mathfrak{e}/d}$, $(1-\mathfrak{c})\sqrt{2\mathfrak{e}} \geq \mathfrak{q}_\eta$ and the second argument of the minimum is at least one, then $\ln J_3 \geq \mathfrak{e}(1 - 4\sqrt{\mathfrak{e}/d}) - \mathfrak{q}_\eta\sqrt{2\mathfrak{e}} + \ln(\eta/4)$.

(e) Drawn seeds. In G_R^{agg} let $B < \infty$ and $1 \leq J_{\mathrm{cd}} \leq \lceil 2^{0.18d} \rceil/2$. Let a drawn-seed certifier of θ^{dev} with N Train calls be sound at level α over every learning function. Some learning function with all coupled pairs equal and $\theta_R^{\mathrm{agg}} = 0$ gives $\Pr[\hat{t} \geq 1 - (1+\gamma)(2J_{\mathrm{cd}} - 1)\bar{\Phi}(B/(2\sigma))] \geq 1 - \alpha - N(2N-1)/J_{\mathrm{cd}}$. There the certificate (b) outputs at most $\delta_{\mathcal{N}}(\varepsilon; \sqrt{R}\Delta/\sigma) + \Upsilon_{\mathfrak{e}}(\ln(1/\alpha)/N')$ with certainty. By the remark of def-certifier the bound binds certifiers of θ_R^{agg}.

Corollary, the price. In G_R^{agg} let $B < \infty$, $B_A > 0$, $\eta \in (0,1)$, $\epsilon_{\mathrm{acc}} \in (0,1]$ and $\alpha + \alpha' < 1$, with $\delta_{\mathcal{N}}(\varepsilon; \sqrt{R}\Delta/\sigma) + \epsilon_{\mathrm{acc}} < (1-\eta/2)\delta_{\mathcal{N}}(\varepsilon; B/\sigma)$. Call a certifier of θ_R^{agg} accurate at L° when it outputs at most $\delta_{\mathcal{N}}(\varepsilon; \sqrt{R}\Delta/\sigma) + \epsilon_{\mathrm{acc}}$ there with probability at least $1 - \alpha'$. Let $N^\star$ be the least number of Train calls of a certifier sound at level α over every learning function and accurate at L°. For $\mathfrak{c}$ with $J_3 \geq 1$, $(1-\alpha-\alpha')J_3 \leq N^\star \leq 4R\lceil \ln(2/\alpha) \exp((\sqrt{\mathfrak{e}} + \sqrt{\ln(1/\epsilon_{\mathrm{acc}})})^2) \rceil$. Hence $\ln N^\star = (1+o(1))\,\mathfrak{e}$ as $\mathfrak{e} \to \infty$ with $\mathfrak{e}/d \to 0$ and $\ln R = o(\mathfrak{e})$, with ϵ_{acc}, α, α' and η fixed. In units of $(R-1)B_A^2/\sigma^2$ the constant is one half.

Label. Proven, (a) to (e) and the corollary.

Reading. With $B_A = B = \sigma$, as for a deviating client in DP-FedAvg with noise multiplier one, $\mathfrak{e} = (R-1)/2$. Fix ε, η, α, α', ϵ_{acc} and $\sqrt{R}\Delta/\sigma$ so that the conditions of the corollary hold. Then $N^\star = e^{(1+o(1))(R-1)/2}$ as R grows with $R/d \to 0$ (proven). At $\varepsilon = 1$ it needs $\epsilon_{\mathrm{acc}} < \delta_{\mathcal{N}}(1;1) \approx 0.127$ (computed).

Without the oracle, a bound against a coalition that moves client j's next starting model through the aggregate needs the trained transformer's law at every reachable starting model. That set is a ball of dimension d around a Gaussian random walk. With the oracle, part (a) reduces every deviation of norm at most B_A in the additive inputs to one Bernoulli parameter of the protocol's own execution. Part (b) certifies it at a cost free of d, of B, of the state and of the architecture. Part (d) and the corollary concern the clipped instance and certifiers sound for every learning function and accurate as in the corollary. Among them the logarithm of the cost of (b) is optimal to first order when $\mathfrak{e} \ll d$ and $\ln R \ll \mathfrak{e}$.

Degenerate oracles. Part (a) holds under both adversary-side oracles, since each release's noise is independent of all that the coalition holds when it sets that round's input. Part (d) holds in both disciplines. At the reference function of (e), where $\theta_R^{\mathrm{agg}} = 0$, the certificate (b) outputs at most $\delta_{\mathcal{N}}(\varepsilon; \sqrt{R}\Delta/\sigma) + \epsilon_{\mathrm{acc}}$ with $2R\lceil \ln(1/\alpha) \exp((\sqrt{\mathfrak{e}} + \sqrt{\ln(1/\epsilon_{\mathrm{acc}})})^2) \rceil$ Train calls. Let $d \geq 6$ and J_{cd} be as in the Reading of thm-dataflow-composition (e)(ii). By (e), a drawn-seed certifier of θ^{dev} with N calls, where $N(2N-1) < \eta J_{\mathrm{cd}}$, outputs at least $1 - (1+\gamma)e^{-B^2/(16\sigma^2)}$ there with probability at least $1 - \alpha - \eta$. Let $\alpha + \eta < 1$ and $\delta_{\mathcal{N}}(\varepsilon; \sqrt{R}\Delta/\sigma) + \epsilon_{\mathrm{acc}} < 1 - (1+\gamma)e^{-B^2/(16\sigma^2)}$. Then no drawn-seed certifier of θ^{dev} with $N(2N-1) < \eta J_{\mathrm{cd}}$ matches the accuracy of (b), for instance at $B_A = 0$, large d and B/σ, $\epsilon_{\mathrm{acc}} < 1$ and small $\sqrt{R}\Delta/\sigma$ (proven).

Proof of (a). Fix a learning function, a B_A-deviating strategy, an adversary that plays it, and $b = 0$. The case $b = 1$ exchanges the hypotheses.

Step 0, no stop. The strategy that plays as the given one until it would stop, and then plays $\mathbf{u}_t = \mathbf{u}_t^{\mathrm{pr}}$ forever, is B_A-deviating and has an exceedance at least as large. So assume no stop.

Step 1, the execution as a function of $(\Omega, y_{<t})$. Work on the space of step 1 of the proof of thm-dataflow-composition (b), and let $\Omega = (\omega, \zeta_1, \ldots, \zeta_R)$, independent of $Z_1, \ldots, Z_R$. Consider Ex_0 under the given strategy and under the protocol's strategy. Three facts make the maps of step 1 of thm-dataflow-composition (b) the same under both, except for the coalition's input nodes and the adversary's state. The input nodes of round t have the single child r_t, so they reach later nodes only through y_t. Every other argument and value that the coalition sets is the protocol's. The randomness of each node is one coordinate of ω, whatever calls the adversary makes on its own. By induction over the order of G, the same measurable maps give c_t, W_t, W'_t, $\bar{o}_t$ and $\mathbf{u}_t^{\mathrm{pr}}$ as functions of $(\Omega, y_{<t})$ under both strategies. Write $\bar{y}_t = \bar{o}_t + f_t(c_t, W_t)$. Under the given strategy $\mathbf{u}_t$ is a function of the adversary's state, hence of $(\Omega, y_{<t})$, with $\|\mathbf{u}_t - \mathbf{u}_t^{\mathrm{pr}}\| \le B_A$. Under the protocol's strategy $\mathbf{u}_t = \mathbf{u}_t^{\mathrm{pr}}$. In both, $y_t = \mathbf{u}_t + \bar{y}_t + Z_t$. The exceedance reads c_t, W_t and W'_t for $t \le R$, and c_R is set before v_R runs. So $\{D_0^\star > \Delta\} = \{(\Omega, y_{<R}) \in \mathbb{A}\}$ for one measurable set $\mathbb{A}$, the same under both strategies.

Step 2, the density. Let $\mathbb{P}^{\mathrm{dev}}$ and $\mathbb{P}^{\mathrm{pr}}$ be the laws of $(\Omega, y_1, \ldots, y_{R-1})$ under the given and the protocol's strategy. Ω has the same law under both. Given $(\Omega, y_{<t})$, a function of $(\Omega, Z_{<t})$, the noise Z_t is independent. So y_t has law $\mathcal{N}(\mathbf{u}_t + \bar{y}_t, \sigma^2 I_d)$ under $\mathbb{P}^{\mathrm{dev}}$ and $\mathcal{N}(\mathbf{u}_t^{\mathrm{pr}} + \bar{y}_t, \sigma^2 I_d)$ under $\mathbb{P}^{\mathrm{pr}}$. Both laws have positive densities with respect to the law of Ω times Lebesgue measure, and $d\mathbb{P}^{\mathrm{dev}}/d\mathbb{P}^{\mathrm{pr}} = \prod_{t<R} \mathfrak{r}_t$ with $\mathfrak{r}_t = \varphi_\sigma(y_t - \bar{y}_t - \mathbf{u}_t)/\varphi_\sigma(y_t - \bar{y}_t - \mathbf{u}_t^{\mathrm{pr}})$.

Step 3, the moment. Write $\mathbb{E}_{\mathrm{pr}}$ for the expectation under $\mathbb{P}^{\mathrm{pr}}$, and let $\mathcal{F}_t^{\mathrm{rel}}$ be generated by Ω and $y_{<t}$. Under $\mathbb{P}^{\mathrm{pr}}$, given $\mathcal{F}_t^{\mathrm{rel}}$, the vector $y_t - \bar{y}_t - \mathbf{u}_t^{\mathrm{pr}}$ is $\mathcal{N}(0, \sigma^2 I_d)$ and $\mathbf{u}_t - \mathbf{u}_t^{\mathrm{pr}}$ is fixed. So (F7) gives $\mathbb{E}_{\mathrm{pr}}[\mathfrak{r}_t^{\varpi} \mid \mathcal{F}_t^{\mathrm{rel}}] \le \exp(\varpi(\varpi - 1)B_A^2/(2\sigma^2))$. The earlier factors are $\mathcal{F}_t^{\mathrm{rel}}$-measurable, so the tower property over $t < R$ gives $\mathbb{E}_{\mathrm{pr}}(d\mathbb{P}^{\mathrm{dev}}/d\mathbb{P}^{\mathrm{pr}})^{\varpi} \le e^{\varpi(\varpi-1)\mathfrak{e}}$.

Step 4, the transfer. $\epsilon_0(\Delta) = \mathbb{P}^{\mathrm{dev}}(\mathbb{A})$ and $\epsilon_0^{\mathrm{pr}}(\Delta) = \mathbb{P}^{\mathrm{pr}}(\mathbb{A})$, and (F7) on $\mathbb{A}$ gives the first bound. For the closed form let $p = \mathbb{P}^{\mathrm{pr}}(\mathbb{A}) > 0$ and $\mathfrak{l} = \ln(1/p)$. The logarithm of the bound is $-\mathfrak{l}(1 - 1/\varpi) + (\varpi - 1)\mathfrak{e}$, with derivative $\mathfrak{e} - \mathfrak{l}/\varpi^2$ in ϖ. If $\mathfrak{l} \le \mathfrak{e}$ the infimum over $\varpi > 1$ is 0. If $\mathfrak{l} > \mathfrak{e} > 0$ the minimum is at $\varpi = \sqrt{\mathfrak{l}/\mathfrak{e}}$, with value $-(\sqrt{\mathfrak{l}} - \sqrt{\mathfrak{e}})^2$. If $\mathfrak{e} = 0$ the infimum is $-\mathfrak{l}$. So the infimum of the bound is $\Upsilon_{\mathfrak{e}}(p)$, and $p = 0$ gives zero. The bounds on the targets are thm-dataflow-composition (b). □

Proof of (b). The experimenter holds every input, runs the other nodes, plays the protocol's strategy, and in round t calls $\mathsf{Train}(x(c_t, D_j^b, W_{t-1}), \perp)$ and then $\mathsf{Train}(x(c_t, D_j^{1-b}, W'_{t-1}), s)$ with the returned seed name s. The runs are independent by fidelity, so (F8) gives $\Pr[\epsilon_b^{\mathrm{pr}}(\Delta) \le \bar{\epsilon}_b] \ge 1 - \alpha$, and $\Upsilon_{\mathfrak{e}}$ is nondecreasing, so (a) gives soundness. With no exceedance, $\bar{\epsilon}_b \le \ln(1/\alpha)/N' \le \exp(-(\sqrt{\mathfrak{e}} + \sqrt{\ln(1/\epsilon_{\mathrm{acc}})})^2)$. Then $\sqrt{\ln(1/\bar{\epsilon}_b)} - \sqrt{\mathfrak{e}} \ge \sqrt{\ln(1/\epsilon_{\mathrm{acc}})}$, so $\Upsilon_{\mathfrak{e}}(\bar{\epsilon}_b) \le \epsilon_{\mathrm{acc}}$. □

Proof of (c). Let L return the starting model except at $x((R, m), D_j^1)$, where it returns $m + 2\Delta\mathbf{e}_1$, with $\mathbf{e}_1$ the first standard basis vector, when $\langle m - m_1, \mathbf{e}_1\rangle > \iota\sigma\sqrt{R-1}$. Take $\mathbf{u}_t = B_A\mathbf{e}_1$ for $t < R$ and $\mathbf{u}_R = 0$. Rounds $t < R$ have no exceedance, and round R has the exceedance $2\Delta > \Delta$ exactly on that event, since $B \ge 2\Delta$. The coordinate $\langle m_R - m_1, \mathbf{e}_1\rangle$ is $\mathcal{N}(0, (R-1)\sigma^2)$ under the protocol's strategy, with $R - 1 \ge 1$, and has mean $\sqrt{2\mathfrak{e}}\,\sigma\sqrt{R-1}$ under the given one. This gives the two probabilities. For the inequality let $p = \bar{\Phi}(\iota)$ and $r = \iota - \sqrt{2\mathfrak{e}}$, so $1 < r \le \iota$. (F6)(iii) gives $\ln(1/p) \ge \iota^2/2 > \mathfrak{e}$, so $\sqrt{\ln(1/p)} - \sqrt{\mathfrak{e}} \ge r/\sqrt{2}$ and $\ln(1/\Upsilon_{\mathfrak{e}}(p)) \ge r^2/2$. The lower bound of (F6)(iv) gives $\ln(1/\bar{\Phi}(r)) \le r^2/2 + \ln(r + 1/r) + \ln\sqrt{2\pi} \le r^2/2 + \ln(\iota + 1) + 1$. The two bounds give the inequality.

Every p in the range is $\bar{\Phi}(\iota)$ for one $\iota > \sqrt{2\mathfrak{e}} + 1$, since $\bar{\Phi}$ is continuous and strictly decreasing. Then $\iota + 1 < 2\iota \le \sqrt{8\ln(1/p)}$. Since $\Upsilon'(p) \ge \epsilon_0(\Delta)$, the inequality gives $\ln(1/\Upsilon'(p)) \le \ln(1/\Upsilon_{\mathfrak{e}}(p)) + \ln\sqrt{8\ln(1/p)} + 1$. The identity is $\ln(1/p) - (\sqrt{\ln(1/p)} - \sqrt{\mathfrak{e}})^2 = 2\sqrt{\mathfrak{e}\ln(1/p)} - \mathfrak{e}$, valid since $\ln(1/p) > \mathfrak{e}$. □

Proof of (d). At L° every release is $\mathbf{u}_t + Z_t$ under both hypotheses, so the view has one

law under both, and the model ignores the dataset. The code. Let $1 \le J \le J_3$. Draw $\mathbf{q}_1, \dots, \mathbf{q}_J$ independently and uniformly from $\{\pm d^{-1/2}\}^d$. For $k \ne l$, $d\langle \mathbf{q}_k, \mathbf{q}_l \rangle$ is a sum of d independent uniform signs, so Hoeffding's inequality gives $\Pr[|\langle \mathbf{q}_k, \mathbf{q}_l \rangle| \ge \mathfrak{c}] \le 2e^{-\mathfrak{c}^2 d/2}$. Over the $J(J-1)/2$ pairs a bad pair has probability below $J^2 e^{-\mathfrak{c}^2 d/2} \le 1$, since $J \le e^{\mathfrak{c}^2 d/4}$. So some draw gives unit vectors with $|\langle \mathbf{q}_k, \mathbf{q}_l \rangle| < \mathfrak{c}$ for all pairs.

The cells. Write $\mathfrak{m} = (R-1)B_A$ and $\mathfrak{v} = \sigma\sqrt{R-1}$, so $\mathfrak{m}/\mathfrak{v} = \sqrt{2\mathfrak{e}}$. Let V_k^{th} be the set of m with $\langle m - m_1, \mathbf{q}_k \rangle > \mathfrak{m} - \mathfrak{q}_\eta \mathfrak{v}$ and $\langle m - m_1, \mathbf{q}_l \rangle \le \mathfrak{m} - \mathfrak{q}_\eta \mathfrak{v}$ for every $l \ne k$. The cells are measurable and pairwise disjoint. For $\mathbf{g} \sim \mathcal{N}(0, \mathfrak{v}^2 I_d)$, the point $m_1 + \mathfrak{m}\mathbf{q}_k + \mathbf{g}$ leaves V_k^{th} only if $\langle \mathbf{g}, \mathbf{q}_k \rangle \le -\mathfrak{q}_\eta \mathfrak{v}$, or if $\langle \mathbf{g}, \mathbf{q}_l \rangle > (1-\mathfrak{c})\mathfrak{m} - \mathfrak{q}_\eta \mathfrak{v}$ for some $l \ne k$. $J_3 \ge 1$ forces $(1-\mathfrak{c})\sqrt{2\mathfrak{e}} > \mathfrak{q}_\eta$. So (F6)(iii) and the choice of J_3 bound $\Pr[m_1 + \mathfrak{m}\mathbf{q}_k + \mathbf{g} \notin V_k^{\mathrm{th}}]$ by $\eta/4 + \frac{J}{2}\exp(-\frac{1}{2}((1-\mathfrak{c})\sqrt{2\mathfrak{e}} - \mathfrak{q}_\eta)^2) \le \eta/2$.

The needles. Let $X_k = \{x((R,m), D_j^1) : m \in V_k^{\mathrm{th}}\}$, disjoint Borel sets since $m \mapsto x((R,m), D_j^1)$ is measurable and injective (Lusin and Souslin). Let L_k agree with L° except that $\mathsf{L}_k(x((R,m), D_j^1); \rho, \xi) = m + B\mathbf{e}_1$ for $m \in V_k^{\mathrm{th}}$. Take $\mathbf{u}_t = B_A \mathbf{q}_k$ for $t < R$, $\mathbf{u}_R = 0$, and the adversary that reads the releases. Rounds $t < R$ carry learner names other than ℓ_R, so $y_t = B_A \mathbf{q}_k + Z_t$ under both hypotheses. Hence $m_R = m_1 + \mathfrak{m}\mathbf{q}_k + \sum_{t<R} Z_t$ is a function of $y_{<R}$, with one law under both. In round R, $y_R = Z_R$ under H_0 and $y_R = B\mathbf{e}_1 \mathbf{1}[m_R \in V_k^{\mathrm{th}}] + Z_R$ under H_1. (F5) with common coordinate $y_{<R}$ and (F6)(i) give $\mathrm{H}_\gamma(P_0 \| P_1) = \Pr[m_R \in V_k^{\mathrm{th}}]\, \delta_{\mathcal{N}}(\varepsilon; B/\sigma) \ge (1 - \eta/2)\delta_{\mathcal{N}}(\varepsilon; B/\sigma)$. lem-needle-lower-bound gives the bound.

The expansion. With $\mathfrak{c} = 2\sqrt{\mathfrak{e}/d}$, $\mathfrak{c}^2 d/4 = \mathfrak{e}$. Let J' be the second argument of the minimum before rounding. Since $((1-\mathfrak{c})\sqrt{2\mathfrak{e}} - \mathfrak{q}_\eta)^2 \ge 2(1-\mathfrak{c})^2\mathfrak{e} - 2\sqrt{2\mathfrak{e}}\,\mathfrak{q}_\eta$, $\ln J' \ge (1-2\mathfrak{c})\mathfrak{e} - \mathfrak{q}_\eta\sqrt{2\mathfrak{e}} + \ln(\eta/2)$. Also $\ln J' < \mathfrak{e}$, so $J_3 = \lfloor J' \rfloor \ge J'/2$. □

Proof of (e). The pair code of thm-dataflow-composition (e)(ii), placed in round 1 and translated by m_1, gives the bound. The function returns the starting model in every later round. The adversary that plays the protocol's strategy and reads y_1 sees the round-1 model minus m_1 plus noise, since $\mathbf{u}_1 = 0$ and the clip acts as the identity on norm B. So (F2) carries the divergence of (e)(ii) to θ^{dev}. At the reference function the model at client j's training inputs does not depend on the dataset. So the view has one law under both hypotheses, $\theta_R^{\mathrm{agg}} = 0$, no run of Ex_b exceeds Δ, and $\bar{\epsilon}_b \le \ln(1/\alpha)/N'$. □

Proof of the corollary. A certifier that outputs below $\vartheta = (1 - \eta/2)\delta_{\mathcal{N}}(\varepsilon; B/\sigma)$ at L° with probability at least $1 - \alpha'$ has $\alpha' \ge 1 - \alpha - N/J_3$ by (d). For the upper bound, take $\max\{\hat{t}_0, \hat{t}_1\}$ with each certificate of (b) at level $\alpha/2$. A union bound makes it sound for θ_R^{agg}. In G_R^{agg} no other node trains, so each run costs $2R$ Train calls, and at L° no run exceeds Δ. The two bounds, the expansion in (d) and $\ln(4R) = o(\mathfrak{e})$ give the asymptotics. □

C.1.10 Replay-audit steering [thm-replay-audit-steer]

Setting. A training input $x = (\ell, m^{\mathrm{st}}, D, \mathcal{H})$ is fixed, and a commitment cm that the auditor can open binds the prover's dataset D. Steering through the choice of D lies outside the theorem. $(\mathcal{M}, \mathrm{dist})$ is a metric space whose closed balls are compact, with its Borel sets. $T \subseteq \mathcal{M}$ is a nonempty closed target set. A random oracle RO maps strings to $\mathcal{R}$ with independent values of law μ. The audit game with tolerance $\tau \ge 0$ and $n_{\mathrm{rep}} \ge 1$ replays has three steps.

1. The prover, with access to RO and $\mathcal{O}$, outputs a model m and a nonce nc.
2. The auditor sets $\rho = \mathsf{RO}(\mathrm{cm}, \mathrm{nc})$, makes n_{rep} calls $\mathsf{Train}(x, \rho)$ with answers $m_1^{\mathrm{rep}}, \dots, m_{n_{\mathrm{rep}}}^{\mathrm{rep}}$, and accepts when $\min_i \mathrm{dist}(m, m_i^{\mathrm{rep}}) \le \tau$.
3. The prover wins when $m \in T$ and the auditor accepts.

This is the goal steer. The prover is unbounded, may know L, and draws fresh hardware randomness in its Train calls. For $\rho \sim \mu$ and independent $\xi, \xi' \sim \pi$ let

$M = \mathsf{L}(x; \rho, \xi)$ and $M' = \mathsf{L}(x; \rho, \xi')$. The *reach*, the *spread* and the *joint reach* are $p_\tau(T) = \Pr[\mathrm{dist}(M, T) \le \tau]$, $\mathrm{sp}_\tau = \Pr[\mathrm{dist}(M, M') > \tau]$ and $p^\circ_\tau(T) = \Pr[\mathrm{dist}(M, T) \le \tau,\ \mathrm{dist}(M, M') \le \tau]$. Drawn-seed calls measure the reach, and coupled calls with one seed name measure the spread and the joint reach.

Statement. (a) Soundness. Every prover that makes q_{RO} queries to RO and any number of Train calls, and that may know L, wins with probability at most $(q_{\mathsf{RO}} + 1)\, n_{\mathrm{rep}}\, p_\tau(T)$.

(b) Completeness. The honest prover, which outputs $\mathsf{Train}(x, \mathsf{RO}(\mathrm{cm}, \mathrm{nc}))$ for a nonce nc, is accepted with probability at least $1 - \mathrm{sp}_\tau$.

(c) Attack. For $n_{\mathrm{nc}} \ge 1$ the grinding prover tries n_{nc} new nonces. It trains at each seed $\rho_k = \mathsf{RO}(\mathrm{cm}, \mathrm{nc}_k)$ and outputs a nearest point of T to the first run M_k within $\tau/2$ of T, with nc_k. With $p = p_{\tau/2}(T) > 0$ it wins with probability at least $\frac{p^\circ_{\tau/2}(T)}{p}(1 - (1-p)^{n_{\mathrm{nc}}}) \ge (1 - \mathrm{sp}_{\tau/2}/p)(1 - (1-p)^{n_{\mathrm{nc}}})$. Under deterministic kernels (a) holds without the factor n_{rep}, and stopping at τ matches it within a factor 3.2 when $n_{\mathrm{nc}} = q_{\mathsf{RO}}$ and $q_{\mathsf{RO}}\, p_\tau(T) \le 1$.

(d) Drawn-seed auditors. On the circle $\mathbb{R}/\mathbb{Z}$ the learning functions $\rho \mapsto \rho + a \bmod 1$ make the replay auditor with $\tau = 0$ complete and sound for point targets. A drawn-seed auditor complete up to λ' against all of them accepts some fixed point with probability at least $1 - \lambda' - \lambda$, for every $\lambda > 0$.

(e) Oblivious provers. If the learning function reads its seed at x through a random table unknown to the prover, q_T Train calls at x win with expected probability at most $(q_T + 1) n_{\mathrm{rep}} p_\tau(T)$. Train calls at other inputs and queries to RO are unrestricted.

Label. Proven, (a) to (e). Open, the gap from τ to $\tau/2$ under hardware randomness (open problem 7).

Without the oracle, an auditor cannot replay at a chosen seed. By (d), a drawn-seed auditor complete up to λ' for every rotation of the circle accepts some fixed point with probability at least $1 - \lambda' - \lambda$, for every $\lambda > 0$. Parts (a) to (c) bound the steering probability from both sides by reach, spread and joint reach. Nobody can write these down for a transformer, and Train calls measure them for any learning function.

Degenerate oracles. The prover holds $\mathsf{RO}(\mathrm{cm}, \mathrm{nc})$ and passes it to Train under the bottom oracle, so the top oracle adds nothing. So the test runs through the auditor, an honest party whose Train calls the theorem designs. In the coupled discipline the auditor has the soundness of (a) and the completeness of (b). Under drawn seeds the statement fails for the rotations of (d). Without the random oracle (a) fails, since a prover that picks its seed picks one whose run lands in T (proven, round-02/builder.md).

Proof of (a). Implement RO by lazy sampling. Let $\rho_1, \dots, \rho_{q_{\mathsf{RO}}+1}$ be independent with law μ. The k-th distinct point the prover queries receives ρ_k, and a point the auditor evaluates without a prior query receives the first unused one. This gives RO its law. Let k^* be the index of the value the auditor uses, and let independent $\xi_{k,i} \sim \pi$, for $k \le q_{\mathsf{RO}} + 1$ and $i \le n_{\mathrm{rep}}$, serve the auditor's i-th replay when $k^* = k$. The prover's transcript does not involve them, so the replays have the prescribed law. A win needs $m \in T$ and $\mathrm{dist}(m, m_i^{\mathrm{rep}}) \le \tau$ for some i, so $\mathrm{dist}(m_i^{\mathrm{rep}}, T) \le \tau$. So the win lies in the union of the events $\{\mathrm{dist}(\mathsf{L}(x; \rho_k, \xi_{k,i}), T) \le \tau\}$, each of probability $p_\tau(T)$ for any prover. A union bound finishes. □

Proof of (b). The seed $\mathsf{RO}(\mathrm{cm}, \mathrm{nc})$ has law μ. The honest model and the first replay are two runs at that seed with independent hardware randomness. The auditor rejects only if they lie more than τ apart, which has probability sp_τ. □

Proofs of (c) to (e). They are in round-02/builder.md, section Replay-audit steering, with the match under deterministic kernels in round-03/builder.md, section Replay-audit steering, corrections, and the notation of (e) in round-04/builder.md. The attack uses independent seeds from new nonces and the triangle inequality between its output, its run and the auditor's replay. □

C.1.11 Further accepted results

Each result below is accepted, and its pointer names the round files that hold its proof.

Perfect substitution [thm-perfect-substitution]. Against an adversary that samples L once per Train call, the real procedure gives the bottom oracle's joint law of honest values and transcript. So bottom-oracle bounds hold for the real procedure (proven, round-01/builder.md with round-02/builder.md).

Reference access [thm-reference-access]. On one node seen through a deterministic channel, the top oracle's best advantage is the coupled divergence under deterministic kernels. Under the bottom oracle no attack reaches the total variation at a rate uniform over learning functions when the views can be atomless (proven, round-01/builder.md and round-02/builder.md).

Atomless two-point lemma [lem-atomless-two-point]. A drawn-seed certifier of TV through a deterministic channel, sound for all learning functions, outputs one with probability at least $1 - \alpha$ on atomless views (proven). Its proof is in round-01/builder.md, corrected in round-02 and round-03.

Model banks [def-model-bank]. A design lists training inputs with seeds, each fixed or drawn, and a bank stores one run per entry. Its empirical oracle answers a Train call with the first unused matching entry and refuses a call that no unused entry matches. It agrees with a learning oracle up to its first refusal [lem-bank-fidelity] (proven, round-01/builder.md). A bank on random halves of a pool estimates the random-subset game for all records at once, for a finite set of tests fixed before the bank is built [thm-bank-soundness]. That game can differ by one from the fixed-dataset game. The estimate (a), the gap (b) and the first half of (d) are proven. That half bounds from below the Train calls that certify every record of a fixed dataset through a Gaussian release. Part (c), on shadow models in the random-subset game, is heuristic (round-01/builder.md with round-02 to round-05).

Active-choice dichotomy [thm-active-choice-dichotomy]. Let the coalition pick choices from admissible sets. Let a certifier be sound for every learning function and accurate where the leakage is zero, to a fixed level in $(0, 1)$, at error probabilities summing below one. In some procedure it then needs Train calls linear in the number of distinct choices, or of choice sequences with state. Coupled certificates achieve this up to a logarithmic factor. No number suffices for an infinite set (proven, round-03/builder.md and round-04/builder.md).

Record-level certificate [thm-record-level-certificate]. One shared base certifies the fraction of a fixed dataset's records whose leakage to a passive coalition exceeds a threshold. Its cost is free of the dataset size and d (proven, round-03/builder.md with round-04/builder.md).

Smoothed-choice certificate [thm-smoothed-choice-certificate]. In G_R^{agg} with $B < \infty$ a covering of smoothed tails certifies a coalition client with inputs of norm at most B_A at a cost exponential in d (proven, round-04/builder.md with round-05/builder.md). It costs less than thm-bounded-deviation-transfer (b) only when $\mathfrak{e}$ exceeds about $d \ln(R^{3/2}\sqrt{d}/\epsilon_{\mathrm{acc}})$ (heuristic comparison of the two costs).

C.1.12 Open problems

A certifier is uniformly accurate when its output lies in $[\theta, \theta + \epsilon_{\mathrm{acc}}]$ with probability at least $1 - \alpha$ at every learning function, at a fixed number of Train calls. `theory/CURRENT.md` states each problem in full.

1. Sum channel. Let $n_h \geq 1$ further honest clients, isolated and stateless, add their clipped updates to the releases of G_R^{agg} with $B_A = 0$. Is there a functional of L with three properties? It bounds every passive adversary as ϵ_0 does in thm-dataflow-composition (b). It is never above ϵ_0 and lies below it at some learning function. Some coupled-discipline certifier is uniformly accurate for it with Train calls fixed by α, ϵ_{acc}, R and n_h.
2. Drawn-seed price above d. When $B^2/\sigma^2 > d$, is there a drawn-seed certifier of the divergence of thm-dataflow-composition (e), uniformly accurate over $\mathfrak{L}_B$, with $e^{O(d)}\,\mathrm{poly}(1/\epsilon_{\mathrm{acc}}, \ln(1/\alpha))$ calls?
3. Shadow lower bound. Does some family of learning functions force every shadow-model attack within ϵ_{acc} of $\mathrm{Adv}^{\mathrm{bb}}(q_E)$ to use order $C^{q_E} \ln|U|/\epsilon_{\mathrm{acc}}^2$ Train calls?
4. Multi-round drawn seeds. In G_R^{agg} with $B_A = 0$ and $B < \infty$, is there a uniformly accurate drawn-seed certifier of $\mathrm{H}_\gamma(P_0 \| P_1)$ whose Train calls depend on R, B/σ, ϵ_{acc} and α and not on d?
5. The price at a fixed ratio. Fix $\mathfrak{a} = \mathfrak{e}/d$, ϵ_{acc}, α, α' and η, and let $\mathfrak{e} \to \infty$ with $\ln R = o(\mathfrak{e})$. Let J^{fb} be the largest number of disjoint Borel sets such that, at L°, some B_A-deviating strategy without oracle calls puts m_R in each with probability at least $1 - \eta/2$. Do $\ln N^\star/\mathfrak{e}$ and $\ln J^{\mathrm{fb}}/\mathfrak{e}$ have the same limit? Known (proven), under the hypotheses of the corollary, $N^\star \geq (1 - \alpha - \alpha')J^{\mathrm{fb}}$ by the proof of thm-bounded-deviation-transfer (d), and $(1 + 2\sqrt{\mathfrak{a}})^{-2} \leq \liminf \ln N^\star/\mathfrak{e} \leq \limsup \ln N^\star/\mathfrak{e} \leq 1$. By Shannon's coding theorem the second limit is the capacity per unit of $\mathfrak{e}$ of the channel from the coalition's inputs to m_R with the releases fed back (heuristic). It exceeds the value $\ln(1 + 2\mathfrak{a})/(2\mathfrak{a})$ without feedback at every ratio computed (heuristic).
6. Threshold CKKS. Does threshold CKKS with smudging realize the ideal releases under RLWE with negligible error, against semi-honest and, with six additions, deviating coalitions?
7. The gap from τ to $\tau/2$. Under hardware randomness, does some prover always win with probability at least a universal constant times $\min\{1, q_{\mathsf{RO}} p_\tau(T)\}$?
8. Seed maps. Under deterministic kernels, with a deterministic channel and atomless views, does some seed map make the coupling of thm-coupling-certificate (c)(ii) exact for $\mathrm{TV}(P_0, P_1)$ in the limit?
9. Finite seeds. Does each impossibility proven here with an atomless seed hold on an n_{bit}-bit seed space, with an additive term that vanishes as n_{bit} grows at fixed N?

C.1.13 Prior work

Program logics build couplings for known code (Barthe et al., LICS 2016), and the coupling certificate samples them for an opaque learning function. The drawn-seed lower bounds are the atomless form of the impossibility of testing differential privacy from samples (Gilbert and McMillan, Allerton 2018, Theorems 10 and 12). The composition rests on Gaussian differential privacy (Dong, Roth and Su, from memory), and the transfer on the Rényi divergence of the Gaussian mechanism (Mironov, CSF 2017). The replay audit applies Fiat and Shamir grinding to proof of learning (Jia et al., IEEE S&P 2021), whose spoofs motivate the random oracle (Zhang et al., IEEE S&P 2022). Novelty is open until the auditing literature (Jagielski et al., NeurIPS 2020, Steinke et al., NeurIPS 2023) is read.